

LPM Manual

Laser Projector LP-HFD2

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Foreword

Dear Customer,

The following user manual is there to help you use your laser projector optimally. Following these instructions will allow you to increase your equipment's durability and avoid possible dangers.

The equipment is state of the art, developed and constructed to the latest technology. It was thoroughly checked prior to delivery, to ensure faultless functioning. It is only to be used for what it was designed to do. Improper use thereof will result in the termination of all warranty claims. The user will be responsible for any hazards or possible damages resulting from improper use. Any persons working on and with the equipment must have read and understood the corresponding chapters of the user manual, especially the "**Safety**" section.

In addition, all staff need to be informed of the risks involved (BGV A 1, 2. Chapter, § 2 and § 4). The operator needs to make sure that the operating instructions have been understood. The operating instructions must be kept ready to hand at the place where the device is used!

Safety

Safety instructions

Safety instructions help to prevent accidents and are there to ensure occupational health and safety among employees. To protect and ensure the safety of yourself and your work colleagues, it is necessary that you observe the instructions carefully. Be cautious when at work; remind yourself that dangers ahead aren't always obvious. It is the user's responsibility to ensure that he or the person working with the equipment, are made aware of the possible health hazards from the exposure to laser radiation.

Accident prevention regulations, as well as this handbook, should be easily and quickly accessible. (BGV A1, chapter 2, § 12).

The symbols of decree and warnings used in this handbook are in compliance with the legal requisite of providing health and safety signs at work, BGV A 8:



Please observe the instructions in the user manual carefully!



Please wear laser protective goggles when laser beam is active.



Disconnect the mains power plug before opening the projector!

Carry out work activities on energised parts of electrical installations and equipment only when in a de-energised state and ensure this condition for the duration of the work!



Warning! General hazard advice!



Warning! Dangerous electrical voltage!



Warning! Laser radiation! Do not look directly into the laser beam or attempt to view it using optical instruments. Laser class 2M.



Warning! Electro-statically sensitive components. Ensure correct earthing. Avoid friction!



Warning! Danger of falling! Ensure sufficient stability and a secure hold during all electrical and mechanical work.

Electronic safety instructions

The following safety instructions are in compliance with the legal accident prevention regulations for electrical systems and operating resources.

The user has the responsibility to ensure that “the electrical systems and operating materials are not modified, serviced or maintained, or operated by any other person, other than a trained electrician“, in accordance to BGV A3, § 3 (**DIN VDE 0105**)

The user has the responsibility to ensure that the electrical systems and operating resources are conducted in compliance with the electro-technical regulations.

In accordance with BGV A3, § 6, no work should be performed on energized (live) parts of electrical installations, equipment or operating resources without complying with the regulations:

Before beginning work on active parts of electrical installations and operating resources, zero-potential status (or de-energised condition) must be ensured beforehand and should be maintained throughout for the duration of the work performed. The user may assign only a qualified operator for working with electrical equipment under voltage. (**DIN VDE 0105**)

In the case of any disturbances in the electrical power supply, please switch off the equipment.

The laser projector contains delicate components that are electro-statically sensitive. Therefore, please do not attempt to repair the projector yourself. Take appropriate measures against electrostatic discharges and electrical fields in accordance with **DIN EN 61340 - 5 -1, ie. VDE 0300 - 5 - 1**.

Take care to ensure correct earthing! Avoid friction!

Safety instructions for lasers

The following safety instructions are in compliance with the legal accident prevention regulations for laser radiation BGV B 2.

Make sure you know which laser class has been allocated to your equipment. The type plate (silver label) on your equipment will verify this. Laser projectors Red / Green / BASIC are categorised as class 2M.

The laser class 2M is defined as follows - according to BGV B 2, § 2, and is in compliance with the following security measures:

The accessible laser radiation lies within the visible spectral range (400 nm to 700 nm).

It is safe for the human eye for short periods of exposure to radiation of max. 0.25s (blink reflex).

Therefore, Class-2 laser equipment may be used without any special protective measures, provided that the application requires neither deliberate eye contact for more than 0.25 s nor repeated eye contact with the direct or reflected laser radiation.

The exposure limit for class 2M lasers with a continuous radiation (GZS) is 1mW.

Safety precautions

Should the laser beam of the Class-2 laser equipment be employed in the working and traffic areas, the user has to ensure that the laser area is made clearly visible and should be permanently marked (BGV B 2, § 7 Accident Prevention Regulations (laser light)).

With laser equipment in Class 2, there is generally no need to further secure the laser area, provided that the exposure to radiation of employees during operation is only occasional and no optical devices are used to look at the light beam.

The requirement for marking is met with the warning sign W 10 „**Caution! Laser radiation**“, and it is equipped with a supplementary sign that corresponds with attachment 2 of the UVV requirements for provision of health and safety signs at work.



Figure 1: Warning sign W10 with supplementary sign

The user has to ensure, with the aid of technical and organisational measures, that any exposure to the laser radiation above the maximum permissible dose, including reflected or strayed light, is prevented in accordance with BGV B 2, § 8. Laser radiation emitted from laser classes 2, 3A, 3B or 4 should only extend as far as necessary for the nature of the task. The laser beam must be limited at the end of the useful range by a diffusely reflecting target area in such a manner, so as to minimise the danger from direct or diffuse reflection.

The area outside the operating range, where the unshielded laser beam radiates, should be reduced as far as possible and should remain out of bounds, particularly in the areas above and below eye level.

For the protection against dangerous reflections, the tools, accessories and calibrating equipment used within the laser area should not feature highly reflective surfaces. Likewise, persons present in the laser area should refrain from carrying items that are highly reflective. Optical components should be unalterably installed, to keep the laser beam within the predetermined optical path. No mirrors are allowed within this optical path, as it could produce dangerous reflections.

Suitable eye shielding, corresponding with the standard DIN EN 207, offer protection against direct, reflected or diffusely scattered laser radiation. However, any viewing directly into the beam of the laser should be avoided, even with the use of protective eye shielding.

Safety guidance

The user has to ensure that any persons working with lasers of the laser classes 2 – 4 are instructed about the correct safety measures. These safety instructions should be given once every year, in accordance with § 7 Abs. 2 UVV "General regulations" (VBG 1)

Accident prevention regulations

The following BG regulations / accident prevention regulations are to be observed:

A – General regulations:

BGV A 1 (VGB 1): Principles of Prevention BGV A 3 (VGB 4):
Electrical systems and equipment

B – Exposures:

BGV B2 (VGB 93): Laser radiation

Applications

The LP-HFD22 laser projector is used for the projection of polygons of any size, generated from manufacturing or production files. These files are exclusively vector files. Pixel graphics, such as bitmaps, cannot be projected by the laser projector.

The projector is able to display the object in the correct position onto various work piece heights. The projected object can also be moved and rotated.

The laser projector offers the possibility of filing a drawing in several plot buffers, which is especially helpful for a quick cyclic projection of different objects one after the other.

Delivery

Unpack your laser projector, check that all parts are present and visually inspect the projector for damage. Try to avoid excessive shock or dirt while unpacking.

The following parts should be included in your delivery:

- 1 Red/Green laser projector
- 1 Power supply cable with European plug
- 1 LPM Operating Software
- 1 Manual
- 4 20° – 40° glass reflectors
- 1 Seven-fold glass reflector disc
- 4 Numbered calibration markers
- 1 Metal drill bit
- 1 Stone/Concrete drill bit
- 1 Strip of reflector foil
- 1 Key for the key switch
- 1 Rittal Lid Key

Note:

The Rittal key is delivered with the power supply cable while the key for the key switch and the reflectors are found in a plastic bag.

If you have any complaints, please contact your supplier immediately.

Optional Delivery Parts

The following parts are optional:

Universal Mount:

- 1 FK40 flange clamp with fit
- 1 Adapter plate for FK40
- 3 M8x40mm cylindrical hexagonal screws
- 3 M8 spring washers for the cylindrical hexagonal screws
- 1 M8x40mm eye bolt
- 4 M8 flat washers
- 4 M6x30mm cylindrical hexagonal screws
- 4 M6 spring washers for the cylindrical hexagonal screws
- 4 M6 flat washers
- 1 300mm steel pipe
- 2 150mm steel pipe
- 1 Flange base piece
- 2 K40 cross clamps
- 2 ISO 8734 hardened dowel pin, d = 6, l = 30

Data Cable:

- 20 meter Ethernet crossover cable
- 2 meter Ethernet patch cable
- 15 meter null modem serial cable

Remote Control:

- 4 button wired remote control and software extension
- Single wireless remote control and software extension
- Triple wireless remote control and software extension
- 3 Ports I/O and software extension

I/Os:

- USB I/O adapter for digital signals
- Bar code reader

Cooling Options:

- Air cooling hose, passive

  Hose, hose clamps, adapter plate and tubing

- Peltier cooler, active

  Peltier cooler for LP-HFD2 as a retrofit kit. Integrated in the housing cover, external power supply.

  Special projector fitted with water cooling aluminium block

  This option is not retro-fitable.

- External interface (PLC):
- Operation of the laser projector has 11 galvanically isolated control lines (3 bit status, 8 bit data) e.g.: PLC or remote control cable.

An overview of our comprehensive software accessories can be found in the "Accessories" section.

Images and Component Descriptions

The following images give you an overview of the structure of your projector:

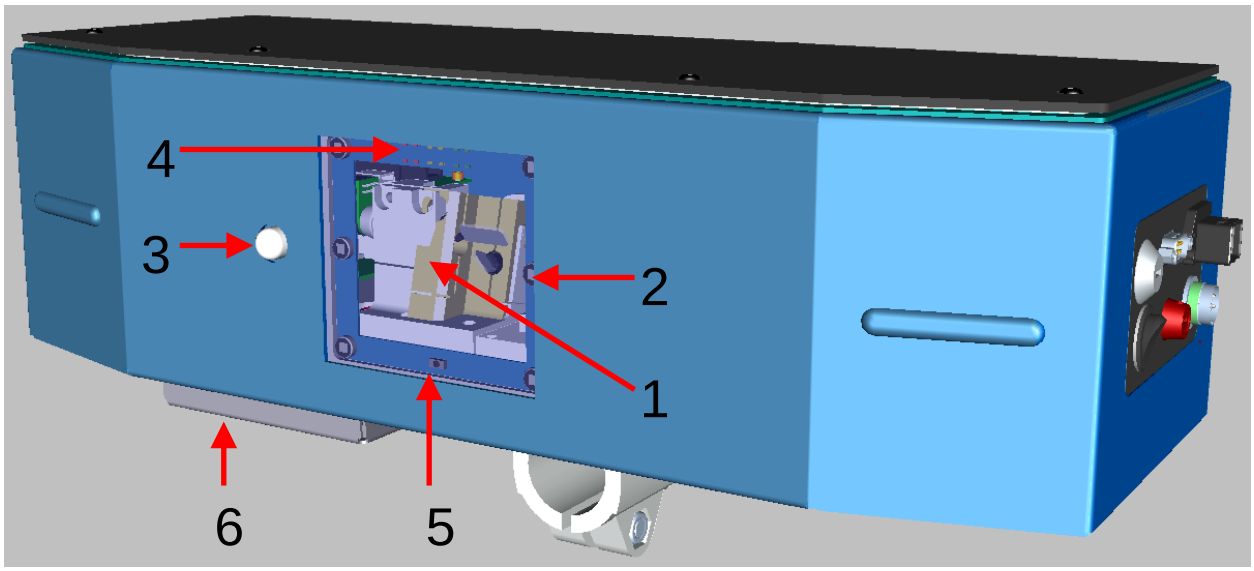


Figure 2: Front view of the laser projector

1. Laser exit window: The laser beam exits through this window.
2. Reflection Sensor: Detects laser reflections, aids in the drift compensation and calibration searches, allows the user menu control with reflective tape.
3. Motion Sensor: Switches laser beam off when it detects movement in the projection area of the projector.
4. Status Lights: Displays the current operating status of the device.
5. Sensor for the IR-remote control
6. Heat Venting Fan: Used to dissipate heat and prevents overheating of the projector. Ensure good air circulation to the fan.



Figure 3: Side view of the laser projector, including the data interface

1. DC-Power Supply connector
2. Fuse holder
3. On/Off switch.
4. Key switch: Used to turn the laser source on or off.
5. RS 232 interface. (serial mode connector)
6. RJ45 CAT 6 E network port: Used for Ethernet communications

Performance test

Check that the new projector is properly functional, before installing it permanently. You can check the performance of the projector by testing if your new laser projector switches to the desired communications mode.



Please ensure correct earthing!



The control computer and the laser projector must be connected to the same earth potential! Take precaution against static discharge, as per DIN EN 61340 – 5 – 1 / VDE 0300 – 5 – 1.

Step 1 – Power supply for the projector



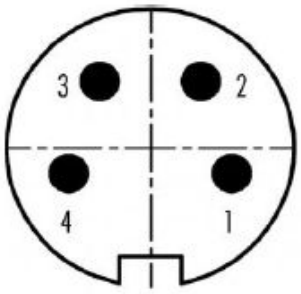
The power adapter for the laser projector has an input voltage of 100-277 V (AC), 47-63 Hz, max 1.8 A. The Output voltage value is 24 V (DC) @ max. 5A.

Connect the components as follows:

1. Power supply for the projector:

The power adapter has a power supply with a wide voltage range, the input voltage being 100-277 V (AC), 47-63 Hz. Output: 24V (DC) and 5A max. Whilst on standby, the laser projector consumes less than 60W, during use a maximum of 120W of electrical output. The mains connection is found on the right hand side of the projector. The device, including the power adapter passed the EMC measures.

Connect the mains switch to the projector's mains plug/connection. Then insert the earthed plug into a power socket. The electricity supply can be interrupted at any time, by removing the mains plug/connection.

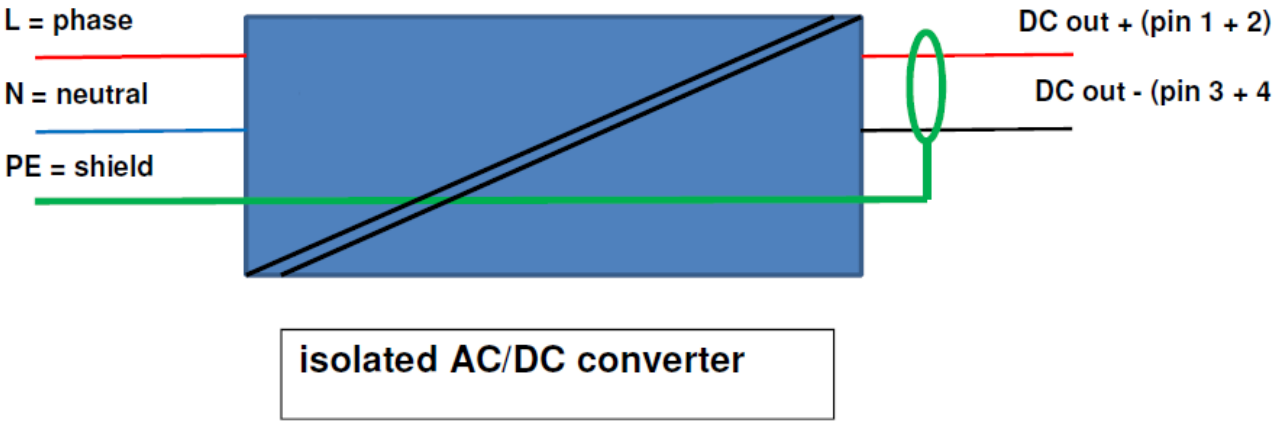


1: +24V DC +/-5%
2: +24V DC +/-5% (optional)
3: GND
4: GND (optional)

Figure 4: Pin assignment of the power supply cable

If using different type of 24V DC power supply, please ensure following:

- The AC ground wire must be galvanic decoupled from DC ground.



- Use a shielded cable between the 24V DC-source and the device.
- The shield of the cable must be connected to the plug of the device.
- The used DC adapter must perform the European EMC guidelines.
- The supply must be in the range of +/-5%

Recommended cable length between 24V DC adapter and device:

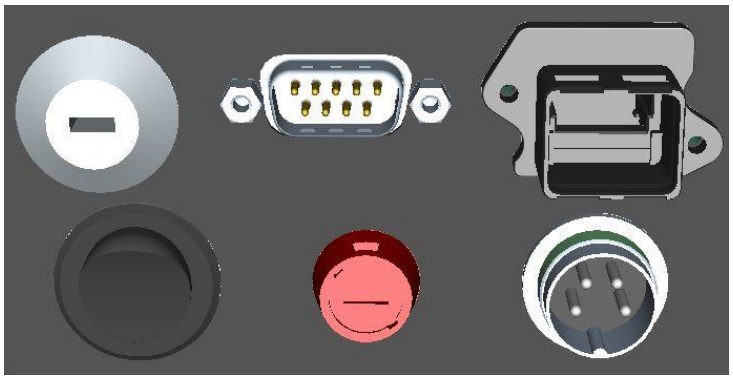
As the DC Voltage is decreasing by the length of the cable, the length and cross section of the cable must be observed in order to ensure the power will be in range of 24V +/- 5%.

Cross section	Cord amount	Maximum cable length *1)
AWG18	2 Leiter (Pin 1 & Pin 3)	6m
AWG18	4 Leiter (Pin1&2; Pin3&4)	12m
AWG16	2 Leiter (Pin 1 & Pin 3)	10m
AWG16	4 Leiter (Pin1&2; Pin3&4)	20m
0,75mm²	2 Leiter (Pin 1 & Pin 3)	5,5m
0,75mm²	4 Leiter (Pin1&2; Pin3&4)	11m
1mm²	2 Leiter (Pin 1 & Pin 3)	7,5m
1mm²	4 Leiter (Pin1&2; Pin3&4)	15m
1,5mm²	2 Leiter (Pin 1 & Pin 3)	11m
1,5mm²	4 Leiter (Pin1&2; Pin3&4)	22m

*1) Für U = 24V; ΔU_{max} = 5%; $K_{(Cu)}$=58S*m/mm²

Step 2 – Data line from PC to projector

A data line cable will help you to connect your control computer through Ethernet or via serial interface on your projector.



1. 9-pole SUB-D-connection (serial interface RS232)
2. Ethernet interface

Figure 5: Projector Interface

Step 3 – Switching on the laser projector for the first time.

Switch on the laser projector.

Ensure that the key switch is turned on, i.e. locked in place. This aids the future on-and-off switching of the laser source.

Observe the status display during the starting process. This is situated at the front of the projector. When the green LED flashes, the laser projector is ready for use and is ready for receiving serial data. When the green LED flashes AND the red LED is steadily on, the projector is ready for use and is ready for receiving data via network. (Meaning of the status indicators are listed in the chapter "Status display on the projector").

The software LPM can now be installed and you can carry out a communications test.

After a successful installation and electrical connection, the control software LPM will now register the projector.

If you want to install the software LPM now, please see chapter "Installing the LPM software" for further details.

Quick guide

This quick guide will help you gain an overview on which steps to follow to reach the final commissioning of your laser projector. For more detailed information, please see the respective links included.

1. Remove the projector from its packaging and check that it is complete.

2. For more information, please see chapter
   "Delivery"

3. Familiarise yourself with the projector, using the illustrations as reference.
   "Images and Components Description"

4. Turn on the projector and conduct a performance test.
   "Performance test"

5. Install the projector
   "Installation of the laser projector"

6. Install the software LPM
   "Installation of the LPM software"

7. Register your projector
   "Projector Identification"

8. Set up your communications mode
   "Serial" or "Network"

9. Calibrate the laser projector with quick calibration
   "Quick Calibration"

10. Register the modules of your LPM software.
    "Software registration"

11. Import your data

12. For more information on importing and extending/upgrading modules, please see chapter

    “**Accessories**”

Installation of the laser projector



The conditions for an exact installation are set out in the following chapters and should be strictly adhered to. Take your time to read through them and with thorough prior consideration.

Installation requirements

The laser projector should be assembled centrally above the working area. The projector's lode point (i.e. the lode) should be placed in the centre of the working area. The lode point of the projector is situated approximately in the centre of the laser emission window, proportional to a perpendicular line. The installation site must be motionless and free of vibration, be able to carry the weight of the laser projector (approx. 10kg) and should also maintain temperature stability throughout. Please observe other factors that could affect the mounting height, such as snow load on the roof.

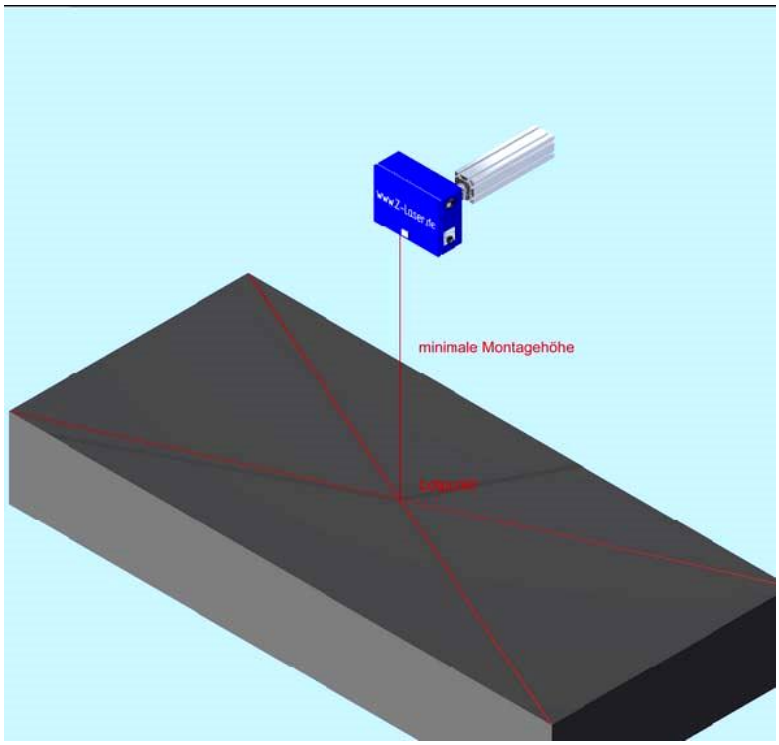


Figure 6: Assembly sketch

The laser projector should be mounted perpendicularly above the centre of the working area. The laser emission window should remain as parallel to the working area as possible. The **minimal mounting height** is identified as the distance between the emission window of the laser beam on the projector, to the top of the projection surface, i.e. the top of the material supports on the projection surface.

Choose your assembly area and mounting area in such a way, that the projection area can be covered completely, even when opened to the maximum projection angle. Please refer to the following diagrams:

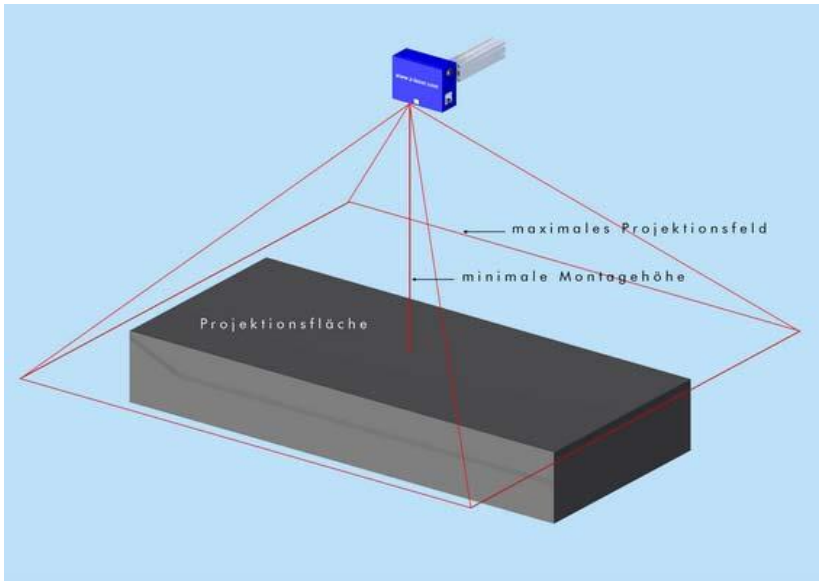


Figure 7: Correct mounting height

The area of projection (grey) will be enclosed completely by the projection field, when opened to the maximum projection angle.

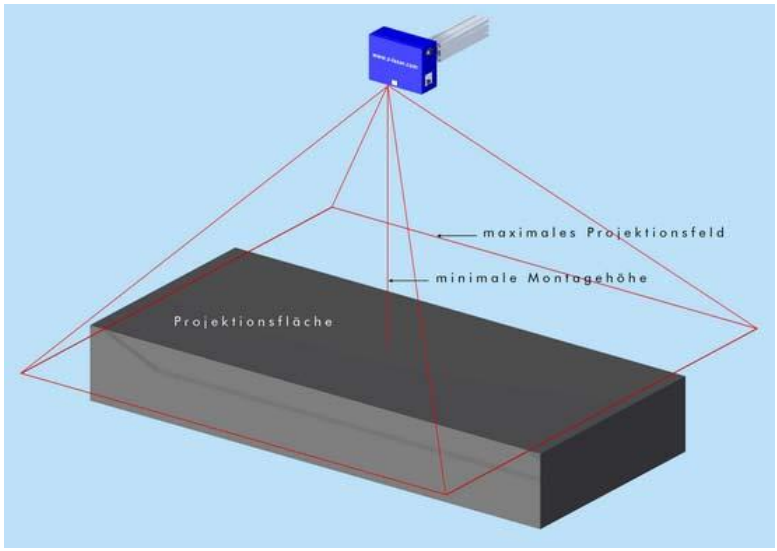


Figure 8: Mounting height too low!

The area of projection (grey) **WILL NOT** be enclosed completely by the projection field, when opened to the maximum projection angle.

The **ideal** mounting height that results solely from perpendicular, centred installations of the laser projector above the projection area can be determined from the following table. You can find the maximal length of your required projection size in the column "**Projection length in an axis direction"** and you can now compare which height is derived from it in the next column "**Mounting Height**".

Table for determining mounting height

Projection length in an axis direction in mm		Mounting Height in mm
at α = 76°	at α = 80°	Mounting Height
1.562	1.678	1000
2.343	2.517	1500
3.125	3.356	2000
3.906	4.195	2500
4.687	5.034	3000
5.469	5.873	3500
6.250	6.712	4000
7.031	7.551	4500
7.812	8.391	5000
8.594	9.230	5500
9.375	10.069	6000
10.156	10.908	6500
10.938	11.747	7000
11.719	12.586	7500
12.500	13.425	8000
13.281	14.264	8500
14.063	15.103	9000

Please note that the data shows the net value. The heights for material supports and the laser projector should be considered separately. You can calculate the exact minimal mounting height of the maximum operating angle with the aid of the following formula:

$h = f * 0{,}5 / \tan(\alpha / 2) - h_p - h_m$

h = Mounting height in [mm], taking all parameters into consideration

f = longest side of the projection surface in [mm]

α = optical opening angle (max. 80°); we recommend an opening angle of 76°

hp = height of projector 320 mm

hm = material height in [mm] (top edge of the strongest material)

Thus, you get a mounting height based on the length of the required projection surface of, for example, 5000mm and an opening angle of 76°:

h = 5000 mm * 0,5 / tan (76° / 2) h = 3.199,85 mm

As the projector is installed on the ceiling, deduct the height of the projector from the height of the support material, to determine the minimal height that should not undercut the **minimal distance** between the working area and the laser emission window.

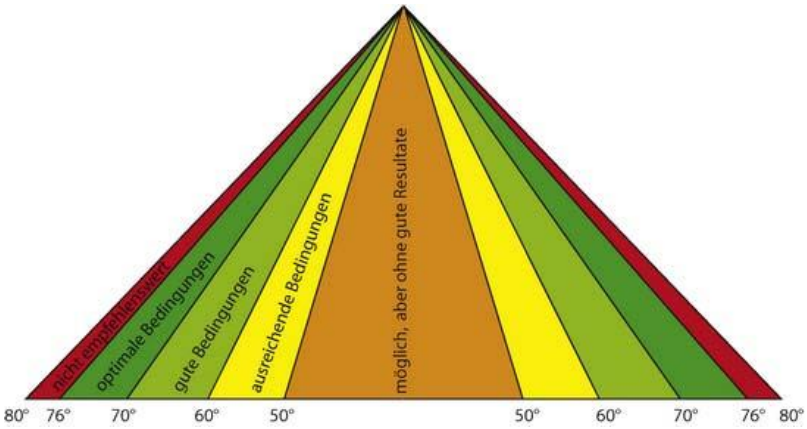


Figure 9: Illustration of projection conditions at different opening angles

80°	- not recommended!
70 - 76°	- optimal conditions
60 - 70°	- good conditions
50 – 60°	- adequate conditions
50°	- possible, but without good results

Based on this scheme, you can see that the **best** conditions for the projector are at an opening angle between 70 – 76°. However, you can expect to find good working conditions even at 60-70°.

You should consider the possibility beforehand, whether to operate your projector via **Ethernet transmission** or via a **serial interface.**

Should you choose to operate your projector via network, an Ethernet connection with RJ45 cabling, with CAT 6/FTP specification (shielded/foiled twisted pair) should be available in close proximity.

To control the laser projector with a serial CAT 6/FTP connection, you will need a cable that is a maximum of 15m long.

Installation/Assembly instructions

The laser projector can be delivered with a universal mount, if requested.



Depending on national regulation, there is an obligation to secure the laser projector during the mounting and/or during long-term operation, to avoid it accidentally falling down.



Please ensure that there is sufficient stability and a secure hold during all electrical and mechanical work! Please adhere to the relevant accident prevention regulations.

Assembly of the universal mounting bracket

To mount the universal suspension, you will need:

- materials to fasten the base clamp
- SW6 hexagonal Allen key
- SW8 hexagonal Allen key

1. After you have established the final installation site, start by securing the base clamp of the universal bracket on to a stable carrying element (e.g. Steel beams, concrete slabs etc.). Please note that the connection has to be strong enough to be able to carry a static weight, hanging perpendicular to the floor, of at least **150kg**. The drilling template provided will help you mark the drilling holes.

Drilling pattern

2. Assemble the first steel pipe (150mm) with the base clamp.
3. Attach the first cross clamp to the first steel pipe and tighten the screws.
4. Assemble the second steel pipe (150mm) with the first cross clamp.
5. Attach the second cross clamp to the second steel pipe and tighten the screws.
6. Assemble the third steel pipe (300mm) with the second cross clamp.
7. Mount the adaptor plate on the top of the unit.
8. Fix the flange clamp FK40 on to the adaptor plate.
9. Hang up the projector with the flange clamp on the third steel pipe (300mm) of the previously installed universal mounting and tighten the screws.

Please take care not to damage the laser emission window when mounting the projector!

10. Now align the projector, so that it runs parallel to the axis of your projection area.



11. Loosen the screws one at a time and tighten back after the position has been corrected.

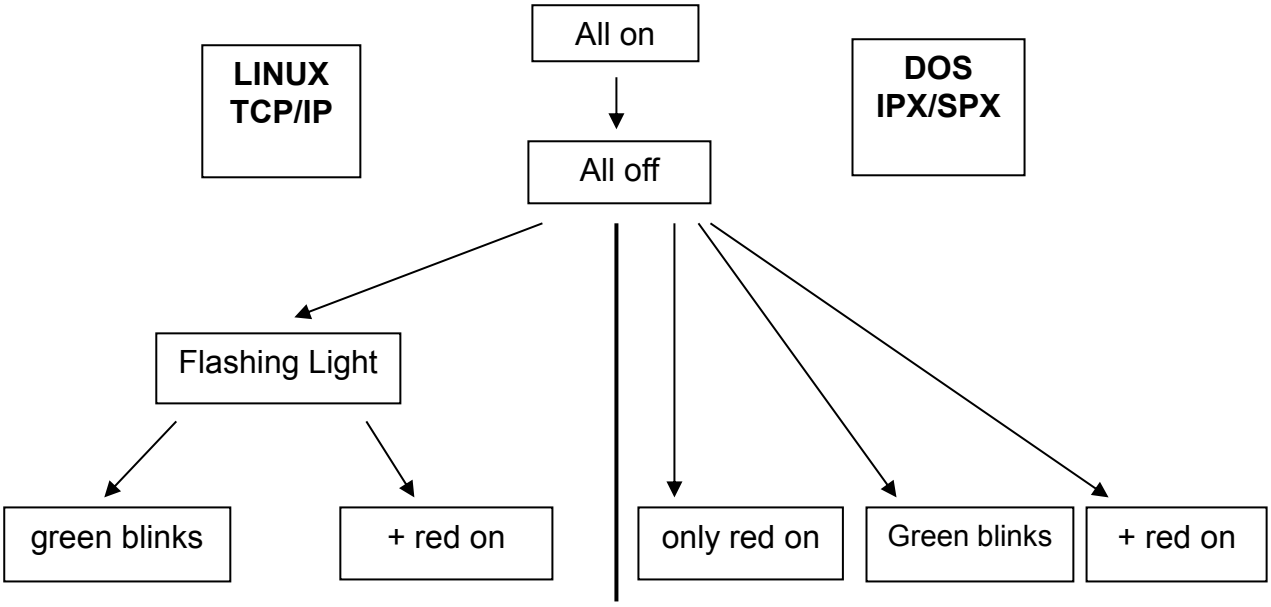


Figure 10: Fully assembled universal mount

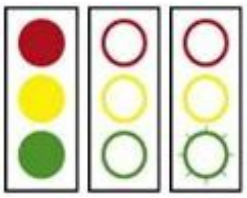
Status display on the projector

The projector has a status display at the front. The various symbols are explained as follows:

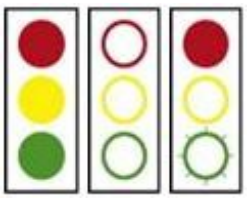
Overview of the various options when switched on:



When switched on:

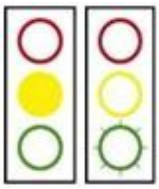


During the initial switching on process all three LEDs will flash. After approx. 30 seconds all the lights will turn off. Then, soon after, the green light should start flashing while the red light will be off. This indicates that the laser projector is ready for use and is awaiting the input of **serial data**.



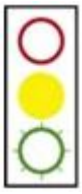
During the initial switching on process all three LEDs will flash. After approx. 30 seconds all the lights will turn off. Then, soon after, the red light should light up statically and the green light should start blinking. This indicates that the laser projector is ready for use and is awaiting the input of **network data**.

Registering the projector via LPM



As soon as the LPM software is opened, it will attempt to detect the projector. While this is happening, the yellow lamp will light up statically for approx. 2 seconds, followed by the blinking of the green lamp. This indicates that the software is now connected and that a projected image can now be sent to the projector.

During projection:



When the yellow lamp flashes statically and the green light is blinking, it indicates that the software is connected to the projector and an image gets projected.

Further displays:



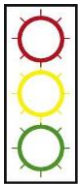
If all 3 LEDs flash cyclically in the order “green → yellow → red → green” etc., it indicates that the motion detector has been activated. No images are projected at this stage.



If all 3 LEDs flash statically, the serial or network client could not start the projector. This could indicate a fault in the hardware. Please contact your regional representative



If the status display does not light up, please check the laser projector's power supply. If necessary, please contact your regional representative

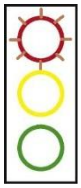


Search for operating point of the green source:
Flashing of all LEDs for a period of 250ms

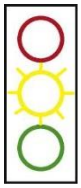
Laser source not at operating temperature:



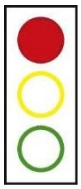
If all LEDs flash for a period of 500ms, it signals that the laser source has not reached its operating temperature (yet); this usually occurs during the projector's booting phase. This status/condition should automatically be lifted after a short while



Overheating (i.e. >40°C AND > 45°C) is signalised with the blinking of the red LED



Yellow lamp flashes for 1 second; idle timer has run out; projection is stopped



Red on; error of communication (i.e. no communications); this lamp flashes also when the projector has no connection to the LPM

Installation of the LPM Software

The laser projection software LPM (Laser Projection Manager) is tasked to prepare the graphical data for the projector from various import modules and transmit it to the laser projector. At the same time, LPM offers a visualisation of the data on-screen and enables interaction with the position, rotation and height of the projection display. Using the new modifiers, LPM has gained the function to manipulate image data and optimally display it.

On the following pages, we will advise you how to use the software. Should you have any questions, please contact your local Z-Laser dealer.

System Requirements

To control the laser projector via the included LPM (Laser Projector Manager) software, your PC should meet the following minimum requirements:

1. Computing power: Intel Pentium / AMD Processor with 300 MHz.
2. Operating system: Windows 7, Windows XP, Windows Vista.
3. Graphics card and monitor with 800 x 600 pixel resolution and 8 bit colour depth
4. CD-ROM / DVD drive
5. 20 MB free hard disk space
6. Keyboard and mouse
7. A free RS232 port or Ethernet network interface card (depending on desired communication)
8. In network systems: Either TCP / IP protocol or IPX/SPX compatible protocol with enabled NetBIOS installed and running

Required preparation for the installation

To use the full functionality of the LPM set-up, you should have already prepared the following:

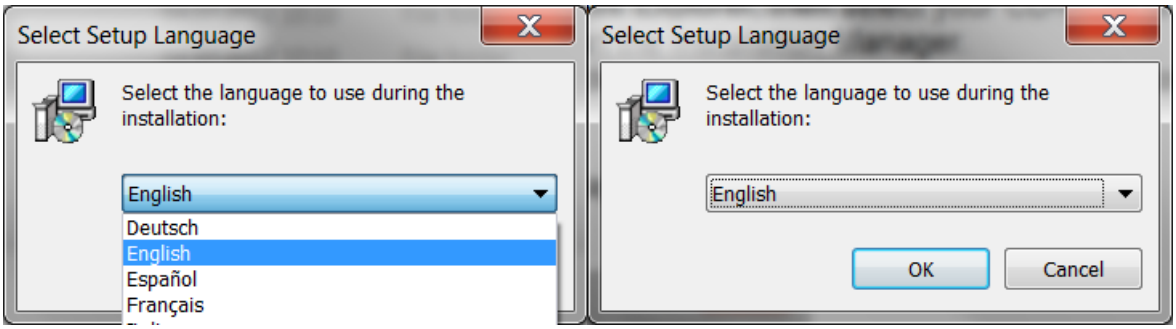
1. Connect the PC and the laser projector together with the communication cable.
2. Apply the supply voltage to the projector and switch the projector on.
3. If the status indicators are flashing green and solid red, then a network connection should be made.
4. If the status indicator is flashing green only, then a serial connection should be made.
5. The auto-run function of your version of Windows should activated.
6. Ensure you have administrator rights on the PC that LPM is to be installed on.

The set-up program, after successful installation, automatically tries to find the laser projector using a serial connection or over Ethernet. Ensure the search process is not influenced by other equipment.

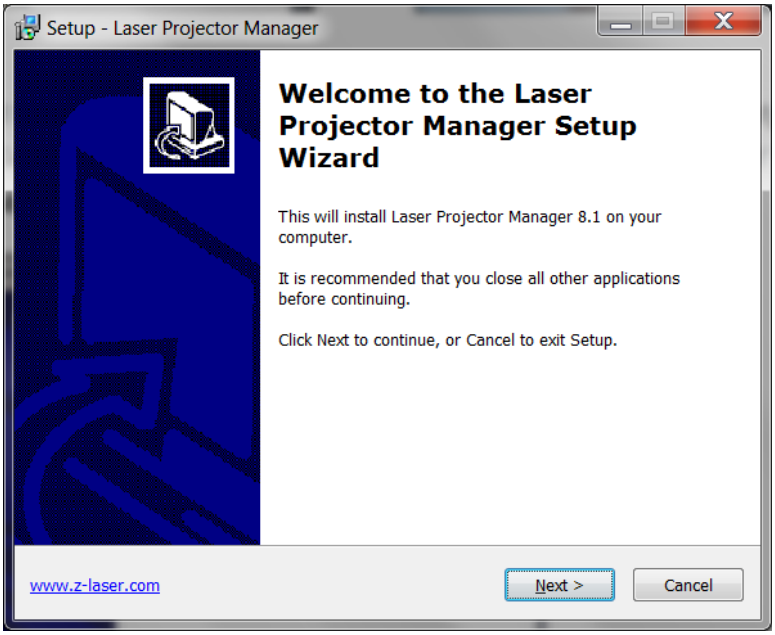
LPM Software Installation

Insert the included CD-ROM into your CD-ROM or DVD drive. After closing the drive, the installation process should start automatically if the auto run feature is enabled. If this is not the case, open Windows Explorer, then select your CD-ROM drive icon. Double-click the file SETUP.EXE to start the Installation Manager.
The set-up wizard will guide you step by step through the installation of LPM:

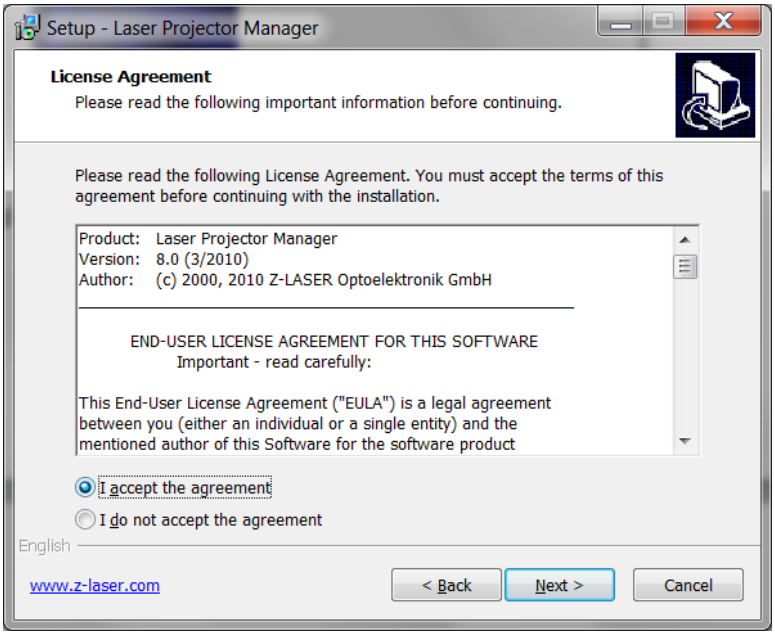
1. Select your language and then click "OK".



2. The LPM set-up wizard will start in the language of your choice. Follow the instructions in the wizard and click either "Continue" or "Cancel."

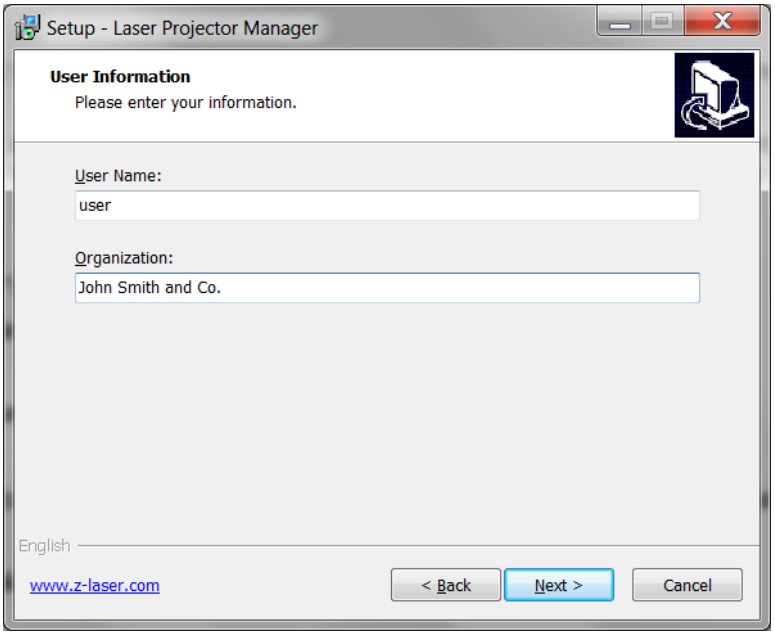


3. Read the license agreement that appears carefully.

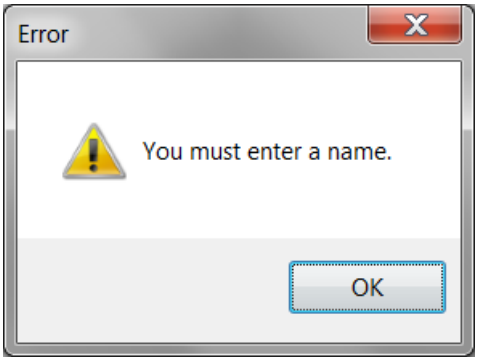


4. If you accept the listed agreements, activate the check box “I accept the agreement” and click “Next”.
5. If you do not accept the listed agreements, activate the check box “I do not accept the agreement”. In this case, the “Next” button is inactive and you will not be able to install the software.

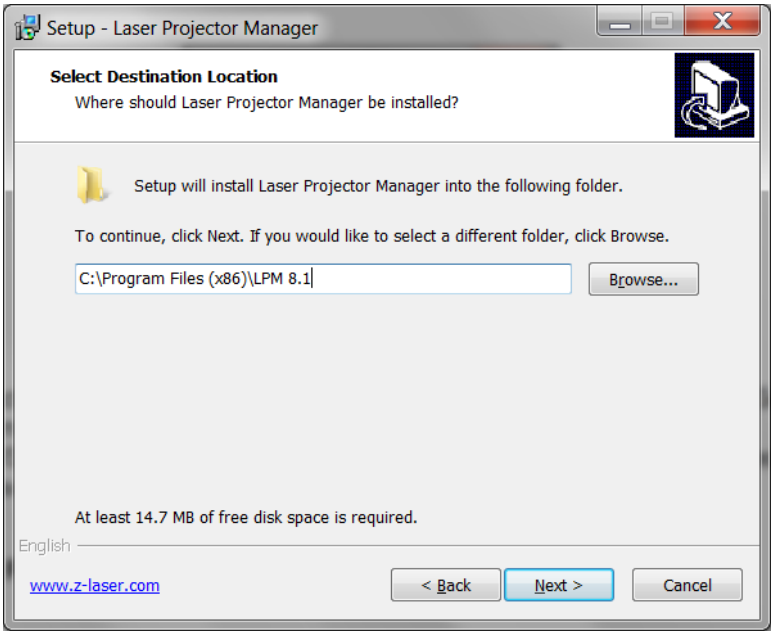
6. The “User Information” window appears next.
7. Enter your name and company name in the fields provided. Click “Next”.



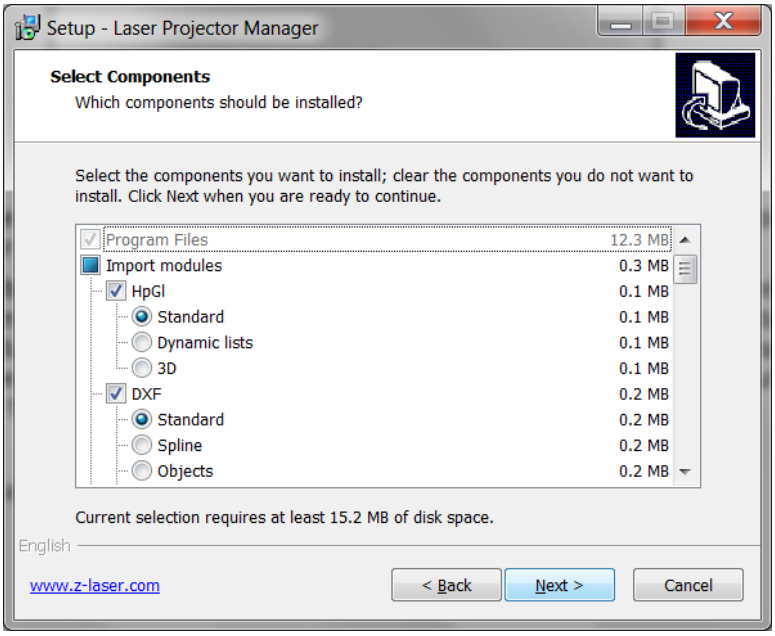
8. A dialogue box will appear as a reminder if you forgot to enter the above information.



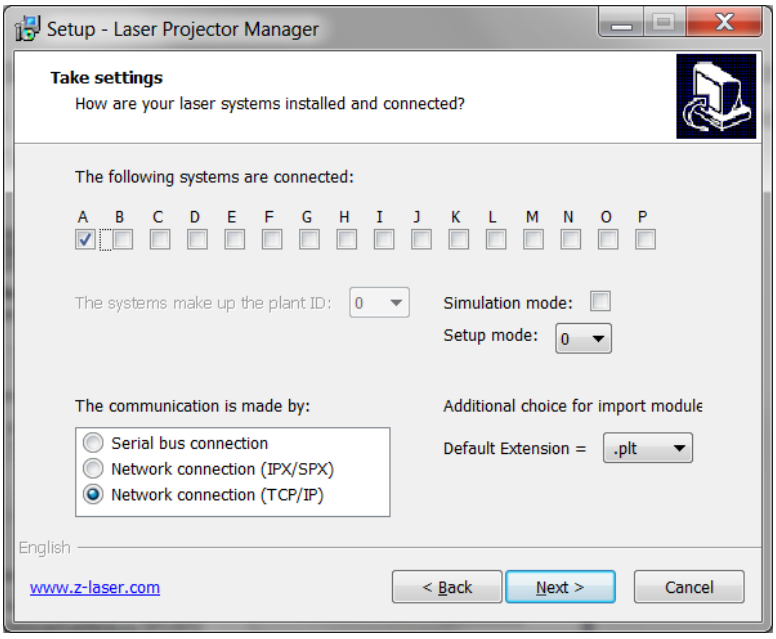
9. Select the destination folder you would like LPM to be installed in. The Set-up Wizard automatically creates a folder "LPM" under the path: C:\WINDOWS\LPM. If you wish to choose another path, select “Browse...”. Select “Next” when you have made your choice.



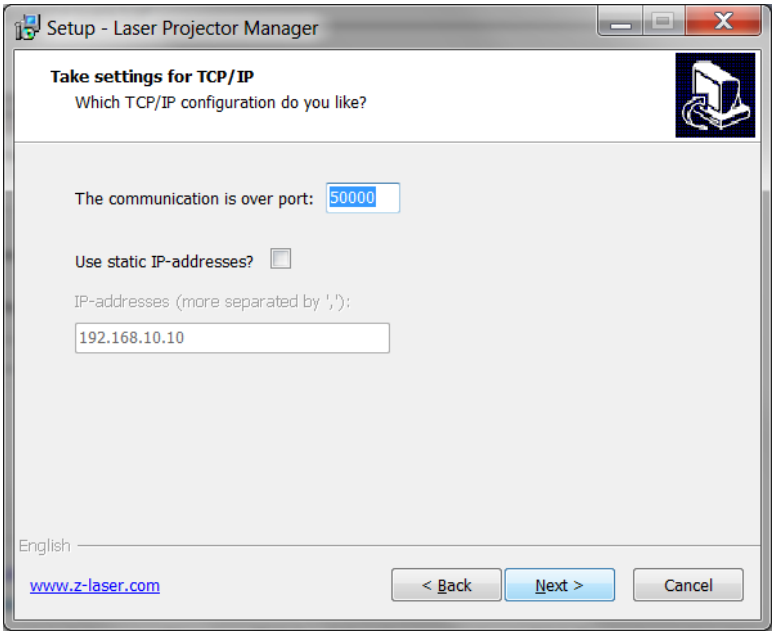
10. In the next window, you can select all the import filters for your application needs. To make your selection, simply activate the filter’s check box.
11. The import filters for DXF and HPGL are already pre-selected. You can make only one selection from any set of import filters and extension modules. If you wish to install multiple filters or modules, then the set-up will need to be started again for each additional module. All import filters already installed will remain after the new set-up. Ensure that sufficient space exists for the desired modules. Then click "Next".



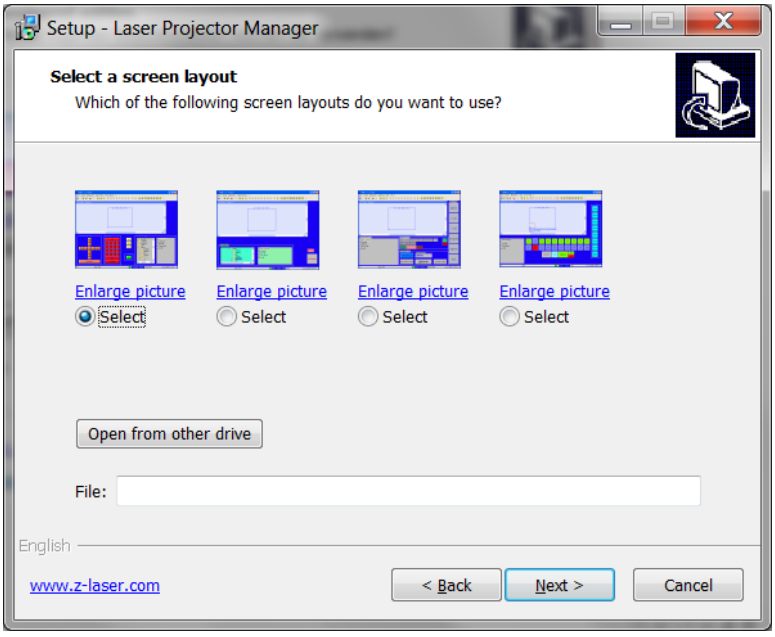
12. Apply your settings as follows:
13. Enter the number of projectors to be connected by selecting the check box of the corresponding character. If you have only one projector, then select the letter “A”. For two projectors, select letters “A” and “B”, for three, select letters “A”, “B” and “C”.
14. You can connect up to 16 projectors.
15. Set the communication type (TCP/IP protocol, IPX/SPX protocol or serial). If you would like to control a group of projectors independently by different computers running LPM, you can insert the group address of the “PlantID”. This setting is only used if you selected the IPX/SPX protocol option.
16. Finally, select an import filter that you would like to open your files with from the “Default Extension =” drop-down menu. When you are finished, select “Next”.



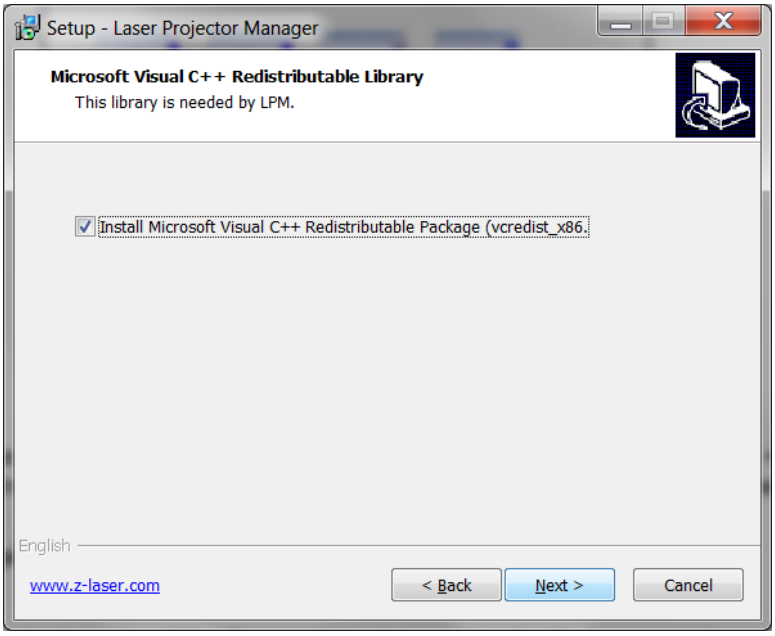
17. If you selected network communication over TCP/IP protocol, an additional window will open which will allow you to adjust the settings of the communication port. The port is pre-initialised as 50000. Allowed values are 50000 – 60000. If you use the same IP address or if there is no DHCP server available, then activate the „Use static IP address?“ check box and enter your IP address. Afterwards, select „Next“.



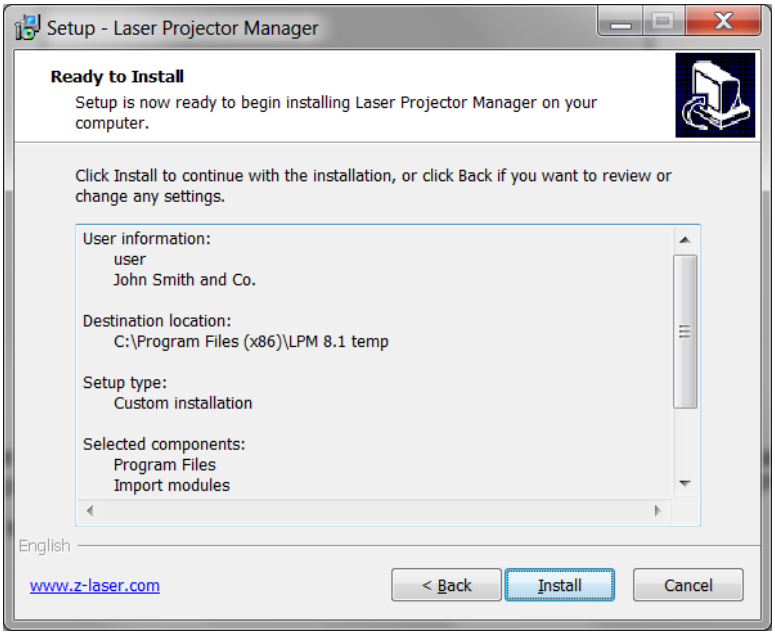
18. In the following menu, you can select the screen layout of the LPM interface. You can enlarge the thumbnails of the user interfaces on display to make your decision. If you already have a specially configured user interface of LPM, you can load it by selecting “Search the hard disk”. Then click "Next".



19. In the next step, you will be prompted to install the Microsoft Visual C + + library. If this library already exists on your computer, remove the check mark from the check box. Click "Next".



20. In the last screen of the Set-up Wizard, you can check all your settings again. Click on "Back" to change your settings. Click "Install" to install on your PC to LPM.



21. The software is now installed and ready for use. You can either call LPM by pressing Ctrl, Alt and L simultaneously on your keyboard or via "Start → Programs → LPM → LPM" or simply via the automatically created shortcut on your desktop. Please note that LPM is copy protected. After installation, your installed modules are free of charge for 30 days. After this period, the modules will stop working and LPM will need to be supplied a registration code from Z-LASER. For information about the registration process, see the chapter “Software Registration”.

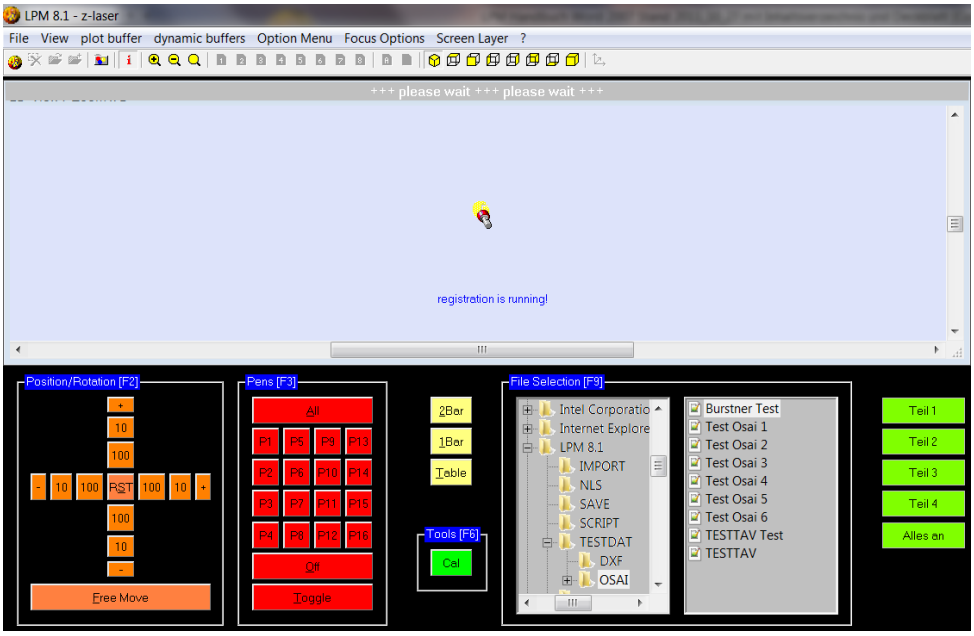
Projector Identification

Identification of a single projector

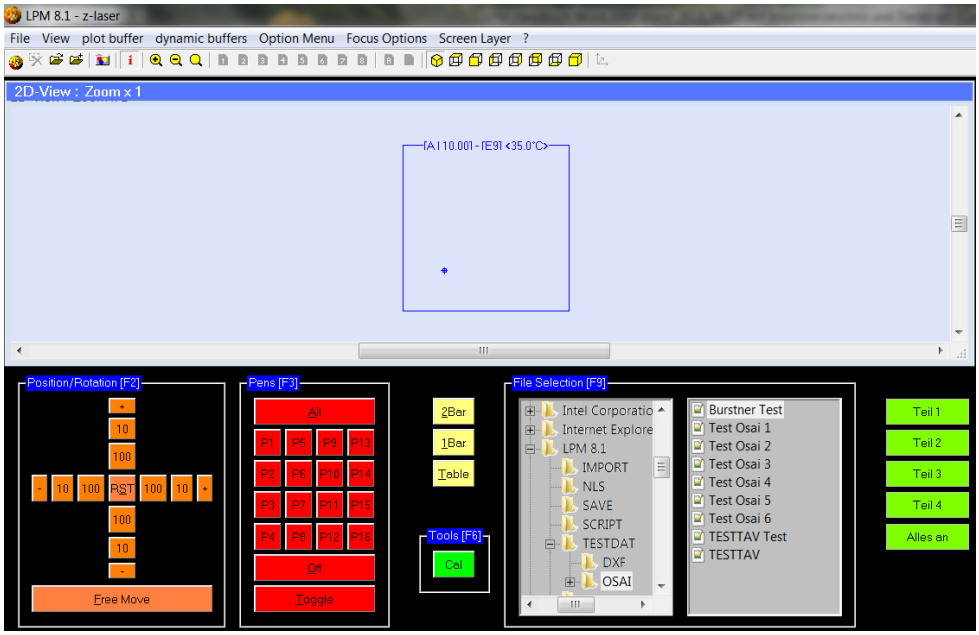
Turn on the projector before starting the control computer, otherwise the network connection to the projector will not be recognised.

If you want to operate several projectors simultaneously, please note that you must assign each projector a different address. All projectors are set to address A during production as default.

The registration of your projector is done automatically as soon as you start LPM. While the projector tries to register your projector, the message “Please Wait” will appear in the dialogue box. You can observe this process by watching the status display of your projector.



If your projector has successfully registered, a blue frame indicating the field border of the projector will appear. If this is not the case, check your communication settings, as described in the next chapter.


LPM 8.1 - z-laser
File View plot buffer dynamic buffers Option Menu Focus Options Screen Layer ?
2D-View : Zoom x 1
[A] 10.001 - [E9] <35.0°C>
Position/Rotation [F2]
Free Move
Pens [F3]
All
Off
Toggle
2Bar
1Bar
Table
Tools [F6]
Cal
File Selection [F9]
Intel Corporatio
Internet Explore
LPM 8.1
IMPORT
NLS
SAVE
SCRIPT
TESTDAT
DXF
OSAI
Burstner Test
Test Osai 1
Test Osai 2
Test Osai 3
Test Osai 4
Test Osai 5
Test Osai 6
TESTTAV Test
TESTTAV
Teil 1
Teil 2
Teil 3
Teil 4
Alles an

Identification of several projectors

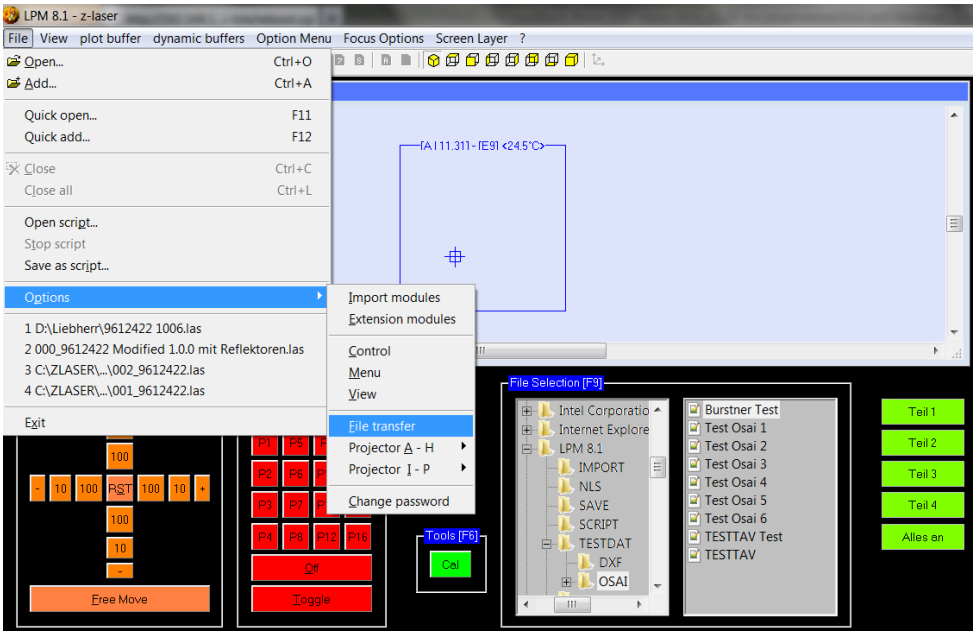
If you use several projectors simultaneously to cover your working surface, you must pay attention to the following details.

No matter if you communicate serially or through the network, you must assign each projector its own identification letter. For each projector, you must later make a quick calibration to measure the field that the projector should cover. All projectors are set to address A during production as default.

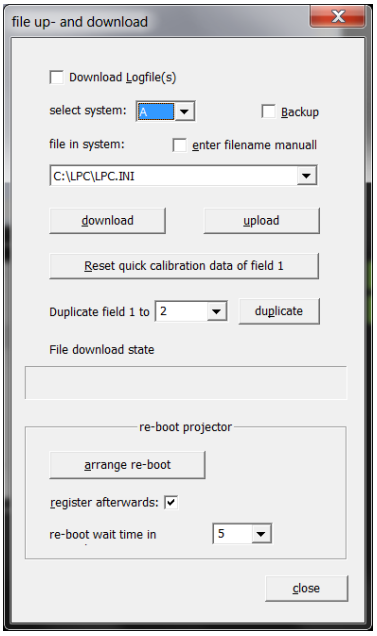
You can change your settings easily by using the web browser. You can find how to do so by reading the chapter "Accessing the laser projector using an internet browser".

1. Open LPM.

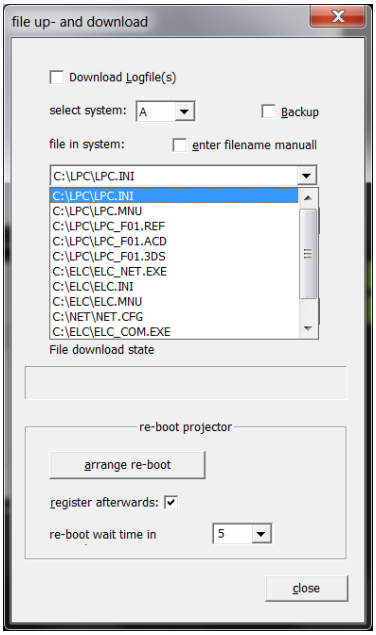
2. Open the “File Transfer” menu by selecting File → Options → File Transfer. Enter the password “cps”.
3. You can change the password at any time by selecting File → Options → Change Password.



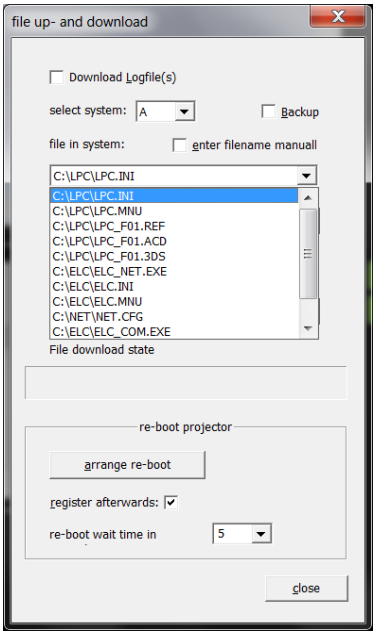
4. You should now see the following menu:



5. Click on the drop-down list marked “file in system” and select the following file:
6. C:\LPC\LPC.INI for Linux or
7. C:\ELC\ELC.INI for DOS



8. After selecting your file, please select the “Download” button to download the file.



9. Enter the path you want to save the file LPC.INI to.

10. Open the LPC.INI file with the text editor. You can now configure the entries. For an overview of the different parameters, see the chapter "LPC.INI".

11. Note that depending on the selected entry, the configuration file LPM.INI will be amended accordingly. Customize your changes under the section [Communication] as follows:

Entry in LPC.INI	**Entry in LPM.INI**
Baud = 9600	Baud = 9600
IPDynamic=yes	StaticIPAdresses = 0
IPPort = 50000	IPPort = 50000
IPDynamic=1	StaticIPAdresses = 1
IPAddrMin = xxx.xxx.xxx.x	IPAdresses = xxx.xxx.xxx.x

12. Save changes to the file under the same path in which you opened it and select the “Upload” button. Enter the path of the saved file. After the file is uploaded, click on the button “arrange re-boot” and the projector will restart under its newly assigned address which you will see in the graphical user interface.

LPC.INI

By default, your laser projector operates under Linux. The associated initialization file stored on the projector is LPC.INI. The LPC.INI file is complimentary to the ELC.INI file in DOS. If your projector runs under DOS mode, you will find the complete ELC.INI in the Appendix. By default, you can change your settings easily via an internet browser. For further information, refer to the chapter "Accessing the laser projector using an internet browser".

Parameter	Default Setting	Description
ClientAddress	A	Change the address of the projector by assigning a new letter from A to P. Note that each letter within a group of projectors can only be assigned once.
Baud	9600	Set the data transfer rate [in bits/sec]. Valid values are 9600, 19200, 38400, 57600 and 115200. This entry is only relevant for serial communication and must match the same entry in the LPM.INI file. **For information about changing this entry, refer to the chapter "Setting up the serial communication".**
Port	03xF8	Specifies the port number of the interface, thus gaining the input and output of data as hexadecimal code. COM 1 0x3F8 COM 2 0x2F8 This entry is only relevant for serial communication mode. **This entry must not be changed.**
IRQ	4	Specifies the interrupt number (in hexadecimal code) for the serial interface. Com 1 IRQ 4 Com 2 IRQ 3 **This entry is only relevant for serial communication mode. This entry must not be changed.**
FontTimer	50	Specifies the output speed of the character of the laser menus.

FontScaling	20	Specifies the size of the letters in the laser menu. The specification, based on the established coordinate system, is measured in millimetres..
RefIntern	0	Allows referencing of the projectors. RefIntern = 0, the referencing is calculated in LPM. RefIntern = 1, the referencing is calculated in the client. The mathematical calculations are different here. Also read the chapter "Referencing".
ControllerBoardRev	D	Specifies the revision number of the controller board; this setting must not be changed!
TimeOut	1500	The TimeOut parameter defines the time, during serial communication, that the projector must wait, after the last byte is received, to start projection of the data. If new data is received during the defined time period, then the data is appended to the existing data and the TimeOut counter is reset.
DAPTolerance	0	Specifies the maximum allowable deviation after multiple searches of the drift compensation points (in increments). This entry should not be changed.
DisablePIR	5	Parameter used to control the motion sensor. 0: Motion activated; 1: Motion activated, the motion sensor is also triggered by a calibration reset. 2: Motion disabled; 3-60: The time in seconds before the projection is restarted, after the motion detector has been activated.
PenCount	1	Specifies whether files with multiple pens are loaded consecutively (1,2,3,4,5) or the the same pens listed in the file (e.g. 1,3,7,14,16).
RemoteOn	BE	Assignment of the buttons for the internal remote control, it can also store other functions!
RemoteOff	BA	
RemoteUp	MR+Y	

RemoteDwn	MR-Y	
RemoteMinus	P-	
RemotePlus	P+	
RemoteM	C	
RemoteEnter	RST	
RemoteCode	15	
RemoteReturn	1	
IdleTimer	3600	Specifies the time in seconds, which is allowed to elapse during inactivity, before the projection is switched off.
CPSBootDelay	0	Used to set a delay in the boot sequence when there are multiple projectors in a network. The delay time in seconds can be set by this parameter.
IPDynamic	yes	Specifies how is the IP address is assigned. Yes = Dynamic IP addressing No = Static IP addressing
IPAddrMin	192.168.10.10	The minimum IP address of a range, in which a free IP address for the projector is to be found.
IPAddrMax	192.168.10.254	The maximum IP address of a range, in which a free IP address for the projector is to be found.
IPPort	50000	Specifies the port on the projector through which the IP protocol is addressed. Valid values are 50000 – 60000.
ShowDAP	3	The current drift compensation point can be displayed for a set time (in sec) after the search for the point is completed, in order to perform a visual inspection of the point. This slows down the process but increases the process reliability, because the user can immediately see whether the drift compensation point was found in the right position.
InitCompatible	0	Different operating modes can be selected for compatibility settings: 0 = Standard 1 = ZLSC mode 2 = as ELC 9.5 (older client versions)

InitBanner	0	This function is used to project operating parameters directly after the start of the projector, such as the IP address and serial number. This can be useful if the projector does not have a connection to the PC. InitBanner = 0 Projection of operating parameters permanently disabled. InitBanner = 1 Projection of operating parameters until the first calibration is made. InitBanner = 2 Projection of operating parameters permanently activated. Note: In the web interface, this parameter is called the initial banner.
ParPortMode	0	Only for internal use to activate the PLC interface.
AutoDeletePlotFiles	0	This parameter configures the behaviour of the plot directory when "projecting via the Plot directory". Note: In the directory \ \<serial_number>\plot and \ \<IPAddress>\plt, the plot file is automatically projected . AutoDeletePlotFile = 0 Automatic deletion in the plot directory disabled. All files copied into the Plot folder will remain until the projector is switched off. AutoDeletePlotFile = 1 A *.plt file copied into the plot directory is deleted immediately after the start of the projection. AutoDeletePlotFiles = 2...100 The plot-directory is counts copies made to the folder and deletes all files when it reaches the amount specified by the user, i.e. after every 2...100 copying processes, the files are deleted.
UseFilenameFromNetCom	0	This parameter allows the user to project HPGL or PLT files that are stored on the projector by sending the appropriate file name to the projector. The data must be stored in the \ \<serial_number>\plot and \

		\<IP Address>\plot folders of the projector. The file name can be sent, depending on the operating mode of the projector, via a serial or network connection. These are the possibilities for configuration: UseFileNameFromNetCom = 0 Mode is disabled (default). UseFileNameFromNetCom = 1 Mode 1 is enabled (Sent file names must start with the ID <ZS> and end with the ID <;>. This is the recommended mode). UseFileNameFromNetCom = 2 Mode 2 is enabled (File names can be sent without the Start and End Id detection)
GreenDriver	0	This parameter configures the driver support for projectors with green laser sources: Green driver = 0 Green driver is disabled (the default value for projectors without a green driver) Green driver = 1 Green driver is enabled (default value for projectors with a green driver)

Setting up the type of communication

In future, only the client version of Linux will be developed further. The DOS client is frozen at the level 9.55. The projectors should be used, wherever possible, in Linux mode.

Laser projectors that have been manufactured since July 2007 are delivered with 2 operating systems. These systems are:

1. Linux clients (Standard) with version numbers greater than 10.00
2. Traditional DOS clients with version numbers less than 9.55

Since July 2008, ***Z-Laser*** Optoelektronik GmbH upgraded the Linux client to version 11.00. This version is almost completely compatible with previous laser projectors.

The latest generation of laser projectors, which have been manufactured since October 2008 have device integrated DIP switches, which are used to switch between Linux and DOS modes and to change factory settings. To find out how do this, read the next section entitled **"Dip Switches"**.

Dip Switches

There are 4 dip switches on the Data Interface Board. The DIP switches are used to determine which operating system (Linux or DOS) and in which communication type (network or serial) to start the laser projector.
If you open the lid, the DIP switches are located on the side wall directly behind the Data Interface. The DIP switches are shown in Figure 11.

Therefore, an external jumper on the serial interface is no longer necessary. If the dip switch is set to "serial", a connection to the communication network through the FTP server or Web-interface is still possible.

By default, the dip switches on the laser projector are set to Linux network (TCP/IP) on delivery.

The settings in the LPM.INI (configuration file of the software LPM) are shown as follows:

```
Use communication=2
StaticIPAdresses=0
IPPort=50000
```

The position of the dip-switches are ON -> ON -> OFF-> Not used.
Please see section on **"Setting up the network connection"**.

If serial communication is desired, set the DIP switch as shown in the table below. Please refer to the chapter **"Setting up the serial communication"**.



Figure 11: Data Interface board with DIP Switches

A table showing how the dip switches must be set for the various types of communication.

Dip Switches	Operating	File

1	2	3	4	System		Name
OFF	OFF	OFF	Not used	DOS	serial	ELC
ON	OFF	OFF	Not used		network	ELC
OFF	ON	OFF	Not used	LINUX	serial	LPC
ON	ON	OFF	Not used		network	LPC
OFF	OFF	ON	Not used	DOS	restore	ELC
ON	OFF	ON	Not used		restore	ELC
OFF	ON	ON	Not used	LINUX	restore	LPC
ON	ON	ON	Not used		restore	LPC

Switch 1: Selects the communication type with the projector
Switch 2: Selects the operating system
Switch 3: Enables or disables the function "RESTORE"
Switch 4: Not used

The following figures show the settings of DIP switches on the various modes.



Figure 12: Linux Network



Figure 13: Linux Serial

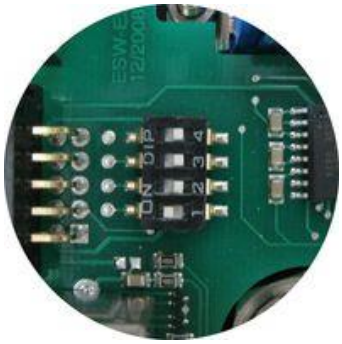


Figure 14: DOS Network

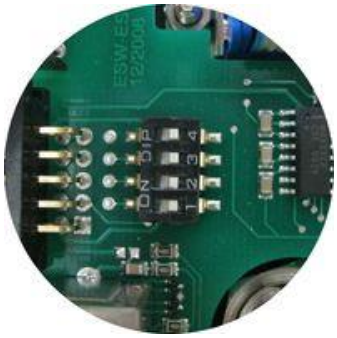


Figure 15: DOS Serial

To disable the dip switches, you must set the controller board revision in the LPC.INI to "C". If "D" is selected, the switches are active.

What happens when you run the RESTORE function?
The projector will reset to the factory state.

When is a RESTORE useful?
A RESTORE is performed when there is no more connection between the PC and the laser projector. This can happen if, for example, the IP address, the port number, a Plant ID or other parameters have been changed to the extent that communication with the projector is no longer possible! Other reasons include the field data has been overwritten and the accuracy of the projection is no longer guaranteed.

Performing the function RESTORE

1. Turn the projector off.
2. Open the lid of the projector with the Rittal key.
3. Switch the 3rd dip switch from off to on.
4. Turn the projector on and watch the status lamps. The laser projector will signal the status of the RESTORE function with a flashing code on the status lamps. It will go:

   Green → Yellow → Red, Red → Yellow → Green

   and then it will repeat.

Note: During this time, the laser projector must not be turned off under any circumstances!

When all 3 lamps blink at the same time, then the RESTORE function is finished and the laser projector may be turned off.

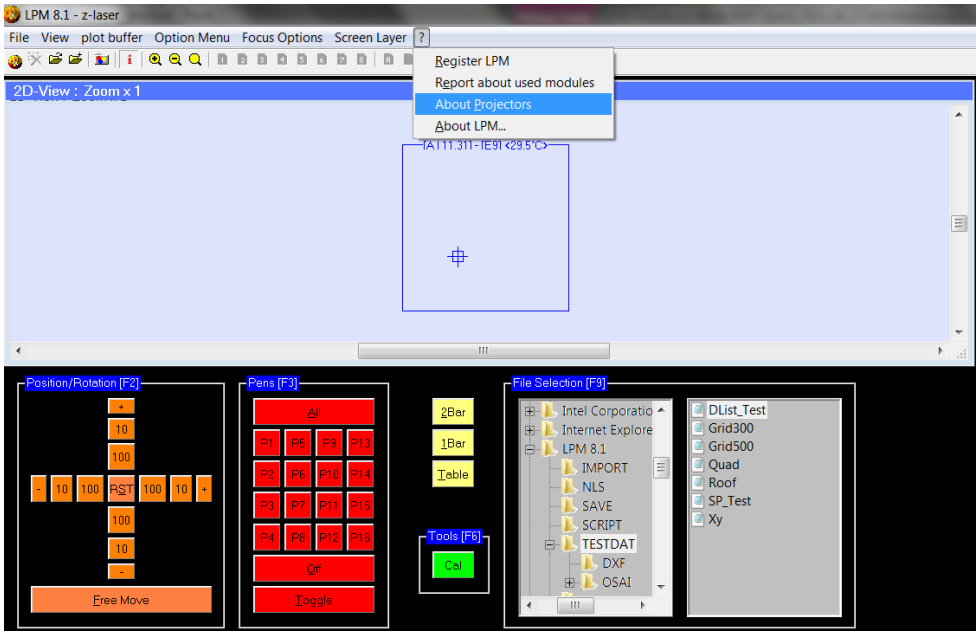
1. Switch the 3rd dip switch from on to off, then close and lock the lid of the projector.
2. You may now switch the projector back on. The projector will start with the most recently saved factory settings (as described above).

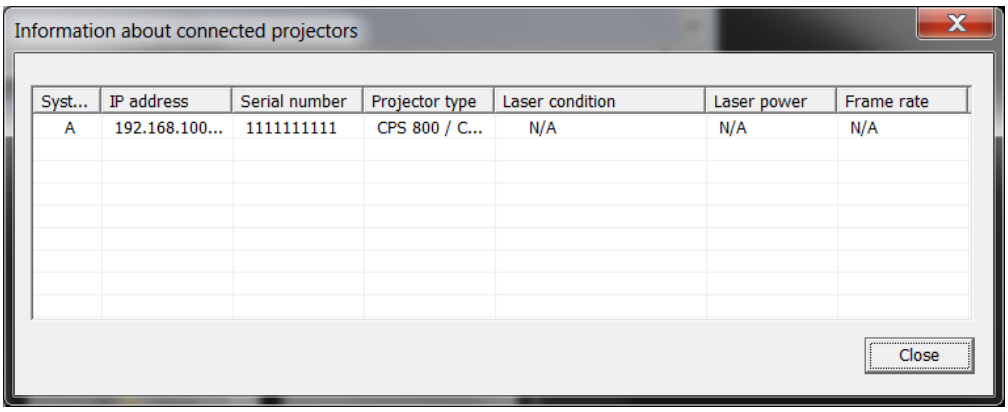
Access to the laser projector using an Internet Browser

The client version 11.00 and later provides the laser projector with a web interface that allows you to adjust the various parameters of the laser projector.

You will require a web browser like Microsoft Internet Explorer or Mozilla Firefox to access the laser projector. You will also need the IP address of the laser projector. If you are using a direct connection between the computer and laser projector or you do not have integrated DHCP server, the projector defaults to the IP address: 192.168.10.10.

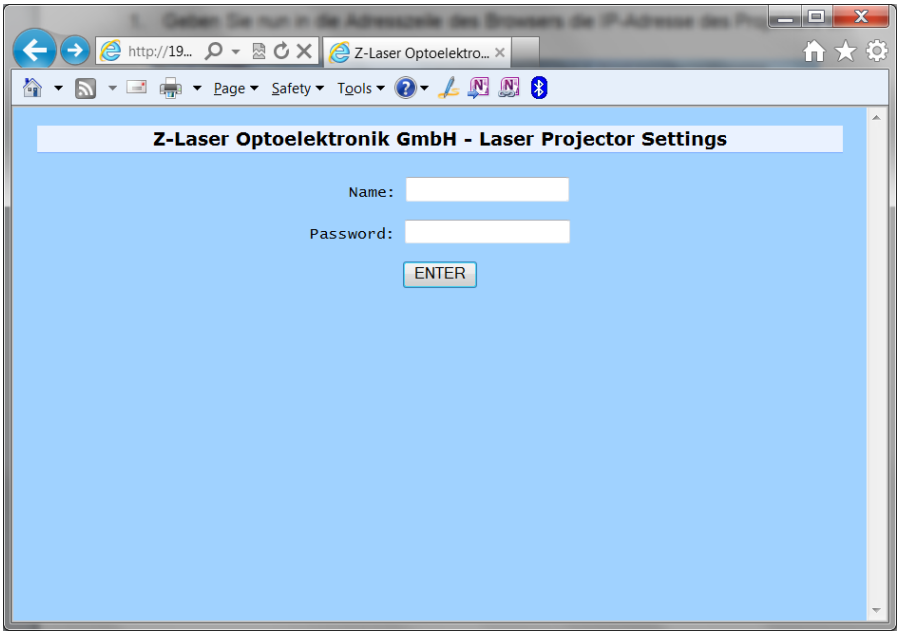
Note: This only works if the projector is operated in Linux mode. The currently assigned IP address of the projector can be found in LPM under “?” -> “Information about the connected projectors”.





Access via the web interface is also possible if the primary connection is established on serial communication. In this case, both the serial and the communication over the network exists.

1. Type the IP address of the projector in the address bar of your browser.



2. You will be prompted to enter a user name and password:
   - Standard User

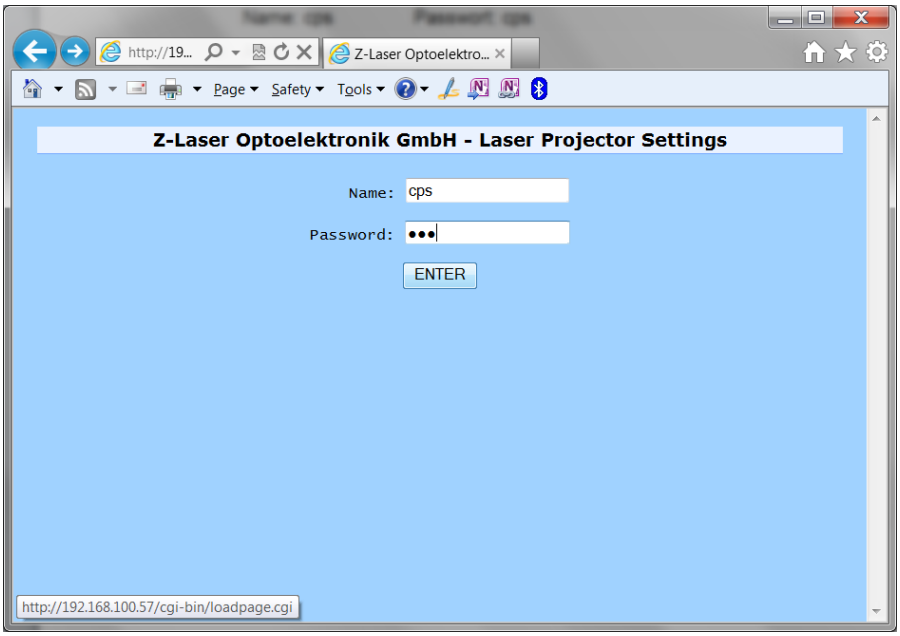
     Username: cps

     Password: cps
   - Administrator

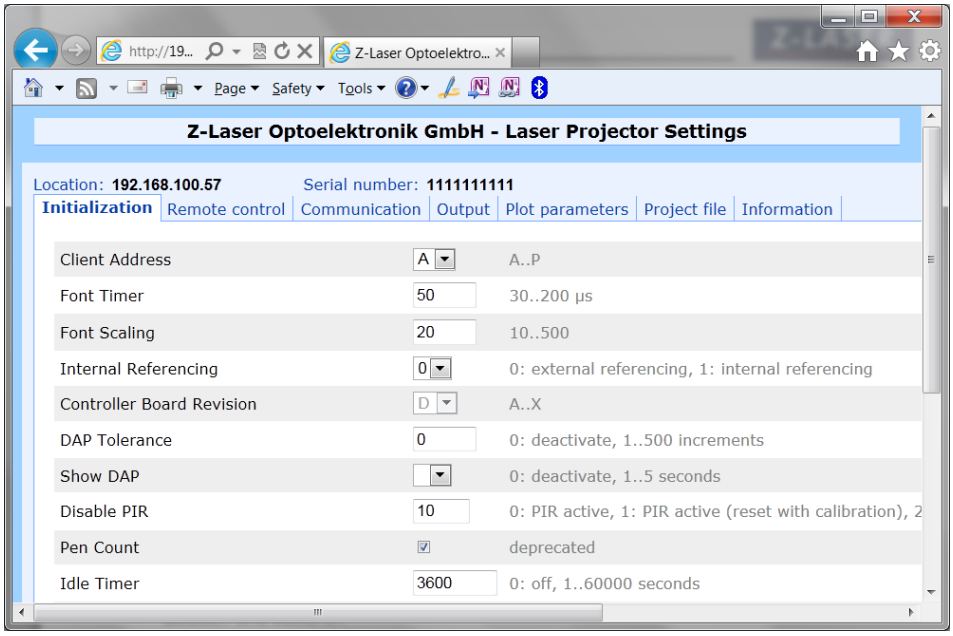
     Username: root

     Password: root

3. Try to avoid logging in as the administrator as you could alter the system settings and permanently damage your projector.



4. Confirm your entry by pressing the „ENTER“ button. The settings of your projector will open in a new window. Here you will notice the serial number of your projector.



5. By clicking on the tabs at the top of the screen, you will be able to read and edit various settings:
   - Initialisation: Setting of the LPC.INI file
   - Remote Control: Settings of the internal remote control
   - Communication: Communication settings
   - Output: LPC_F01.CFG
   - Plot Parameters: LPC_F01.CFG
   - Project File: Upload a PLOT file to the projector

6. The displayed information is largely derived from the LPC.INI.

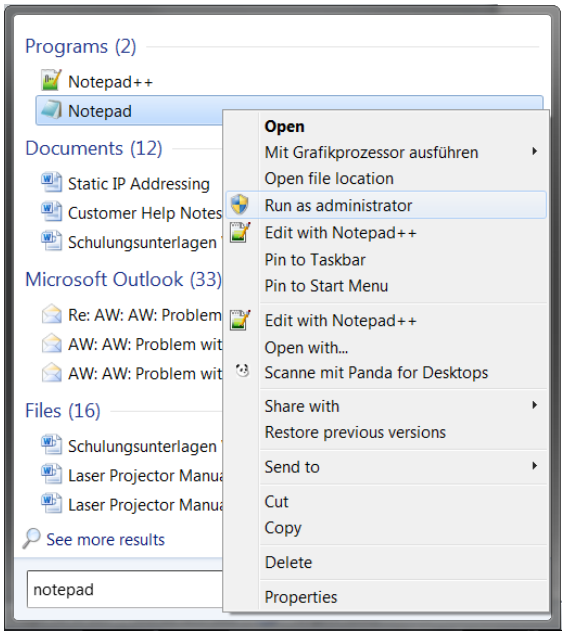
7. You can also use the button mounted at the bottom of the page:
   - Save values — Save any entry made
   - Back to save values — Return to the last saved values
   - Back to initial values — Return to the initial values
   - Restart the projector — Restart the projector for saved values to take effect
   - Enter normal mode — Leave administrator mode to enter standard user mode.

Set-up of the network connection

To establish the network connection between the LPM software and your Windows operating system, you need to change some settings. This is particularly the case if your projector is not detected automatically when the LPM software is started.

Please close all applications before continuing.

1. Open the LPM.INI file which you will find under Start -> All Programs -> LPM 8 -> LPM.INI.
    a. In Windows 7, it is not easy to edit the LPM.INI file. Locate the “Search Programs / Files” field in the Windows Start Menu and search for “Notepad.EXE”. Click on the file with the right mouse button and select “Run as Administrator”.



    Open LPM.INI by selecting File -> Open and locating the INI file. Now you will be able to edit the file and save it without any problems.

    b. You can also gain access to the INI file through LPM by selecting “Option Menu” → “Direct Command”. Enter the command “/EDITINI”.

2. Initial values of LPC.INI of the laser projector:

ClientAddress	A
Baud	9600
Port	0x3F8
IRQ	4
FontTimer	50

FontScaling	20
RefIntern	0
ControllerBoardRev	D
TimeOut	1500
DAPTolerance	0
DisablePir	0
PenCount	1
RemoteOn	BE
RemoteOff	BA
RemoteUp	MR+Y
RemoteDwn	MR-Y
RemoteMinus	P-
RemotePlus	P+
RemoteM	C
RemoteEnter	RST
RemoteCode	15
RemoteReturn	1
IdleTimer	3600
IPDynamic	yes
IPAddrMin	192.168.10.10
IPAddrMax	192.168.10.254
IPPort	50000
RGBMode	0
ShowDAP	0
CPSBootDelay	0
IntCompatibility	0
InitBanner	0
GreenDriver	0 = Red, 1 = Green
AutoDeletePlotFiles	1
ParPortMode	0

At the beginning of the LPM.INI file under the heading [Communication], you will find the following entries:

[Communication]	
Simulation=0	Allows you to work with LPM without a connected projector 0 = Simulation off

	1 = Simulation on
UseCommunication=2	0 = Serial communication mode 1 = Network communication mode over NetBIOS / IPX/SPX Protocol 2 = Network communication over TCP/IP protocol
StaticIPAddresses=0	Informs LPM to search for projectors using dynamic or static IP addresses. 0 = Searches for all projectors in the network. 1 = Searches only for projectors with IP addresses stated in the entry IPAddresses.
IPPort=50000	Specifies the port on the projector through which the IP protocol is addressed. Valid values are 50000-60000.
IPAddresses=192.168.100.190	Specifies the IP address from which a connection should be established on the projectors. Multiple IP addresses must be separated by commas.
NetbiosLana=99	Network address for network communication over NetBIOS / IPX/SPX protocol
NetbiosPlantId=0	Here is the address for a group of laser projectors to be used independently of each other in a network. This is used only over the NetBIOS / IPX/SPX protocol
Port=0	Serial communication interface. 0 = automatic COM port search Not relevant for network connections!
Baud=0	Speed of transmission through the serial interface (bit rate). 0 = automatic detection of the bit rate. Not relevant for network connections!
CommandAtProgEnd = /P& 11111111 /BS	Defines the last command when the program terminates. Stop the image projection in all 8 image buffers
ShowTxErrorInt=1	1 = Error message activated by incoming data communication problems. 0 = Disables error messages.
ShowTxErrorExt=1	1 = Error message activated by outgoing data communication problems. 0 = Disables error message
Slaves=1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0	Every attached laser projector represents a "1" in order to register that projector. It can connect up to 16 projectors, which corresponds to a numerical sequence of A. .. P.

	Slaves = 1,0,0,1 means that only the laser projectors with the client identifier of A and D will be sought for and registered.
CREFSequenz=A1B2	Uses Projector A with a measured reference point 1 and Projector B with a measured reference point 2 for referencing with the direct command *CREF. The reference points must have been measured previously in the set-up of the projector. Valid values are for the projectors A...P and the reference points 1 and 2.
IdlePollingEnable=1	1 = Enable querying of the laser menu. 0 = Disables querying of the laser menu.
ExpertMode=0	This value should not change under any circumstances, otherwise there is a risk that the factory calibration will be deleted.
; Slave_A ... Slave_P=ClippingArea: Xmin,Ymin,Xmax,Ymax for all **Slaves set to 1**	This entry is commented out by the semicolon at the start of the line(;) and it will not be considered as a command. This is an information to the field boundaries in the simulation mode, which reflects the laser work area in 1/10 [mm]. Only relevant if you operate in the LPM simulation mode.
Slave_A=0,0,50000,40000	Shows the coordinates of the laser field border of projector A in 1/10 [mm]: the corners of the rectangle are located at (0/0), (50000/0), (50000/40000) and (0/40000). Only relevant if you operate LPM 8 in simulation mode.
Slave_B=50000,0,100000,40000	Shows the coordinates of the laser field border of projector B in 1/10 [mm]: the corners of the rectangle are located at (50000/0), (100000/0), (100000/40000) and (50000/40000). Only relevant if you operate LPM 8 in simulation mode.

3. Should your laser projector not have been detected by the automatic registration process, you can change the parameters manually or reset them at any time. After you have made your changes, save and close the file.

4. Start LPM. When the connection is established between the LPM and the projector should be in the graphics window, a blue box will be seen.

5. If no connection could be established, close LPM, and open the file LPM.INI. Configure your settings.

After each change, LPM.INI must be saved and LPM must be restarted.

Trouble Shooting:

1. Samba Server: check whether a connection exists.
2. Check the connection through the Internet browser (See section **"Access to the laser projector using an Internet Browser"**).
3. Serial: Download LPC.INI, possibly exchange the Null-Modem cable.
4. 192.168.10.10 to restart the projector without a network connection.

Setting up the network protocols

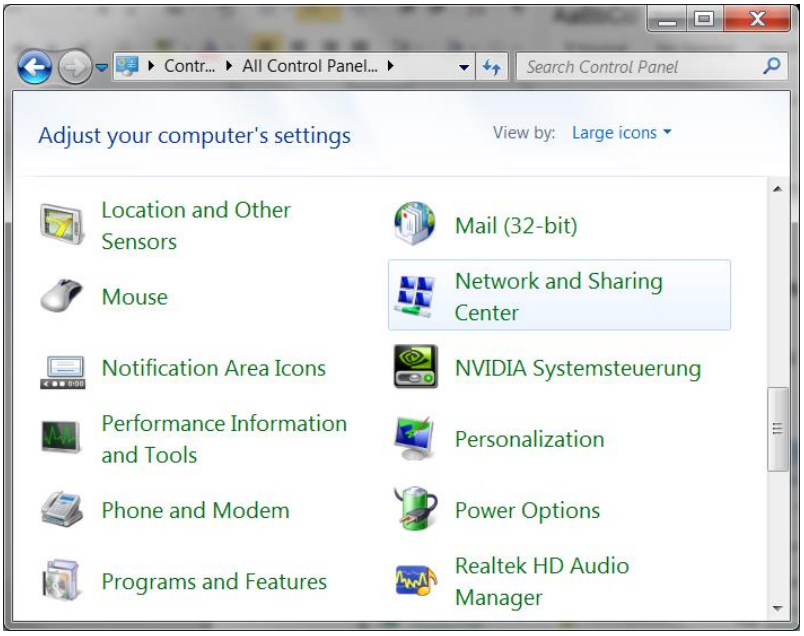
The network protocols govern the flow of data between devices. The communication between PC and laser projector comes with TCP/IP as standard.

- **Case 1**: describes the procedure setting up the internet protocol TCP/IP version 4 under Windows 7.
- **Case 2**: describes the procedure setting up the IPX/SPX protocol on Windows XP and older versions.

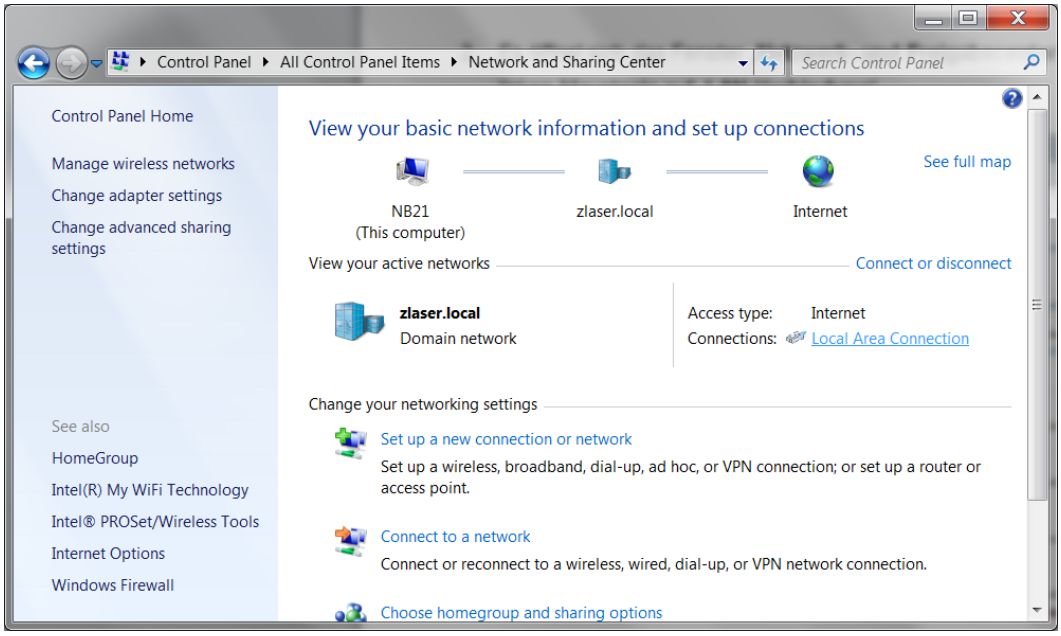
Close all applications before continuing.

Case 1: Setting up TCP / IP in Windows 7

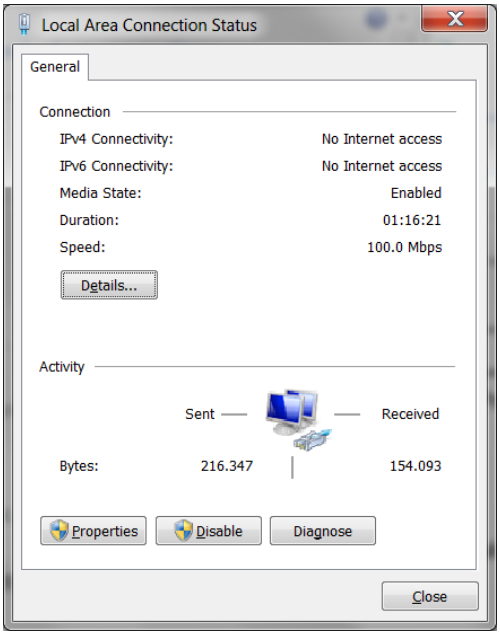
1. Click on the Control Panel and select the "Network and Sharing Center".



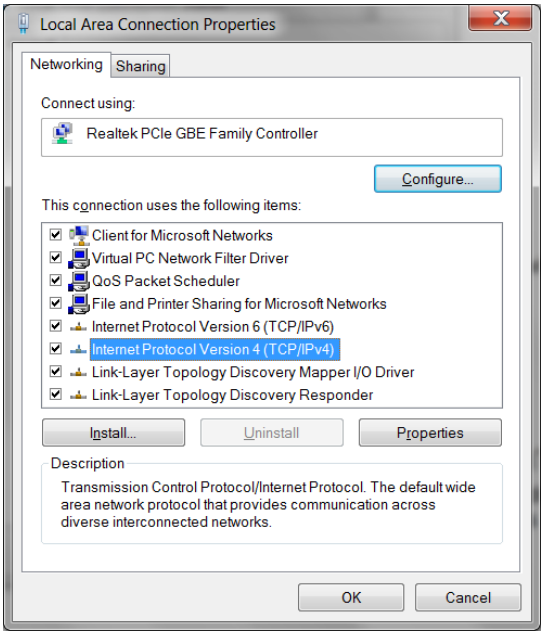
2. This opens the window "Network and Sharing Center". Click with the left mouse button on "Local Area Connection".



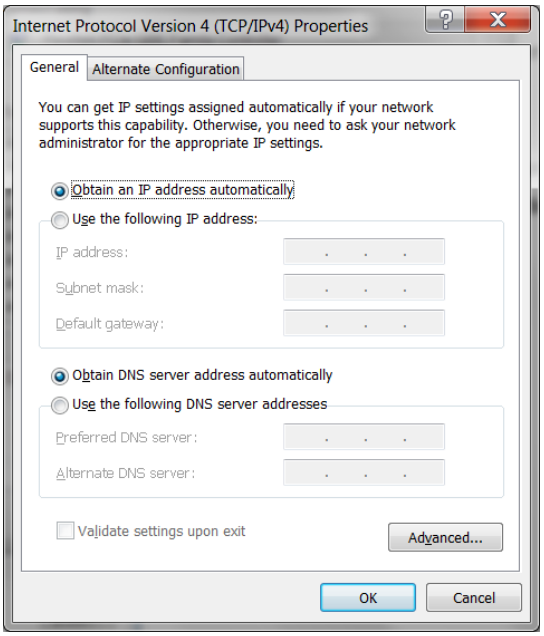
3. The "Local Area Connection Status" window opens. Click the "Properties" button.



4. The "Local Area Connection Properties" is now open. You can also see which protocols are already installed on your PC.



5. Select the entry "Internet Protocol Version 4 (TCP/IPv4)". Then click the "Properties" button. This opens the window "Internet Protocol Version 4 (TCP/IPv4)".
    a. If the PC is connected to a DHCP server, it receives its network address automatically. In this case, select the option "Obtain an IP address automatically" in the window "Local Area Connection Properties".



    b. If the PC is not connected to a DHCP server, but with a switch, proceed as follows. Select the radio button "Use the following IP address". Now enter in the "Use the following IP address" field a free IP address that is otherwise not already in use in the rest of the network.

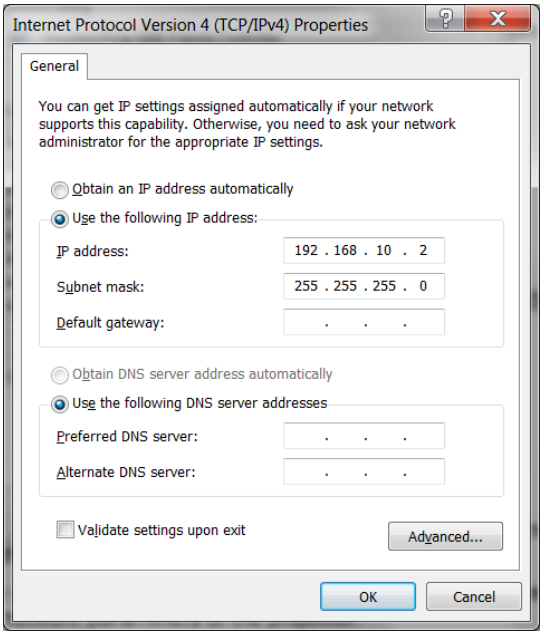
       For example: 192.168.10.2 ... 192.168.10.9, because this IP address range is rarely used.

If the laser projector is not integrated into a DHCP network or if it is not connected

to a Router, then the projector will start in this address range.

Please note: This only applies in the delivered state, as long as there has been no changes made to the network parameters of the projector.

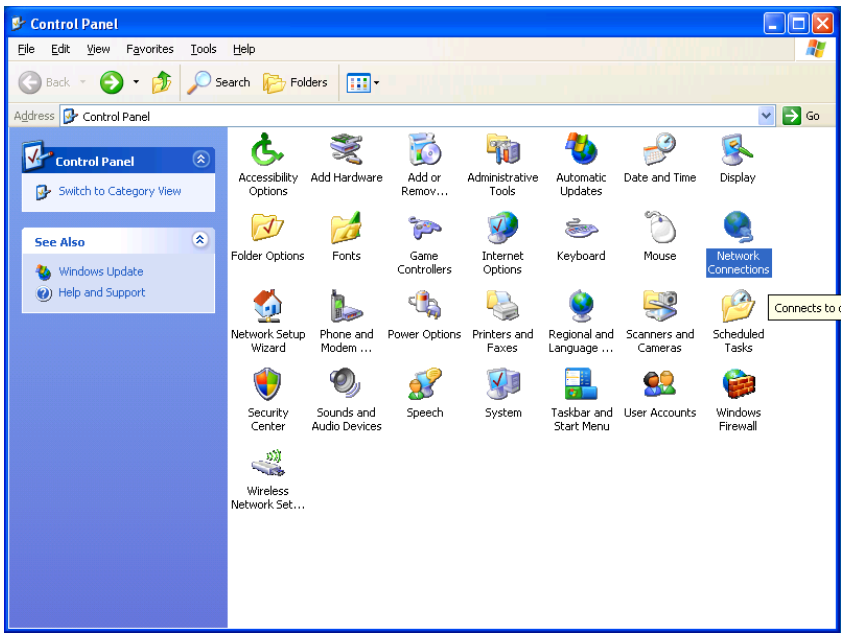
If unsure, ask your system administrator.
In the "Subnet Mask" field, the value is 255.255.255.0.



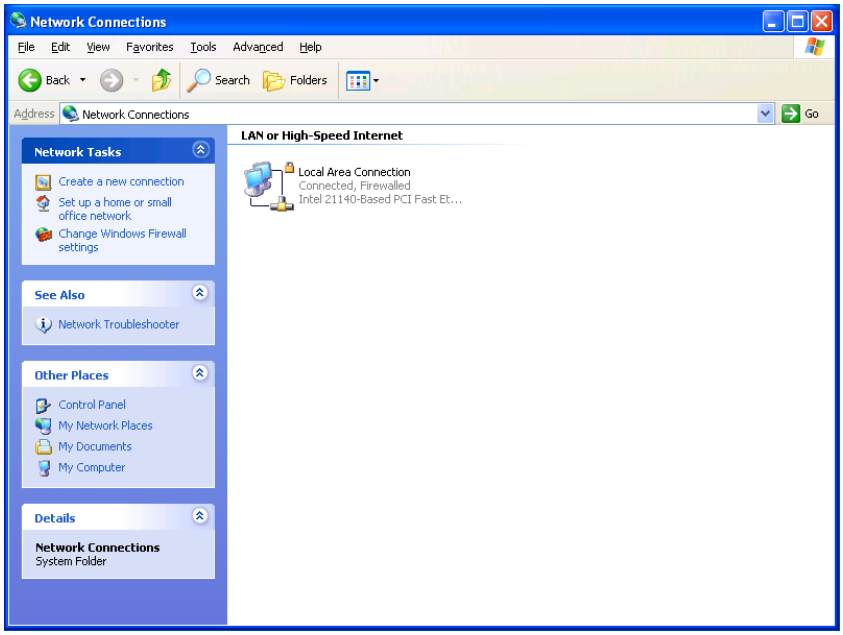
6. Confirm your entry by clicking "OK". Close all other windows. The TCP/IP network connection is now set up.

Case 2: IPX/SPX protocol with Windows XP

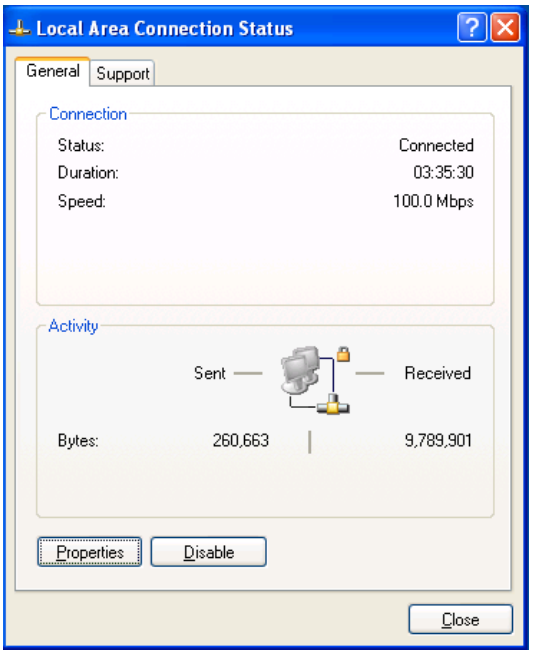
1. Go to Start -> Control Panel and click on “Network Connections”



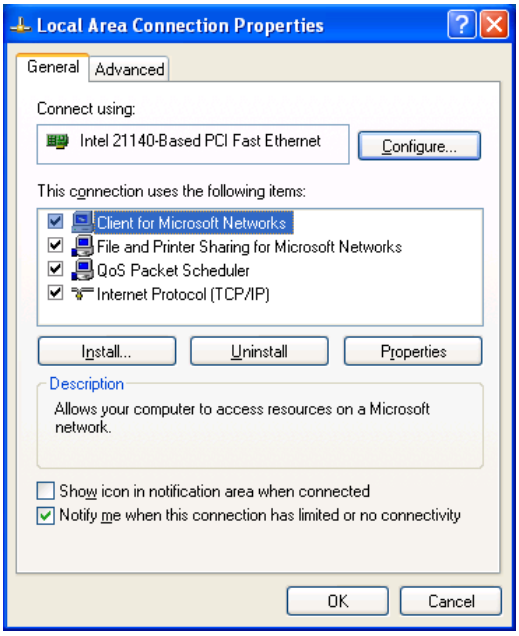
2. Click on the icon “Local Area Connection”.



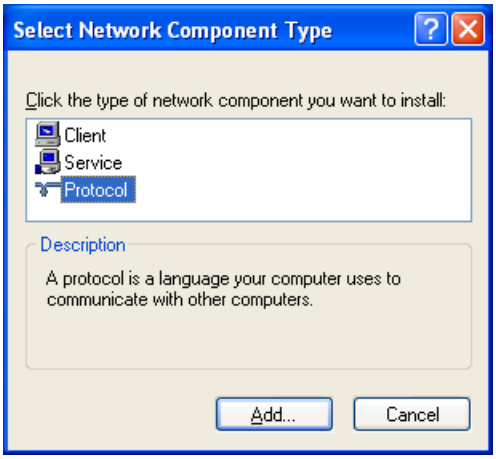
3. The "Local Area Connection Status" window opens. Click the "Properties" button.



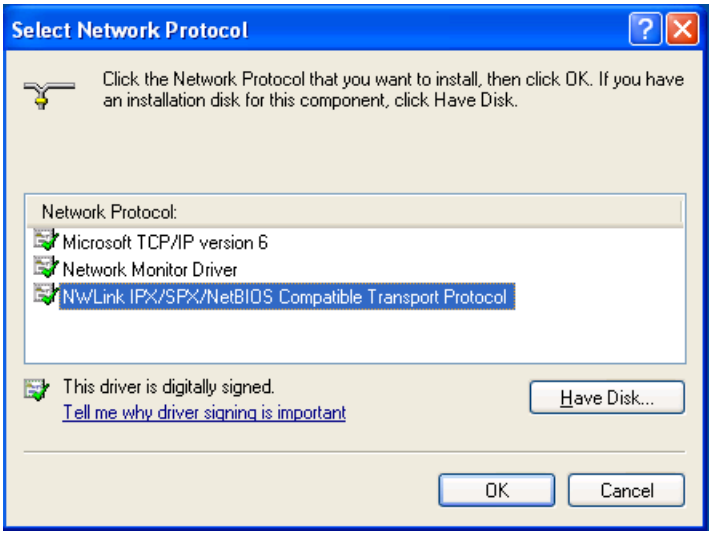
4. The window "Local Area Connection Properties" is now opened. You can now see which protocols on your PC are already installed. If the IPX / SPX protocol is already installed, click the "OK" to close the application.



5. To add the protocol, click on the "Install" button. It opens a window where you can select the network component that you want to install. Select "Protocol" and click on the "Add ..." button.



6. In the next window, select the "NWLink IPX/SPX/NetBIOS-compatible Transport Protocol" and click on the "OK" button.



7. Afterwards, check in the protocol window that the "NWLink IPX/SPX/NetBIOS-

compatible Transport Protocol" has been added and activated with a check mark.

Setting up the serial connection

To establish a serial connection between the LPM software and your Windows operating system, you need to make some adjustments. This is in particular the case if your projector does not automatically open when the LPM software was detected. Please close all applications before proceeding.

1. Open the LPM.INI file, which you will find, depending on the instillation path, under Start -> All Programs -> LPM 8 -> LPM.INI.
2. At the beginning of the LPM.INI file under the heading [Communication], you will find the following entries:

[Communication]	
Simulation=0	Allows you to work with LPM without a connected projector 0 = Simulation off 1 = Simulation on
UseCommunication=0	0 = Serial communication mode 1 = Network communication mode over NetBIOS / IPX/SPX Protocol 2 = Network communication over TCP/IP protocol
StaticIPAddresses=0	Informs LPM to search for projectors using dynamic or static IP addresses. 0 = Searches for all projectors in the network. 1 = Searches only for projectors with IP addresses stated in the entry IPAddresses.
IPPort=50000	Specifies the port on the projector through which the IP protocol is addressed. Valid values are 50000-60000.
IPAddresses=192.168.100.190	Specifies the IP address from which a connection should be established on the projectors. Multiple IP addresses must be separated by commas.
NetbiosLana=99	Network address for network communication over NetBIOS / IPX/SPX protocol
NetbiosPlantId=0	Here is the address for a group of laser projectors to be used independently of each other in a network. This is used only over the NetBIOS / IPX/SPX protocol
Port=0	Serial communication interface. 0 = automatic COM port search When the search was unsuccessful, you

	can enter manually the COM port number. You can check what interfaces are available by checking in → Control Panel → System → Hardware → Device Manager connections you what interfaces are available. In general, however, this is "1" or "2"
Baud=0	Speed of transmission through the serial interface (bit rate). 0 = automatic detection of the bit rate. Valid values are 9600, 19200, 38400, 57600 and 115200, by default, it should be 9600.
CommandAtProgEnd = /P& 11111111 /BS	Defines the last command when the program terminates. Stop the image projection in all 8 image buffers
ShowTxErrorInt=1	1 = Error message activated by incoming data communication problems. 0 = Disables error messages.
ShowTxErrorExt=1	1 = Error message activated by outgoing data communication problems. 0 = Disables error message
Slaves=1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0	Every attached laser projector represents a "1" in order to register that projector. It can connect up to 16 projectors, which corresponds to a numerical sequence of A. .. P. Slaves = 1,0,0,1 means that only the laser projectors with the client identifier of A and D will be sought for and registered.
CREFSequenz=A1B2	Uses Projector A with a measured reference point 1 and Projector B with a measured reference point 2 for referencing with the direct command *CREF. The reference points must have been measured previously in the setup of the projector. Valid values are for the projectors A...P and the reference points 1 and 2.
IdlePollingEnable=1	1 = Enable querying of the laser menu. 0 = Disables querying of the laser menu.
ExpertMode=0	This value should not change under any circumstances, otherwise there is a risk that the factory calibration will be deleted.
; Slave_A ... Slave_P=ClippingArea: Xmin,Ymin,Xmax,Ymax for all **Slaves set to 1**	This entry is commented out by the semicolon at the start of the line(;) and it will not be considered as a command. This is an information to the field boundaries in the simulation mode, which

	reflects the laser work area in 1/10 [mm]. Only relevant if you operate in the LPM simulation mode.
Slave_A=0,0,50000,40000	Shows the coordinates of the laser field border of projector A in 1/10 [mm]: the corners of the rectangle are located at (0/0), (50000/0), (50000/40000) and (0/40000). Only relevant if you operate LPM 8 in simulation mode.
Slave_B=50000,0,100000,40000	Shows the coordinates of the laser field border of projector B in 1/10 [mm]: the corners of the rectangle are located at (50000/0), (100000/0), (100000/40000) and (50000/40000). Only relevant if you operate LPM 8 in simulation mode.

3. If your laser projector has not been detected with the automatic registration process, you can change the parameters manually or reset them at any time. After you have made your changes, save and close the file.

4. Start LPM. When the connection is established between the LPM and the projector, a blue box will be seen in the graphics window.

5. If no connection could be established, close LPM, and open the file LPM.INI. Configure your settings. Make sure that the entry "UseNetbios" to 0 has been set.

After each change, LPM.INI must be saved and LPM must be restarted.

Interfaces

Hyper terminal

There is the option to use your laser projector serially, without the LPM software, by applying the Windows standard programme “**Hyper Terminal**”. However, a quick calibration of the projector will not be possible with Hyper Terminal. To define the points of reference and drift compensation that are necessary for measuring out your working space, it is essential that you use the Z-Easy guide for quick calibration!



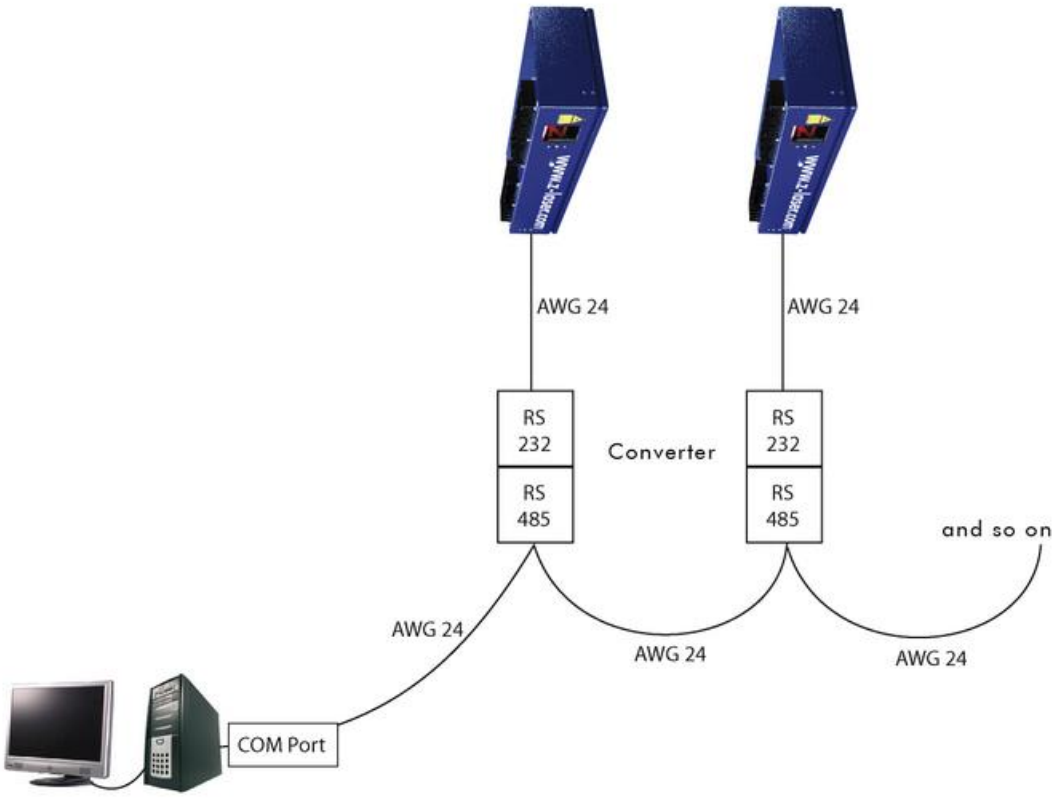
Please note that you will not be able to communicate with your projector, i.e. no exchange in data, when using this application. You will merely be able to send data to it in **HPGL-format**, for further projections.

Requirements

To use your projector/s via Hyper Terminal, you will need the following accessories:

· **RS-232 / RS-485 –** converters in accordance with the number of projectors you have.
· **AWG 24** twisted pair cable, shielded and with at least two conductors.

With the aid of an AWG 24-cable, connect the RS-232 interface of the respective converters, with the 9-pole SUB-D connection, on your projectors.
With the aid of an AWG 24-cable, connect the RS-485 interface of a converter with a free serial COM port on your PC, as well as the RS-485 interface of a second converter (when using two projectors). If you are using three projectors, please connect the RS-485 interface of the second converter with the RS-485 interface of the third converter, etc.



Installation

The following explains the installation of the Hyper Terminal:

1. Call up the Hyper Terminal via Start → Programs → Accessories → Communication → Hyper Terminal.

2. The “link description” window is displayed.



   Add a new connection name under “Name” and click on the telephone symbol. Then press “OK”.

3. The “Connected with...” window is displayed.

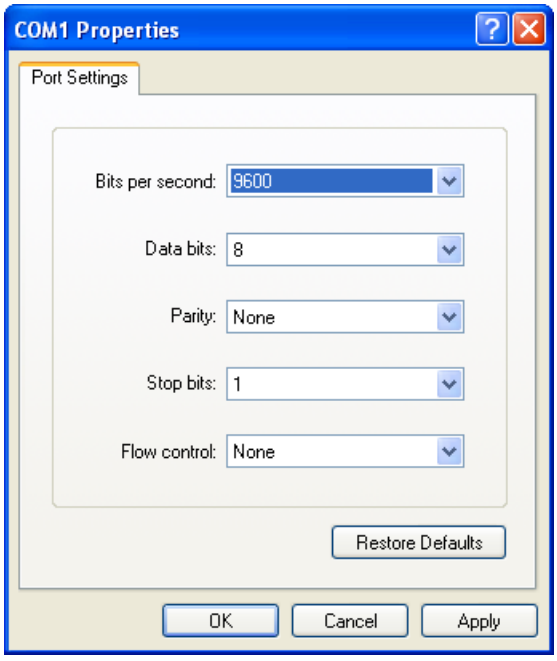


   Under “Receive connection via”, select the COM port on your PC that will be connected to the projector.



Then press “OK”.

4. The “COM1 Connection Properties" window is displayed.



5. Set up as follows:

Bits/second (Baud):	9600
Data bits:	8
Parity:	none
Stop bits:	1
Flow-control:	none

Please note:

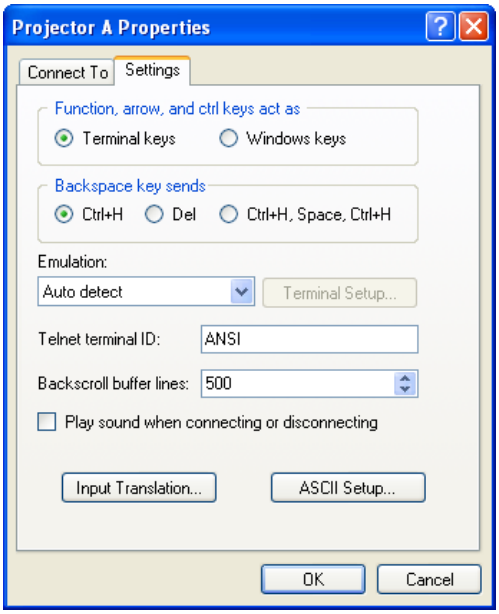
The baud rate, which is installed under “Bits per second”, should meet with the baud rate that is recorded in the projector internal ELC.INI data. The default factory

setting is normally 9600 bps. You can change this at any time and using the LPM software being essential for this.

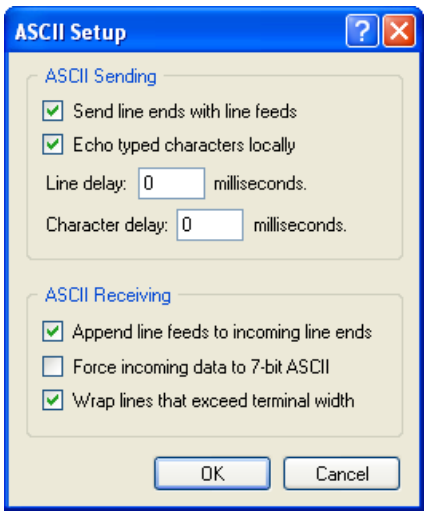
Click “Accept” and then “OK”.

6. The Hyper Terminal window should now be displayed on your pc screen.

   Via Data → Settings, call up the properties window of your connection. Click on the directory settings. Do not change the pre-settings. Select the button “ASCII-Configuration...”



7. The ASCII-Configuration window is displayed.



Activate the following check box:

- Send line ends with line feeds
- Local echo for input text
- Upon receipt, attach a line feed at the end of the line
- Wrap long lines in the terminal window

To change the settings, click “OK”.

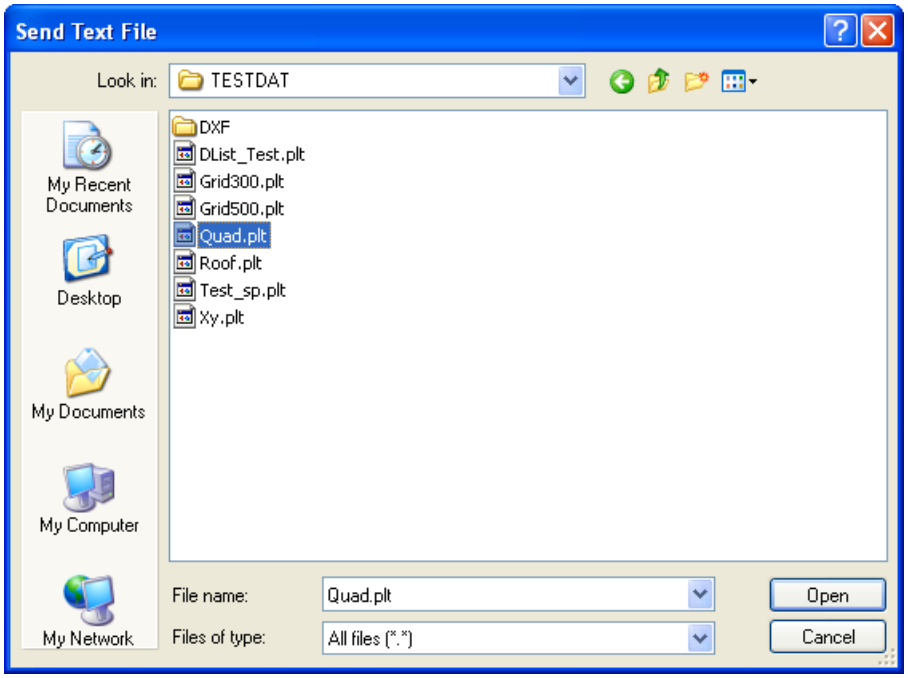
8. Choose the symbol “Call” on the toolbar, or open the menu bar and click on the entry “Call”.

9. Hyper-Terminal will now provide a connection between your pc and the projector.

Data transfer

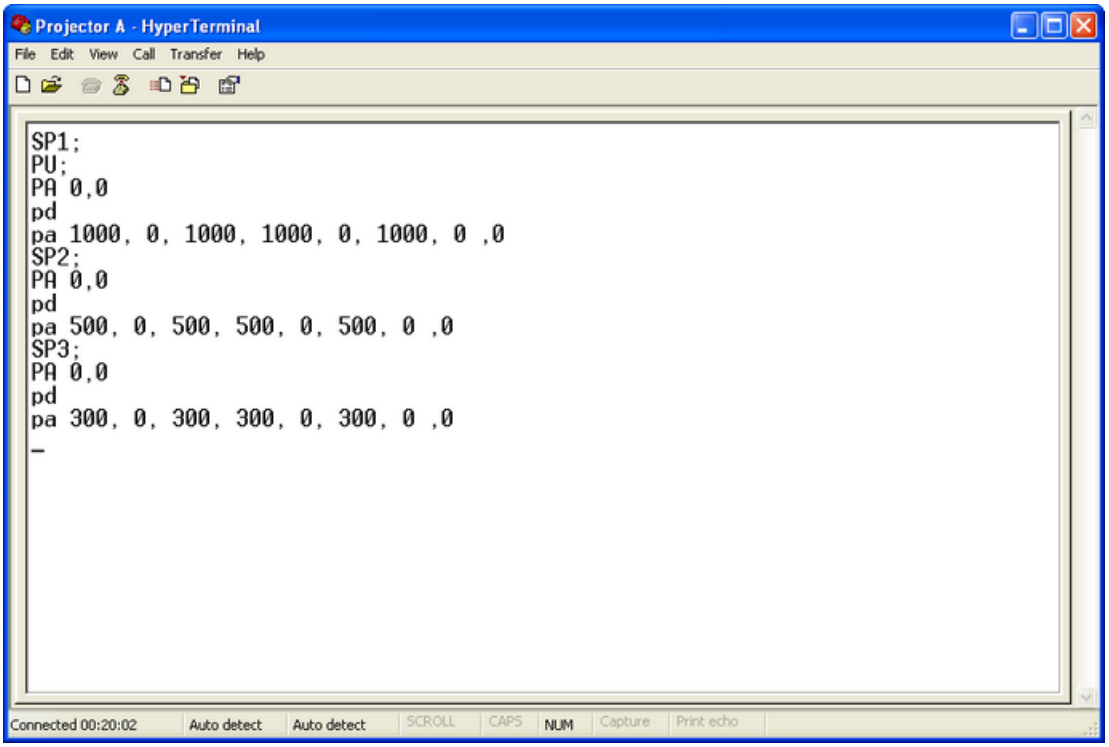
In the top menu bar of the Hyper Terminal window, click on “**Transfer**” and then the sub-command “**Send text file...**”

An Explorer-window will open, which will allow you to select data via the path to your files. To ensure that the data can be displayed in the Explorer-window, please select “All files (*.*)” under “File type”.



After selecting the file type, click “**Open**”.

The data (i.e. commands) found in the file Quad.plt should now appear in the Hyper Terminal window.



At the same time, your laser projector should now project the Quad.plt test image. You can project any data in the **HPGL format** this way.
You can also give all HPGL instructions directly through the Hyper terminal window.
To end the projection, please click the "**Disconnect"** symbol on the Hyper Terminal window toolbar, or navigate to the menu bar and go on to the item "**Call**" and click on the sub-command "**Disconnect**". Alternatively, you can disconnect and/or activate the projection via the Hyper terminal window, using "**ZF**" to disconnect and **"ZN**" to activate.

A list of all the HPGL instructions, as well as additional Z-instructions, is found in the chapter HPGL- and Z-commands**.**

Samba share folder

The client has a Network drive (“Samba-Share”) which contains the following directory:

- **Plot**

In this directory, copied HPGL data (.plt) will be projected automatically.

All data can be erased, as soon as the projection is in progress.

Note: Drag & drop is possible.

- **Manual**

In future, the projector manual will be stored here. The directory is available for reading only.

- **LPM**

In future, the installation programme for LPM will be stored here. This directory is available for reading only.

- **Data**

Data (*.plt and CSV) used for the operation via SPS navigation are stored here (Multitec). This includes information regarding the projector. This directory is available for reading only.

The network drive is accessible as follows:

- Type two back slashes (“\\”) in the address bar of Windows-Explorer, followed by the IP address, eg. \\192.168.100.22
- Via Windows-Explorer → network environment → entire network → Microsoft Windows network → zlaser → laser projector (serial number)

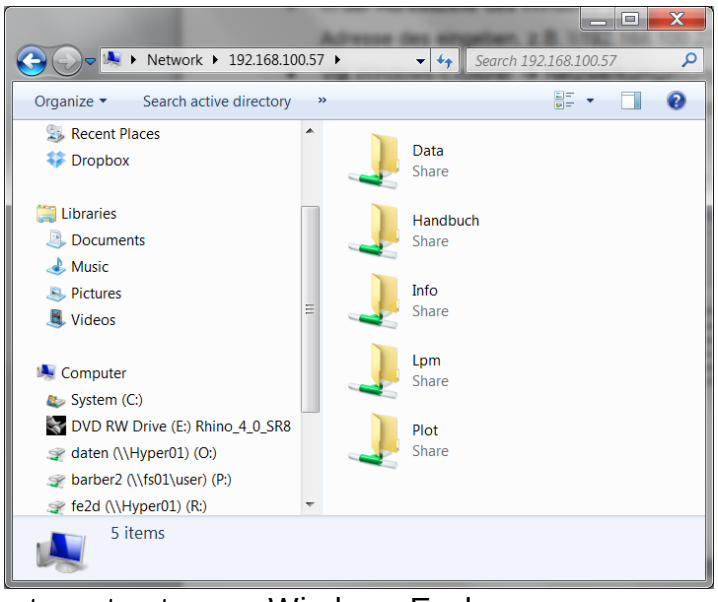
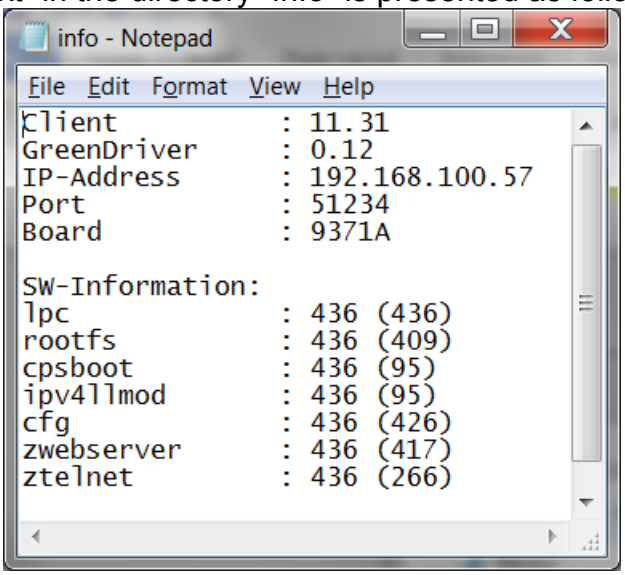


Illustration of the directory structure on Windows-Explorer:

The data content “info.txt” in the directory “Info” is presented as follows:

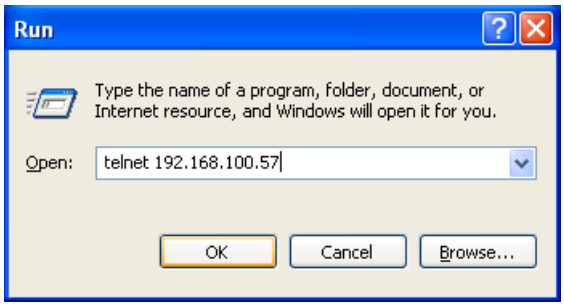


Telnet

Telnet connection allows for a network link to the laser projector, using an IP address. The standard Telnet port is 23. This only works for Windows XP or earlier versions of Windows.

The connection allows for the projection of HPGL commands as well as specific Z-Laser commands. “Client Version” and “Temperature” information can be shown.

Go to the “Start” menu, type or click on “Run”. In the “Open” field, enter the following: telnet, space, the laser projector's IP address.



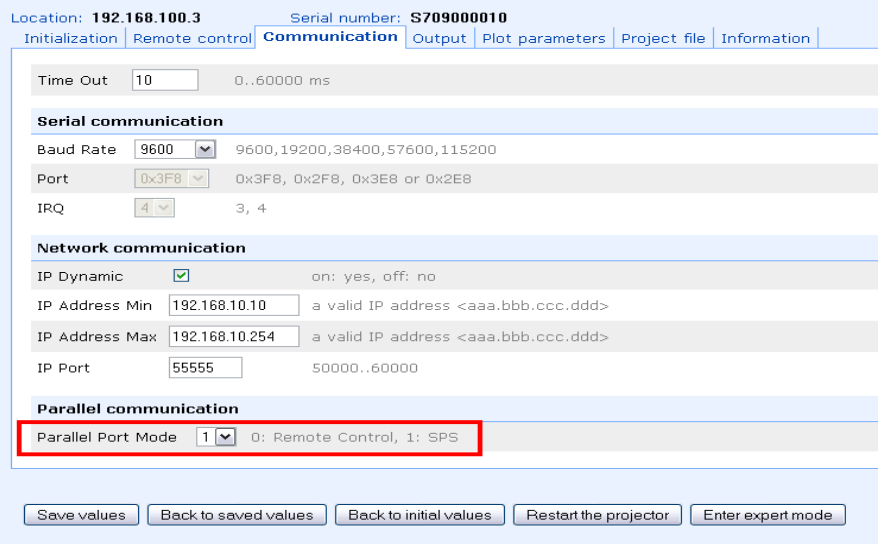
Press “OK”. The following dialogue box will appear:



Projections of drawings stored on the projector

1. SPS interface (selection by bit pattern)

- The projector is connected via a SPS interface
- Parameter PartPortMode =1



- CSV file structure:

<D0..D7>, “<command [parameter]>“,,,,

bit pattern command + parameter not used

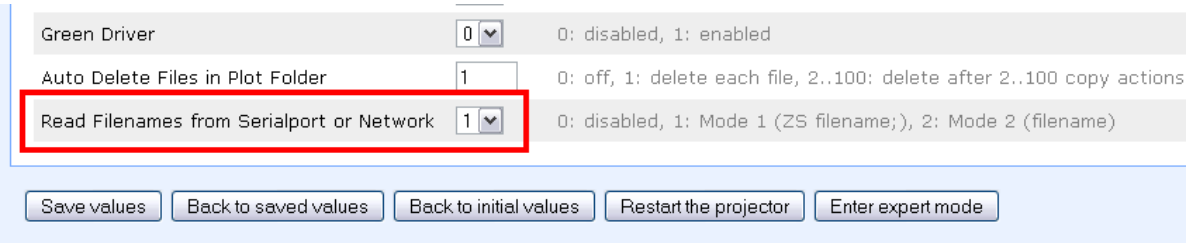
- Save the CSV files in the projector list:

\\<IP-address>\data or

\\<SerialNumber>\data

2. Serial interface (selection by file name)

Parameter **UseFileNamesFromNetCom**



Select the projection data by sending the following to the serial port:

ZS < file name_without_extension>;

Start file name without extension end marker

3. Network (selection by file name)

- The projector is connected to the network.
- Parameter UseFileNameFromNetCom=1
- Save the files in the projector list “Data”.

  \\<IP-address>\data or

  \\<SerialNumber>\data
- Start/select the projection by sending the file name via network link.

Driver for Green Laser Sources

Z-Laser offers a newly developed laser driver technology for green projectors. A stable operation of green DPSS laser sources could only be ensured for a limited time in the past, neither the life nor the failure mode could be predicted reliably.

With intelligent electronics and special control software, Z-Laser can now enable much longer lifetimes for green laser sources in its projectors. Rather than replace the highly complex drivers, they are continuously monitored during operation and adjusted according to its changing properties so that they continue to operate stably in the specified range. This is totally self-sufficient; the user is only informed about the on-going operations and has needs-based intervention.

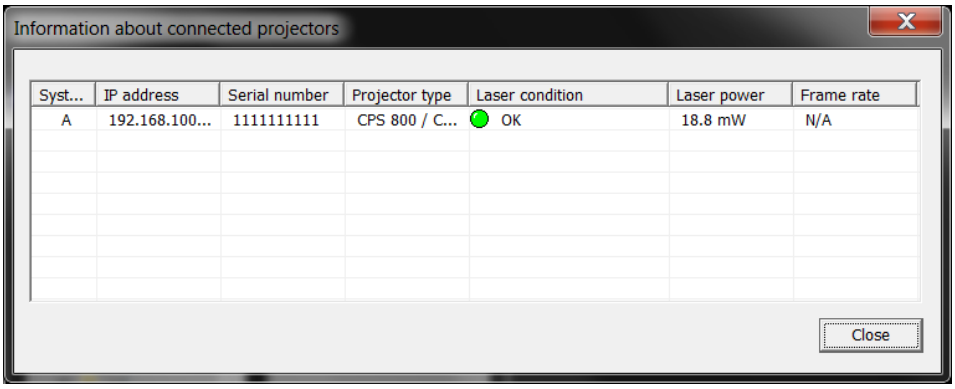
There is constant monitoring of the condition of the laser sources on the status display on the projector. This can be viewed in the LPM software by selecting “?” -> “About Projectors”.

The main properties of the Green driver are:

- Control of laser power through a special control loop.
- Signalling of the state "Power too low" (and proposing an operating point search).
- Independent search for a new operating point of the laser source.
- Signalling the state of the laser source (End-Of-Life or EOL)

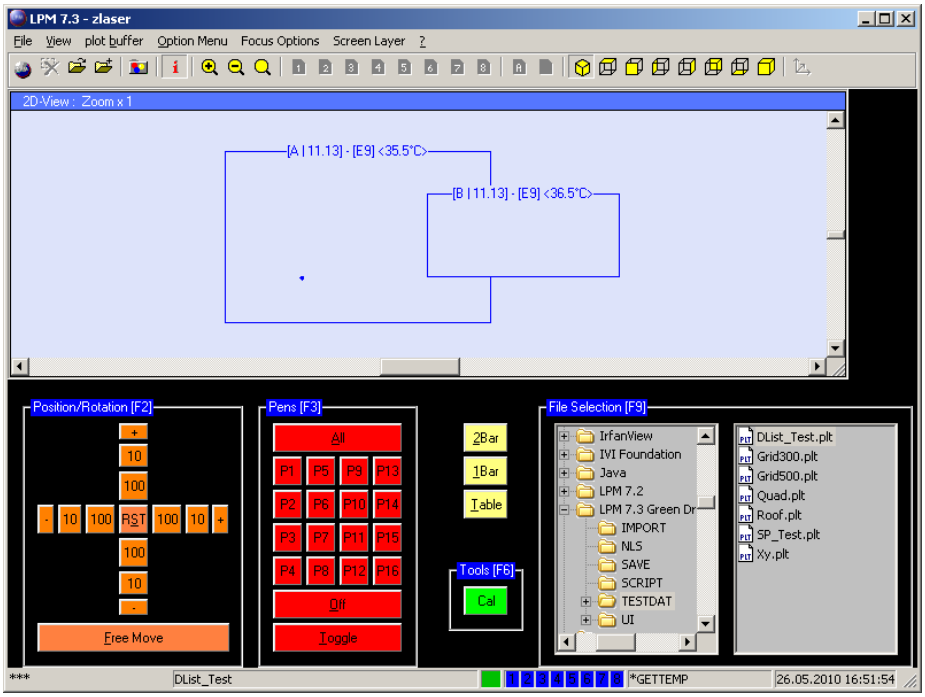
Status displays of the Green Laser Source:

- LPM -> “?” -> “About Projectors”.
- Green: long life expectancy
- Yellow: short life expectancy
- Red: source needs to be replaced



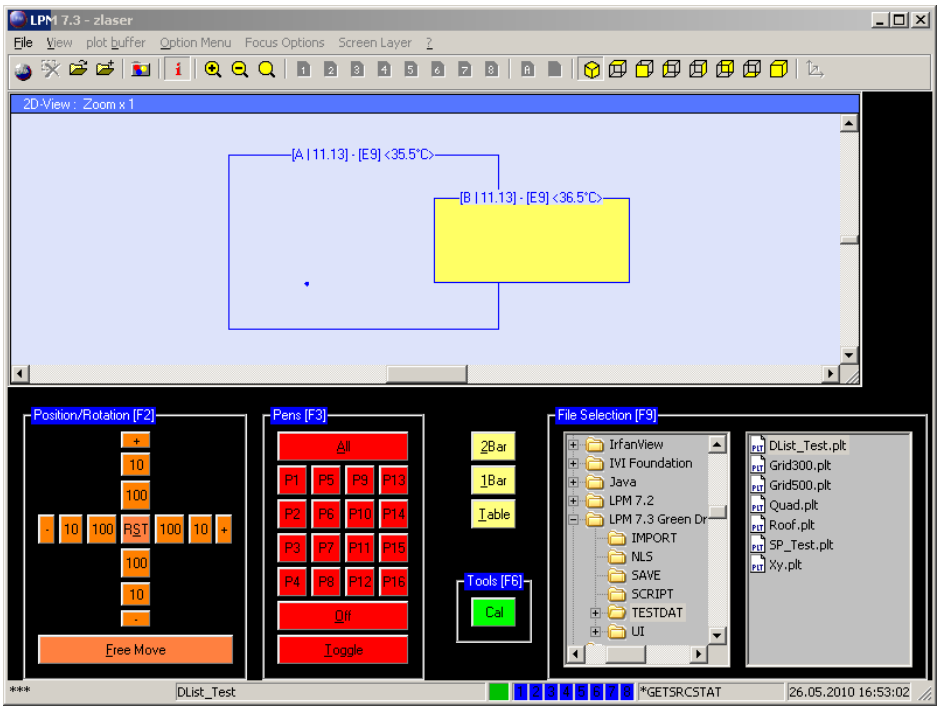
Implementation in LPM:

1. User interface during normal operation.



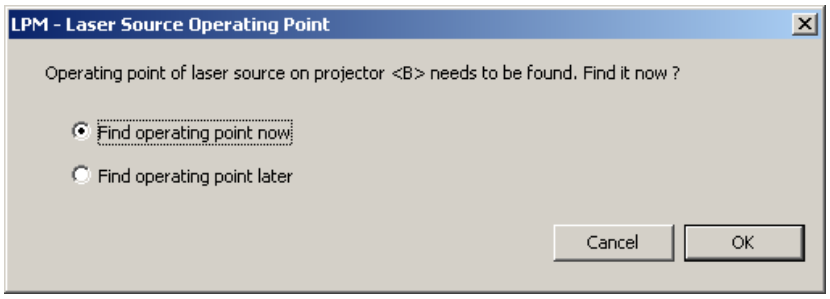
2. In LPM.INI, under the heading [MainFrame], there is a parameter called "getSource". The default setting is 60 (unit = minutes). This means that after a period of 60 minutes, the status of the source is requested. If the field of that projector turns yellow in the graphical interface after the driver reports back (output power cannot be reached), an operating point search is recommended.

   An operating point search can take up to 3-4 hours. They should only be started if the projector is not required during this period. The laser projector projects a test pattern during this period, so direct visual contact with the laser projection should always be avoided!

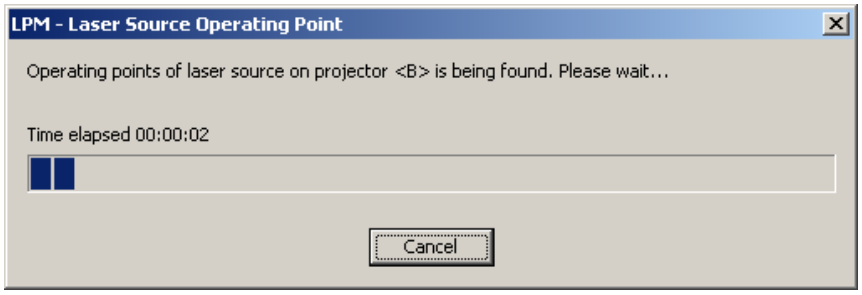


Operating point search:

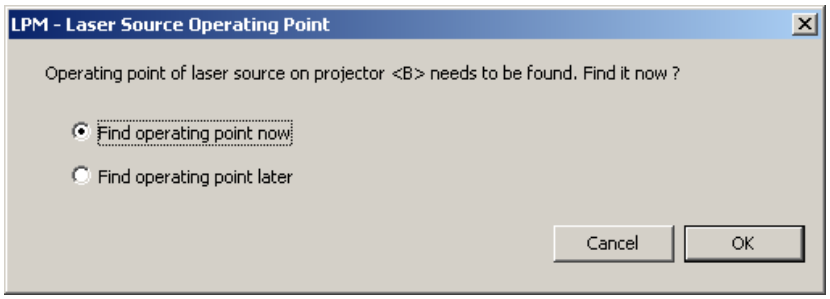
a. Find the operating point now: Select the option "Find operating point now" and click "OK".



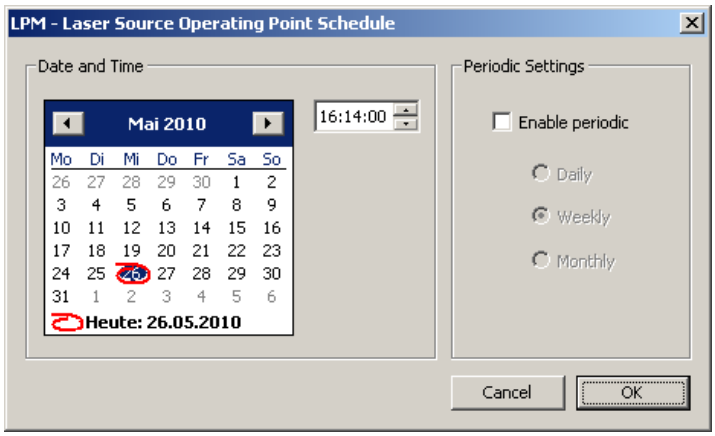
The operating point search will begin.



b. Find operating point later: Select the option "Find operating point later" and click "OK".

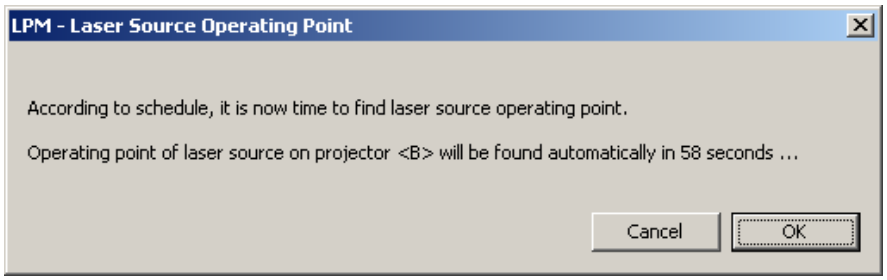


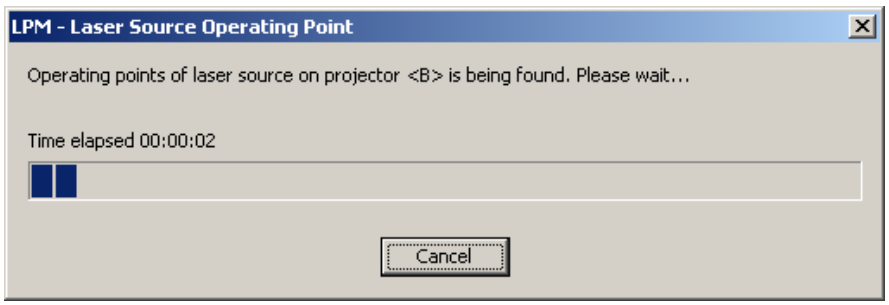
A dialogue window will open with a calendar. Now set when you want to start the operating point search. Confirm your entries with "OK".



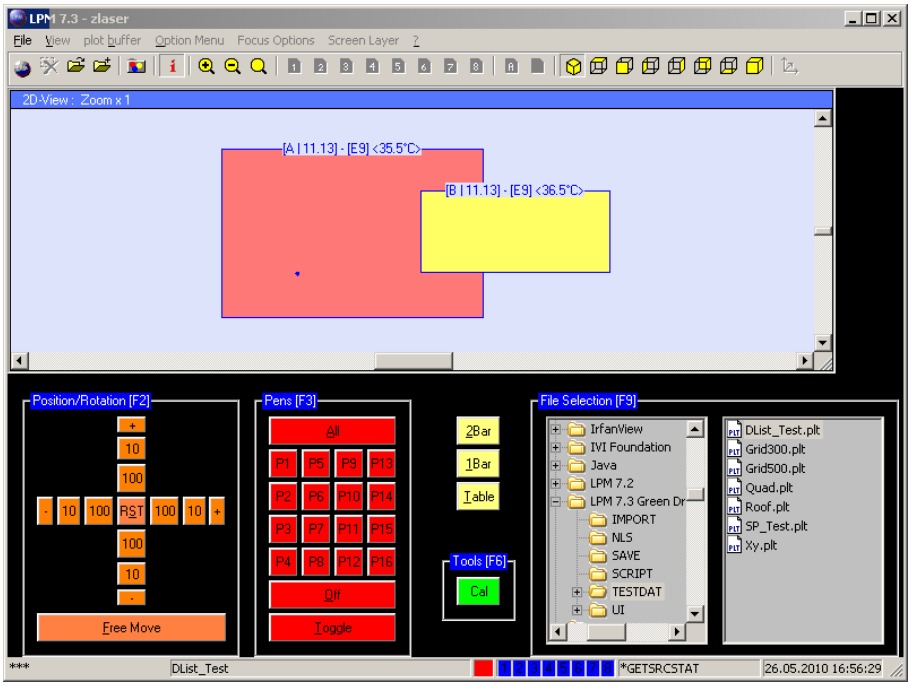
At the specified time, the operating point search is then automatically

started. This appears in the following dialogue box.





3. Red light: laser source is not ready. This means that the driver does not respond. In these cases, the projector will no longer project! Contact a service representative.

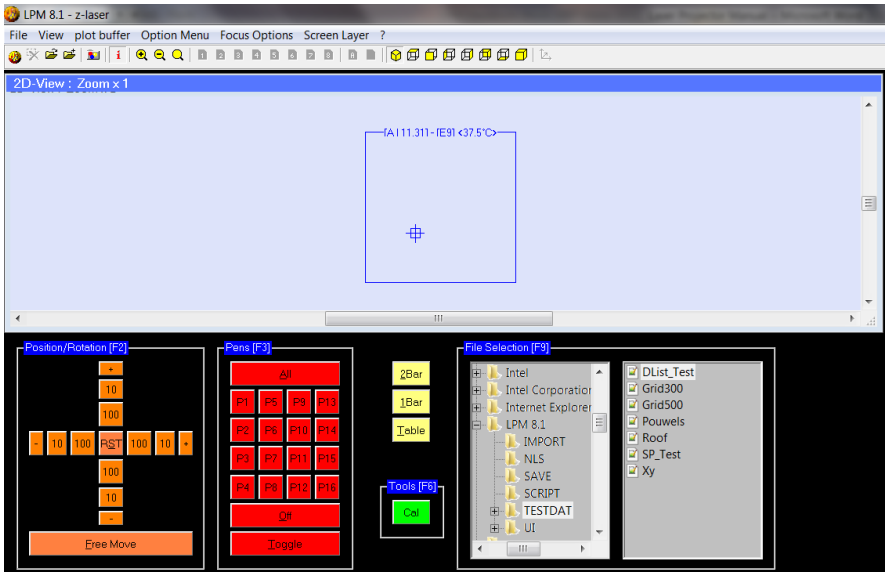


4. Benefits for the customer
5. Constant output power by APC (Automatic Power Control).
6. Operating point search (search bias) can be performed by customers themselves.
7. Current state of the laser source is shown.
8. Higher quality of the projection as extensively tested operating points are used

Quick Calibration

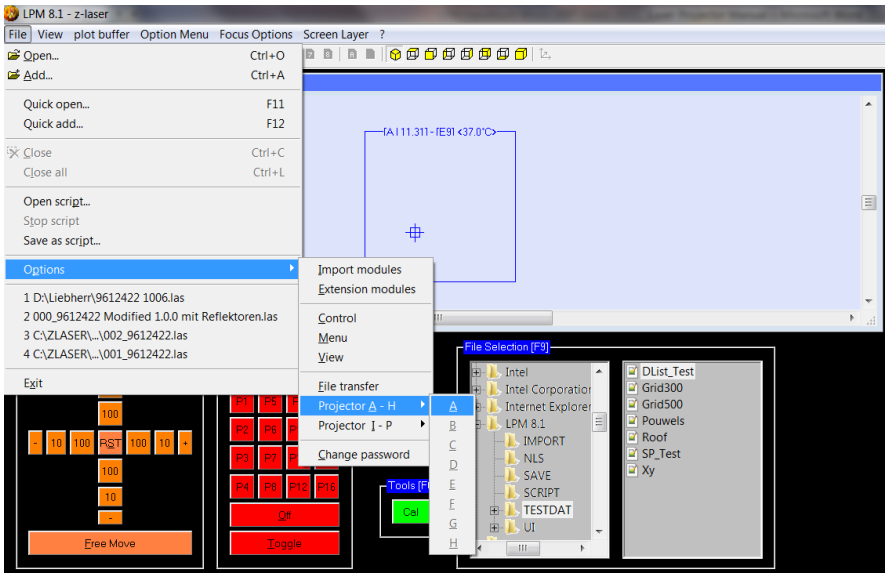
The quick calibration process allows your projector to learn about its projection environment. The coordinates of the working environment, as well as your projector's position within this environment, will be transferred to the projector during the process.

1. Open LPM and wait for your projector to be registered (as seen by the appearance of the blue field border in the graphical user interface).



2. You can start the quick calibration process by going to:

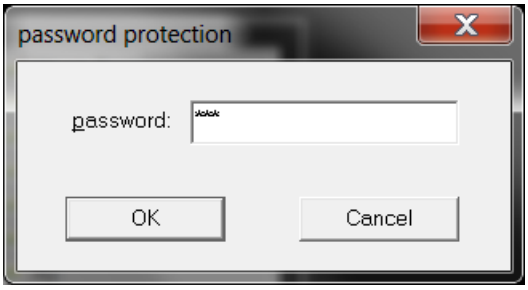
   FILE → OPTIONS → PROJECTOR A – H → PROJECTOR A



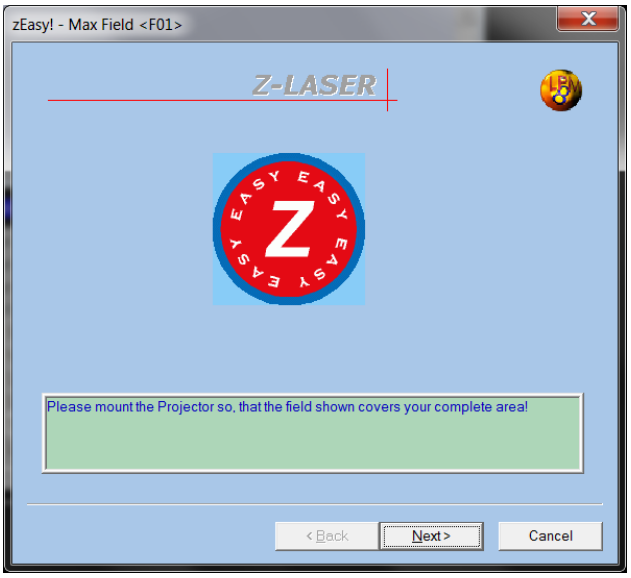
3. Insert your password. The default password is cps.

   You may change the password by going to:

   FILE → OPTIONS → CHANGE PASSWORD



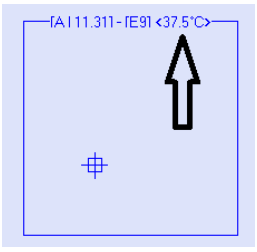
4. The Z-Easy assistant is now started and it will now guide you step by step through the quick calibration process of your laser projector.



Your laser projector will now display the maximum field border. This is the maximum possible area into which a projection can be displayed. The projector should be repositioned to include as much of the working area inside of the maximum field border.

At this time, it is best to allow the projector to display the maximum field border for at least 15 minutes to allow the projector to warm up and achieve thermal stability for optimal accuracy during the quick calibration process.

The optimal temperature is 33° Celsius. The current temperature of your projector can be found in the blue field border of your graphical user interface in LPM. Proceed after the optimal temperature has been reached.



Continue after approximately 15 minutes.

5. Define the location of your drift compensation points. These are the glass reflectors included with your projector.

   These reflectors should be located at a minimum distance of 200mm to the maximum field border but outside of the working area. They should also be mounted at a similar height to the working area if possible.

The reflectors should be placed in a fixed position and should be visible to the projector at all times. Do not cover them with tools, cloth, etc. It is possible to mount the reflectors in brackets by drilling 8mm holes at the correct positions.

The drift compensation points are used to correct any future offsets caused by thermal or mechanical drift.

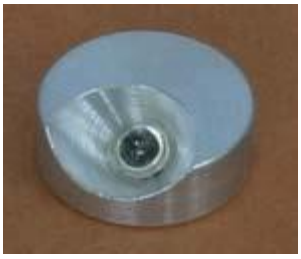
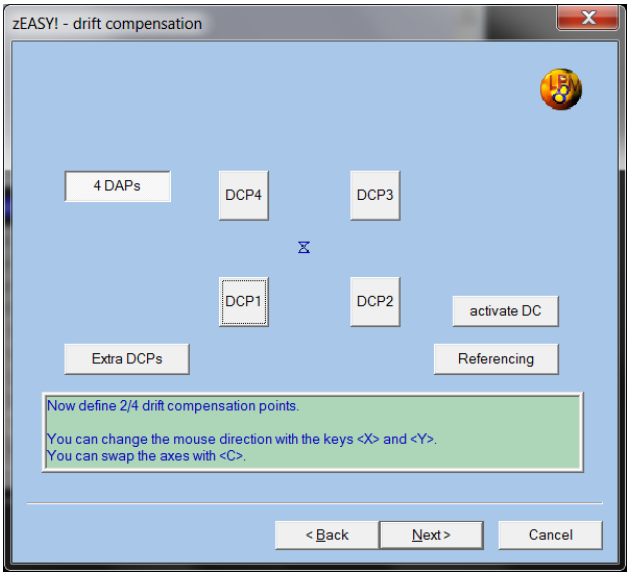


Figure 16: Drift Compensation Point (DCP)

6. The Z-Easy – Drift compensation window will now appear.

   Ensure the buttons “4 DCPs” and “Activate DC” have been selected. If the “Activate DC” button has not been selected, then the projector will search for the drift compensation points at the factory defined positions.



   Alternatively, you can define and search for only two DCPs by unselecting the “4 DCPs” button. You must be position these DCPs diagonally to each other.

   However, for reasons of accuracy, we strongly recommend you use a minimum of 4 DCPs.

7. Select the button “DCP1” to measure the first drift compensation point.

   Move the projected cross to the position of the first DCP.

   The following keys will allow you greater control over the movement of the cross:

Cursor	Fine tuning
1, 2 or 3	Mouse speed
X	Swap X axis movement
Y	Swap Y axis movement
C	Swap X and Y axes

As soon as the cross lies directly over the point, press the left mouse button or the return key to initiate a search of the reflector. The projector will now search for the reflector using horizontal and vertical lines.

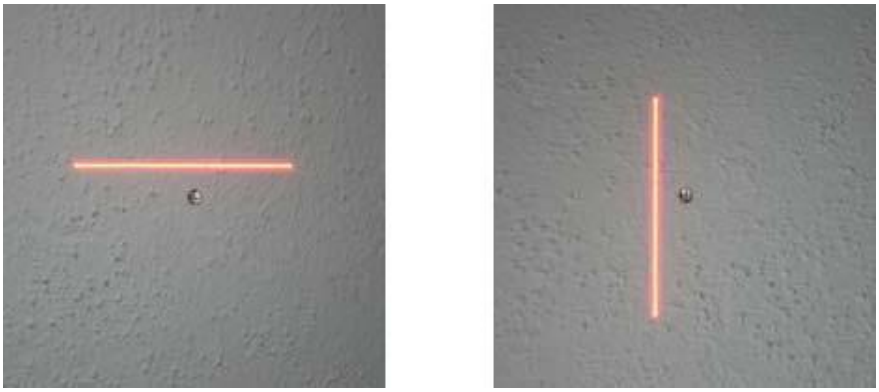
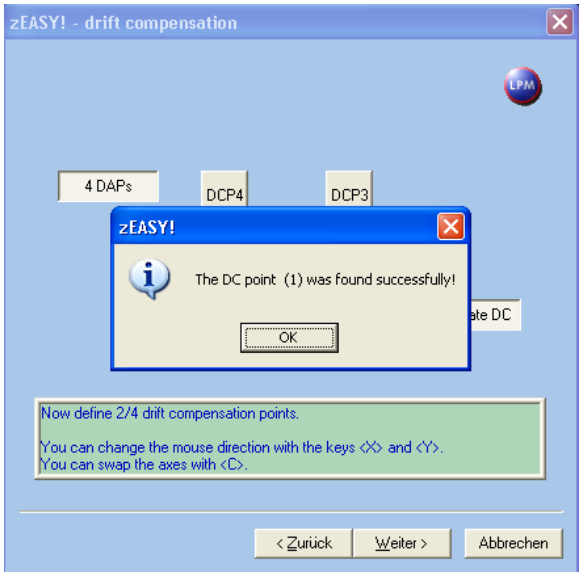


Figure 17: Selecting the DCP

8. A message box will appear informing you of the success or failure of the search. If the search was successful, check by eye that the cross is located in the middle of the DCP before clicking “OK”.

   If the reflector was not found, ensure there are no objects are in the path of the laser beam. Additionally, check that the reflector is correctly fixed in place. Repeat the DCP search.



   Repeat the process for DCP2 → DCP4. Ensure all four reflectors have been found once and select “Continue”.

9. The next step is to familiarize your projector to the system of coordinates of its working environment. To do this, make four points within the working area where the connected lines of these points create a rectangle and the coordinates match those of the area.

   To achieve the greatest possible accuracy, position these points as close to the corners of the working area as possible. For accuracy, it is absolutely necessary to create the rectangle without tolerances.

   If you drill or mark the rectangle with a CNC machine, please use whole numbers.

Pay attention to the chosen coordinate system and the accuracy of the points. If the rectangle is measured manually, make sure that all 4 points lie on the same plane. Check the perpendicularity of the rectangle by measuring the two diagonals of the rectangle. Both values must be equal. To determine the desired length of the diagonals, you can use Pythagoras' theorem ($a^2 + b^2 = c^2$).

Mark your four points with the included calibration markers numbered 1 – 4 and note the corresponding X and Y axis coordinates in the installation protocol included in the appendix.



Figure 18: Calibration markers

This procedure is used to restore the coordinate system to the projector if the projector becomes defective.

However, please note the following:

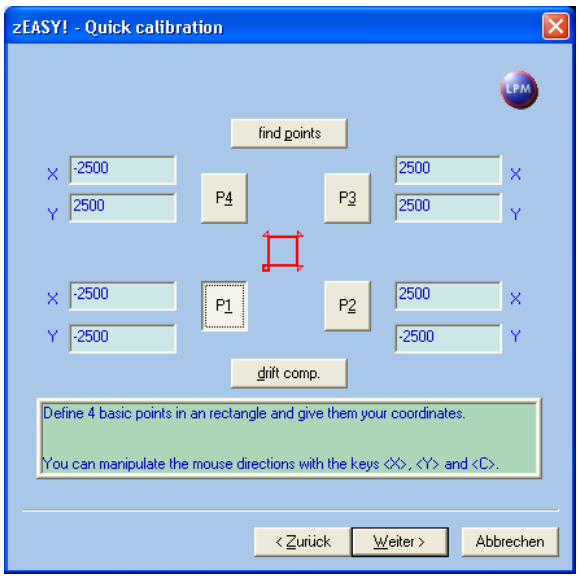
If the calibration markers are used to calibrate the projector, the calibration height is increased by 4mm.

The height offset must be therefore decreased by 4mm when setting up the field border.

If the calibration markers are counter-sunk to the same level of the table, then the height offset is zero.

Please use the stone or metal drill bits which are included with your delivery to drill the working area for the calibration markers.

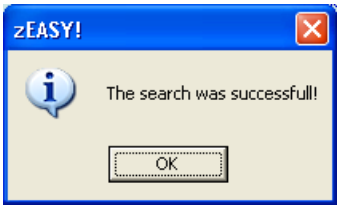
10. After defining the four points P1 – P4, select the button “P1” to measure the first point.



Move the laser cross that is projected to the nearest point. You can move the cross to its rough position with the mouse and fine tune it with the cursor keys of the keyboard. As soon as you have exactly positioned the cross over the point, click the left hand button on your mouse or press the return key to confirm its position. Insert the X and Y coordinates. The values must be whole numbers.

11. Repeat the process for P2 – P4.

    If you used reflectors for marking the points, you can search for the points by roughly move the cross over the points and by selecting the button “Search Points”. The projector will now search for all four points by itself and set their positions. A message box will inform you of its success or failure.



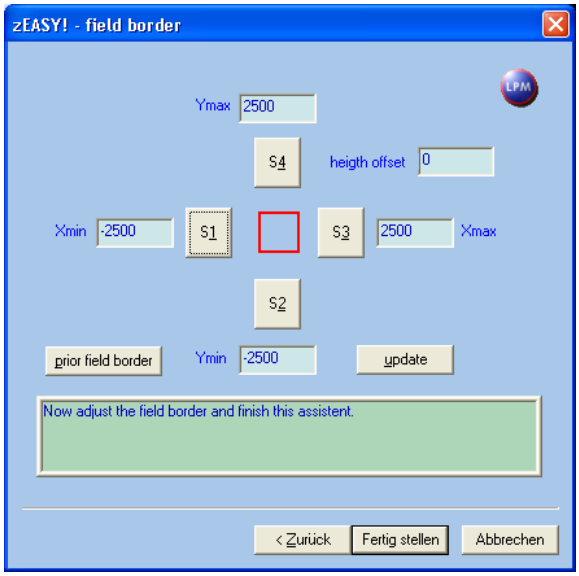
    Select the “OK” button.



However, we would recommend setting the positions manually to achieve higher accuracy. If there has been a long period of time between the measuring and the actual setting of the points, you must compensate for any drift by selecting the “Drift Compensation” button.

Select “Continue” once you have set all four points.

12. In the next step, the projector projects a rectangle called the field border. The software displays the field border for the points defined in the last step. The displayed field border edges should therefore project directly on top of the points. If this is not the case, go back to the previous window and ensure the coordinates and point positions are correct and that the height offset is set to zero. The field borders should be adjusted in such a way that a projection can be displayed anywhere in the desired working area but not outside of it.



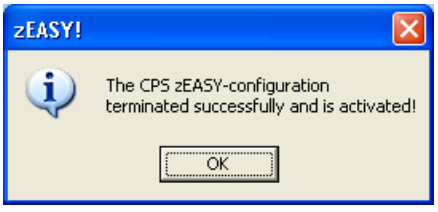
If you want the points made in the step 14 to define the field border, ensure the edges of the field border lay on these points and check that the coordinates of the X and Y axes correspond to the coordinates of the points.

13. If you wish to define the field border manually, you can select each side individually by pressing the buttons “S1 – S4” and move the mouse or use the direction keys of the keyboard until the side is in its preferred position. Using the left click on the mouse or the return key on the keyboard to confirm the position. Once that is complete, the new coordinate of that side will now appear in the field which corresponds to the side you selected. Alternatively, you can enter the values in manually and press “Actualize” to confirm the field border's position.

    If the field border collapses to the size of the original field border, then you have exceeded the maximum field borders. Redefine the field borders so that all sides are within the maximum field border.

14. If the height at which you calibrated was at the height of the table, but above or below the required zero level, you can insert the height difference in the field marked “Height Offset”. As soon as you have set the field border and height offset correctly, select “Continue”.

15. The quick calibration is now finished and an information window will inform about the success of the Z-Easy calibration. Your projector is now ready and you will be able to project your first projection file.



16. To prevent the loss of your calibration data, please also read the chapter “Data Backup”.

Referencing

This chapter will introduce referencing with the following example. This deals with the set-up of one or more laser projectors on a pallet rotation system in a pre-cast concrete plant.

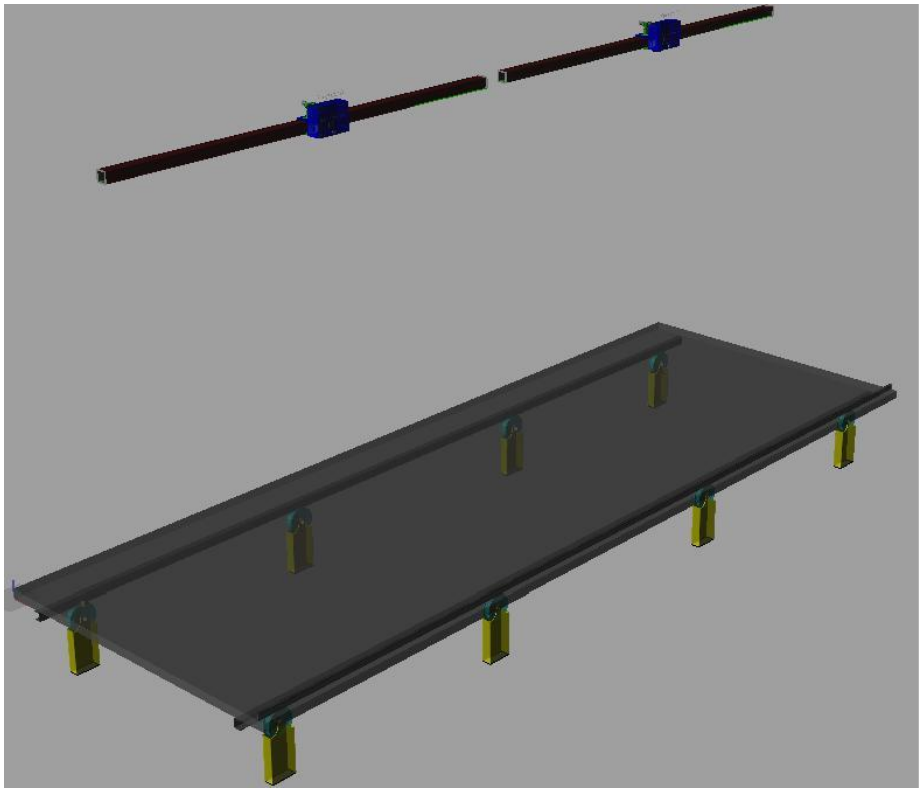


Figure 19: View of a two-projector system

1. Two laser projectors are installed above the pallet. The projectors should be mounted stress and vibration free.

2. Assembling the drift compensation points:

   The drift compensation points are secured on the roller conveyor underneath the pallets. This serves to protect the reflectors!

Where possible, the four drift compensation points should be mounted next to the pallet.

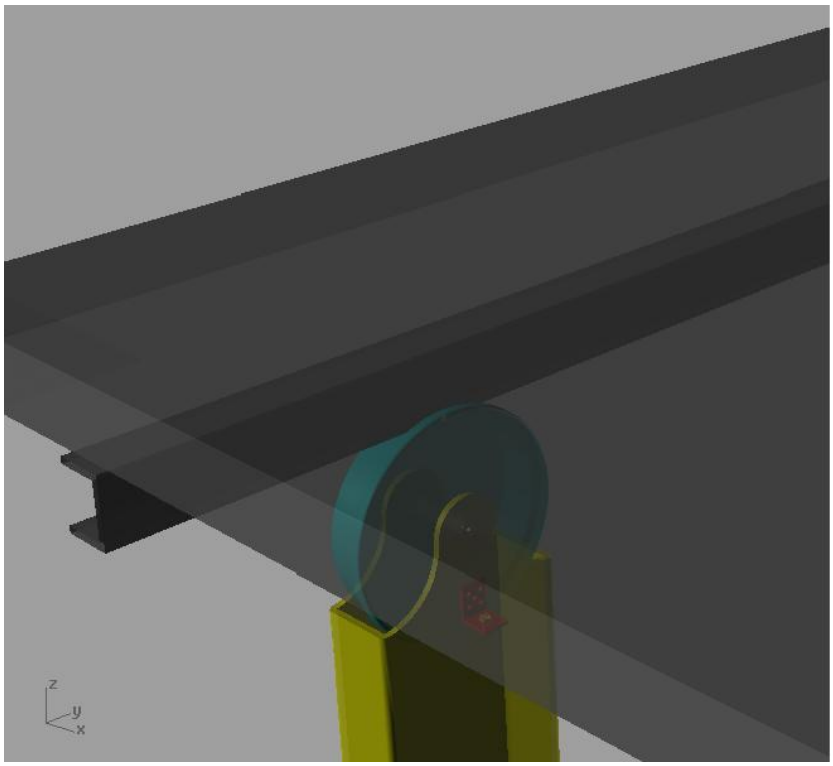


Figure 20: Assembly of the drift compensation points

The drift compensation points are situated in a reflector holder and look like small glass beads. Weld small angles onto the roller conveyor and drill 8mm holes for the reflectors. When the holders are fixed, align the reflector with the projector!



Figure 21: Reflector holder with drift compensation points

A word of caution: each projector needs four reflectors. Only in very rare cases can you use one reflector from two projectors. However, where the projectors meet, in the centre, the reflectors can be positioned close to each other. As the drift compensation points are aligned to their respective projectors, there is no risk that one projector would disrupt another. If the roller conveyor allocation is not symmetric, the positions can be crossed.

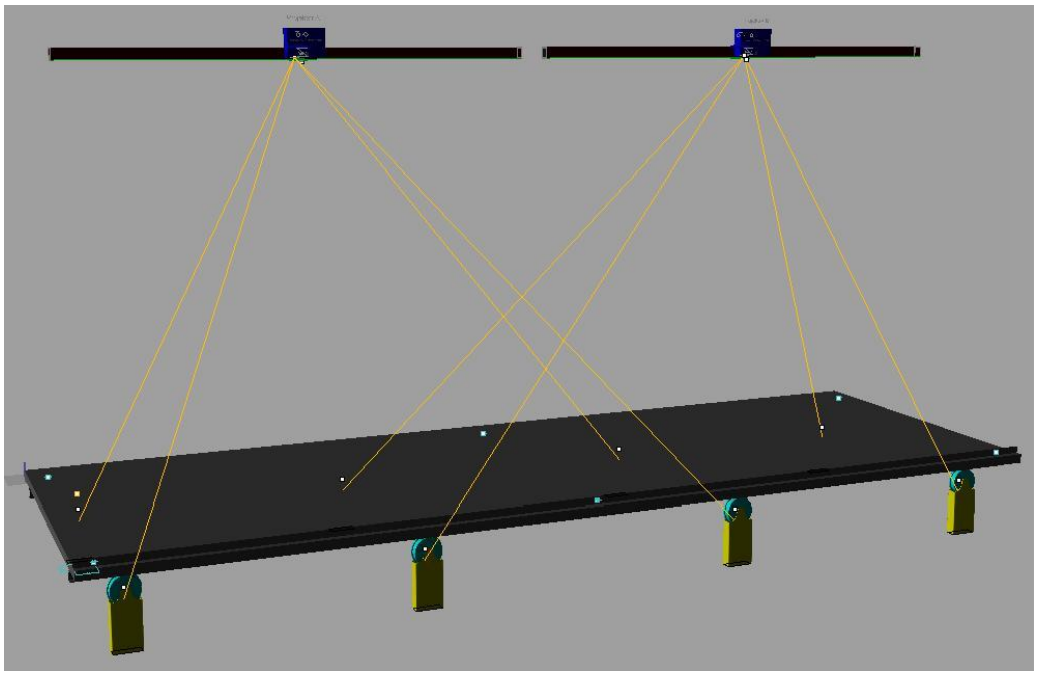


Figure 22: Roller conveyor with two laser projectors

When the reflectors are fixed in place, open the LPM and go to the projector set up, via:

“File → Set up → Projector A-H → Projector A”.

Make sure that the future working area lies within the indicated projection area!

1. Click on “Continue” and you will be taken to the drift compensation points set-up.

Click on the “DCP 1 Button” and, using the mouse, you can move the laser cross around.

Please note: The laser projects a cross, that needs to be moved from the lode point to the next reflector point. There is NO prescribed sequence in which to do this! The projector sorts these points independently!

Please run this process on all four reflectors. You can now simply click through the quick calibration to recalibrate the drift compensation points for the second projector.

2. Reference Points

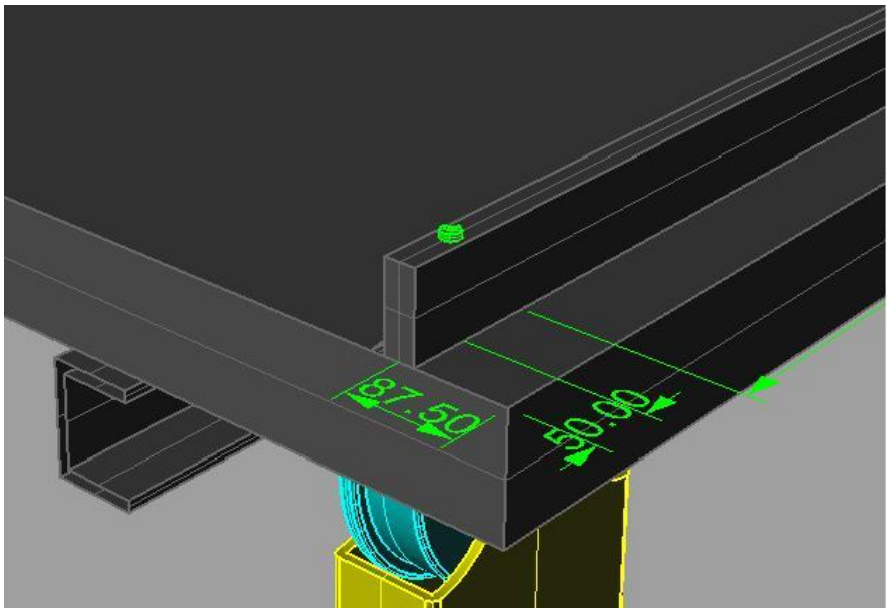


Figure 23: Fixing the reference points on the unit

The reference positions help to align the laser projection to their respective pallets.

Feature: By referencing, the point that is adjusted first is used as an offset. This results in a shift in the X and Y axes towards the pallet.

The second point is responsible solely for the rotation to the axes. Calculating the rotation is accomplished through measuring the discrepancy between the “master pallet” and the pallet currently in use.

Please pay attention to the parameter in

LPM.INI

CREFSequence=A1B1

(These are the projectors, which represent the points that are now actually measured in the projector.)

LPC.INI

RefIntern=0

3. The reference points:

Now the projector is adapted to suit local conditions. A clear, previously arranged geometry should be provided for the projector. The clear geometry should consist of one PERFECT rectangle. This rectangle is the core centrepiece for the entire set-up. Now move a pallet to the station.

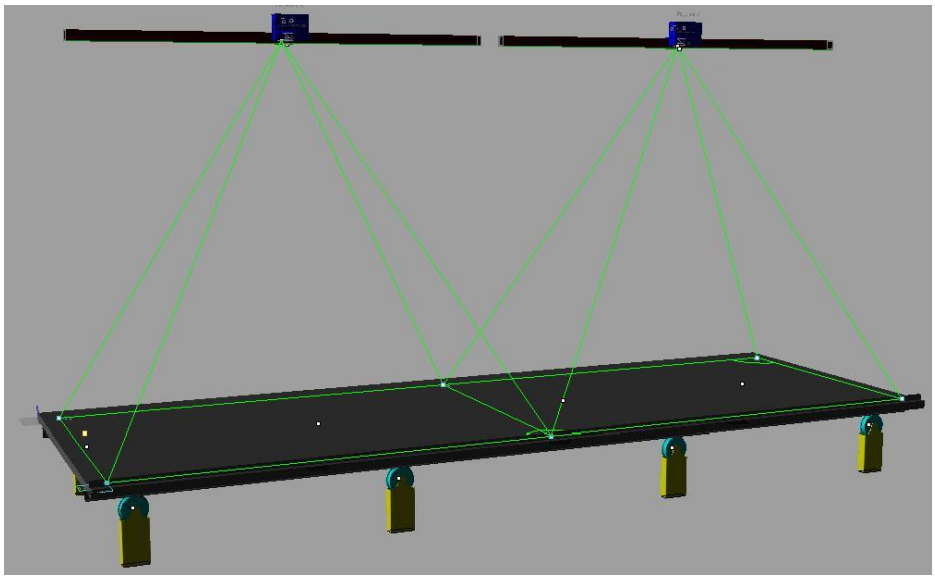


Figure 24: Reference points

Some background information on how to generate a rectangle:

It is advantageous if fixed form boards are in place on the pallet. About 200mm from the form board towards the centre of the pallet, draw a mark on either side. Use a lay rope to mark a perfect, straight line on the form. You can also use a taut string and a magnet to keep the string tight. Please check to what extent the string is parallel to the stopper/form board. Sometimes steel fabricators are negligent in this area! Now measure out two more marks from the form board, with a distance of 3200mm between them, (if possible) parallel from the form board and mark a second parallel line, as before, using a lay rope or a second taut string. Now test whether the lines really have a 3000mm gap between them.

The theory to define a perfect rectangle:

From the middle of the form board, measure a line 4000mm in both directions on the bottom 200mm line. You now have set the basis for your first orthogonal (perpendicular) line. Place the tape measure on the last measured point and measure again that the second diagonal line (at 3000mm) is exactly 5000mm.

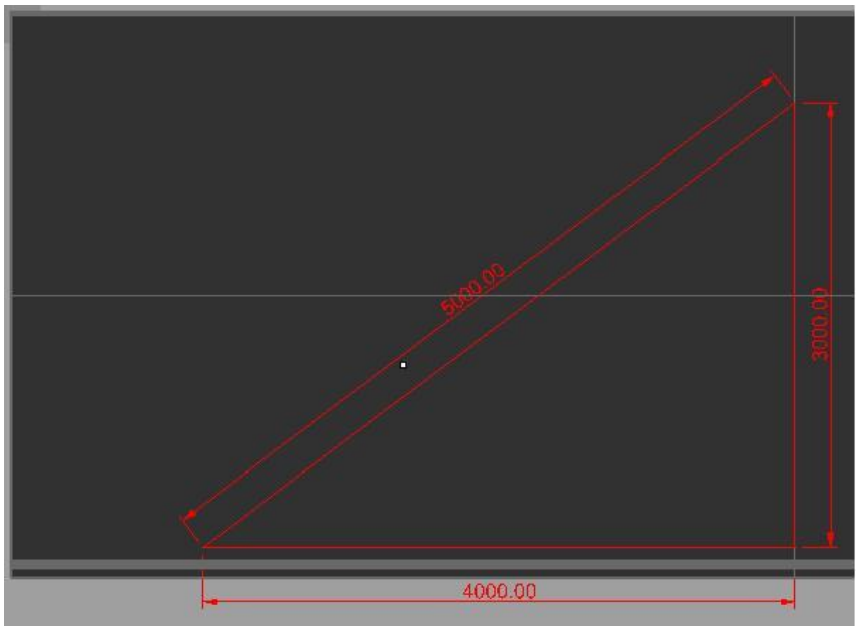


Figure 25: Defining a perfect rectangle

Please then measure again. It's really important to measure the points several times and ensure the points as close as possible to the correct positions!

Now go back to the Quick Calibration and go directly (without searching for the drift compensation points again) in the referencing window.

Now click on P1 and steer the laser cross with the mouse to the first point. Use the arrow keys to place the cross over the point with finer accuracy. Again, try to be as accurate as possible. If you are are doing this from the control PC, get help from someone who can show you where to move the point. You could also use a laptop!

Note: The screen mask describes how you are able to change the orientation of the mouse. Use C to rotate the axes and X or Y to reverse the direction of motion in the axes.

It is MANDATORY to ensure that the order is complied with and the points are in the correct order. E.g. bottom left is the point P1, and enter the coordinates X = 200 and Y = 200. Remember that the zero point was created at the bottom left corner and directly on the form board. There is only one rule; to use coordinates that can be displayed later in the graphical data.

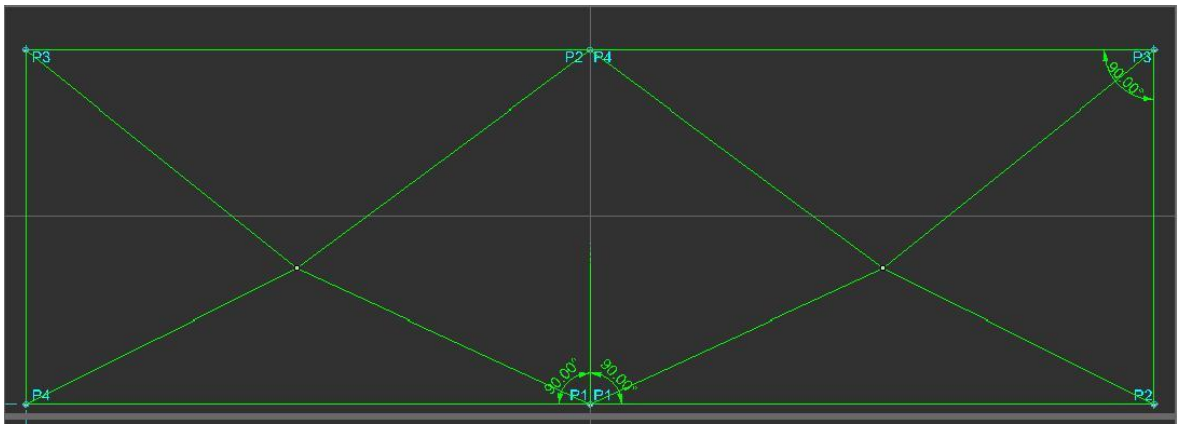


Figure 26: Set-up points in plan view

Note: In order to achieve the highest accuracy, the set-up points should be arranged as shown in the figure above: P1 is in the middle of the bottom form board and next point is then counter-clockwise from the point before.

If the four points have been measured and the correct coordinates given, then click on “Continue”. Searching is not necessary or possible.

Now you are in the “Set field borders” window. If everything is correct, then a rectangle should be displayed that is drawn exactly over the 4 set-up points! Otherwise, check the data one more time when setting up the set-up rectangle. Usually there is a incorrectly measured point or improperly entered coordinates.

This example is only for the left hand projector:

P4 lower left 200/200,

P1 centre below 5300/200,

P2 centre above 5300/3200,

P3 left top 200/3200.

Adjust the field boundaries manually by entering the values. (The field borders are the limits for the projection and should be a little larger than the subsequent projection), e.g. Xmin -200 Ymin -200 Xmax 5300 and Ymax 4000!

Then click on "Finish" and the projector is set up.

In the second projector, do the same, only the values are then:

P1 lower left 5300/200

P2 lower right 10600/200

P3 upper left 10600/3200

P4 upper left 5300/3200

Now go to the field borders and test whether all four corners of the field border lie on the

actual measured points. Now set the field boundaries for Xmin to 5301, so that the 5300mm line will not be displayed simultaneously by two projectors.

For the Y axis, set Xmax 10800 and Ymax 4000 again. After finishing you will see a change in the graphical view. This is now adjusted to the real coordinates.

For the first test, it is recommended that you open a test file from the directory \LPM 8.3\ TESTDAT, namely the grid500.PLT file. It now projects a big image, which displays grid lines every 500mm on the pallet. Each displayed line should now lie on every 500mm marking on a measuring tape! If the entire area is not covered, you can click on the image in the graphical window in LPM. The picture will then appear black. With "shift" and "right cursor" buttons, you can move the image 100mm to the right.

4. The Referencing:

Referencing is the function which adjusts the laser projection to the newly stowed pallet (in movement and rotation). Do not yet drive the pallet out, very important! You can now define a measurement for two points on each pallet. Always attempt to mount them on a permanently fixed form board and place the first point at the beginning of the first projection.

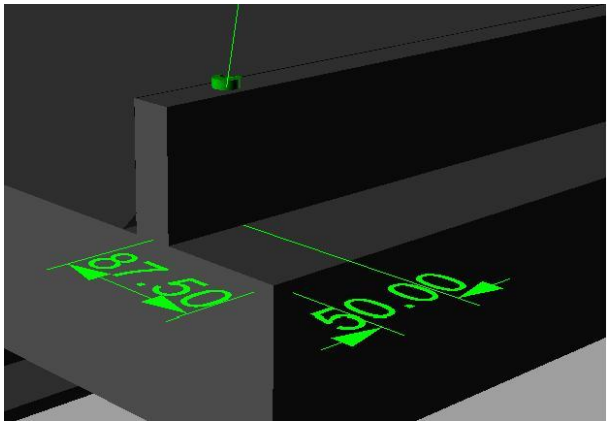


Figure 27: Reference point

All pallets must have these reference points in the same place!

It is not necessary to know the coordinates exactly, but it is necessary that the points are drilled identically on each pallet and equipped with reflectors. In the quick calibration for the drift compensation points, you will also find a button for setting the reference points. Click it and you choose the first point. Move your cursor to the first point and confirm this. If the reference points are at a different level than the previously established level, enter the height in millimetres. Then continue through the quick calibration to the field border section. Click the button "prior field borders" and select "Finish". Repeat for the second projector.

Tweaking

Background

Tweaking of the laser projector allows an experienced user to perfectly adapt the projection to the working environment.

Tweaking can be made when projecting on the target work piece or integrated into the installation process.

Function

The tweaking process is tied to the coordinates of the drift compensation points. The tweaking of the projector has a linear effect on the projected field. This results in a simple orientation. The modification of a tweaked point affects the opposite points linearly – 100% at the drift compensation point, and 0% directly opposite it.

Limitations

The maximum adjustment range of the tweak points is basically unlimited. However, this function should not be used if the error is greater than 15mm (with a mounting height of 3.5m).

It is best to confirm if other causes are responsible for the projection being inaccurate before beginning the tweaking process.

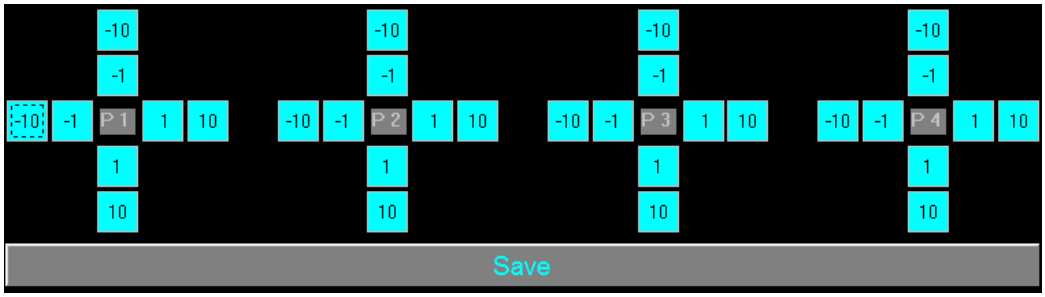
Availability

The version of tweaking described here is valid only for Linux client versions from 11.07 and onwards.

Tweaking has been in use since DOS client version 9.54. For projectors with DOS version 9.40 or 9.50, update features are available but older projectors cannot be retrofitted.

Operation

All calculations of the tweaking function are made directly inside the laser projector. LPM only provides the functions to send the commands to the laser projector. A specially adapted user interface for LPM is available (see The User Interface below). The following screen shot shows the 4 point tweaking interface.



Tweaking has no unit of measurement

The measurement of tweaking can be roughly determined as follows:

65500 increments divided by the maximum field border in mm. Using the factory calibrated maximum field border of 6010 by 6010mm, this corresponds to approximately 10 increments per mm.

The tweaking points P1...P4 have the same order as the drift compensation points DCP1...DCP4.

Caution: The user interface arrangement of the tweaking points is not the same as the actual search order of the projector!

Tweaking is based on a virtual displacement of the drift compensation points. Please note that the drift compensation points are sorted by the laser projector itself!

By calling the drift compensation search (the CAL button), the order of the drift compensation points are shown. As seen from the perspective of the LP-HFD2 laser projector, the order is typically:

- Bottom right DCP 1
- Top Left DCP 2
- Top Right DCP 3
- Bottom Left DCP 4

During the set-up of the quick calibration rectangle, P1 is the most accurate point in the projection. Please do not confuse P1 with drift compensation point 1 (DCP1) or Tweak point 1 (P1 on the tweaking user interface).

To use the tweaking procedure effectively, the user must program a rectangle which is as close to the drift compensation points as possible and has known dimensions. By pressing the buttons marked 1, -1, 10 or -10, an offset command is passed to the projector and then a drift compensation search is performed. The projection of the rectangle should now have moved at the corner closest to the DCP which was modified. By modifying all the corners of the projected rectangle to match the corners of the measured rectangle of known coordinates, the tweaked projection becomes accurate to the working area.

It is recommended to start the projection of the rectangle before using the tweaking function. This way, it is easier to see any change in the image made during tweaking. Please note the input is not measured in mm.

If the user is satisfied with the tweaked image, then the SAVE button should be pressed to maintain the change after the projector has been rebooted or switched off.

If the tweaked image appears to be less accurate than before, simply close LPM and reboot the projector. Do not press the SAVE button.

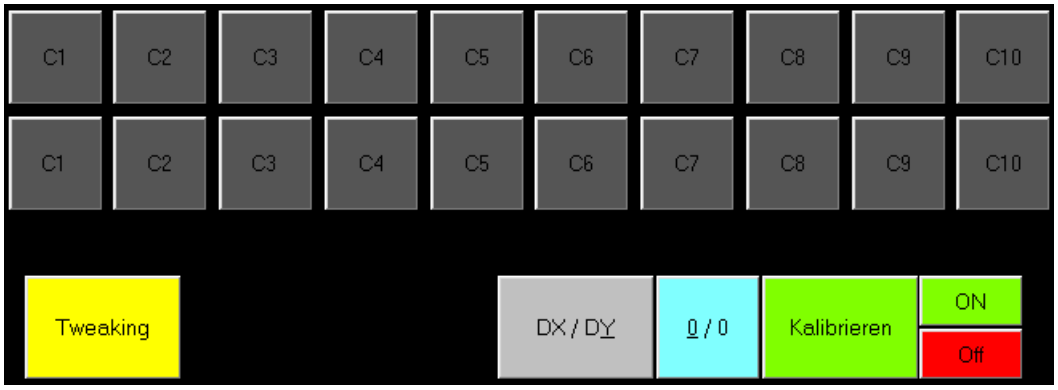
The User Interface

The file LPM_TWEAK.CTL can be installed using the LPM 7.2 (or later) installation

program or by manually inserting the file into the UI folder and modifying the following line in the LPM.INI file:

[General]

CtrlFile=UI\LPM_Tweak.CTL



When using more than one projector, it is necessary to change the entry for addressing the projectors in the LPM.INI. Each projector must be tweaked separately.

Projector A Slaves=1,0,0,0,0,0,0,0,0,0,0,0,0,0,0

Projector B Slaves=0,1,0,0,0,0,0,0,0,0,0,0,0,0,0

…….

Projector F Slaves=0,0,0,0,0,1,0,0,0,0,0,0,0,0,0

Activate only the projector on which you wish to make changes. All other projectors should be disconnected from the network and switched off at the mains for safety reasons.

After starting LPM, only the selected projector and a standard user interface will appear. By selecting the “Tweaking” button, a different user interface will be displayed with the controls for the tweaking process. If the graphical interface is too small, the user can click on and drag the lower right hand corner to suit your requirements. Below, one can see the functions which have been described above.

Picture of tweaking user interface with P1 to P4 crosses

Tweaking Commands

Tweaking can be controlled with direct commands. This should only be used by an experienced operator for faster and more effective use. The commands for tweaking are as follows:

/TRANSP3 V n dl ds

where

n is the number of the drift compensation point.

dl is the movement of the drift compensation point in the long axis in increments.

ds is the movement of the drift compensation point in the short axis in increments.

The change will be immediately visible after giving the command to the projector.

To address the correct projector, the tweak command should be prefixed by the command *ADR #, where # is the letter of the projector to be tweaked.

For example:

*ADR C /TRANSP3 V 1 1 0

will tweak DCP1 one increment in the long axis of Projector C.

The changes will remain temporary until the values have been saved by the command /TRANSP3 VW. After sending the /TRANSP3 VW command, the drift compensation search will be started and the data will be saved in the projector.

Working with multiple projection areas

As a rule, LPM operates with only one projection area. Some applications require two or more projection areas.

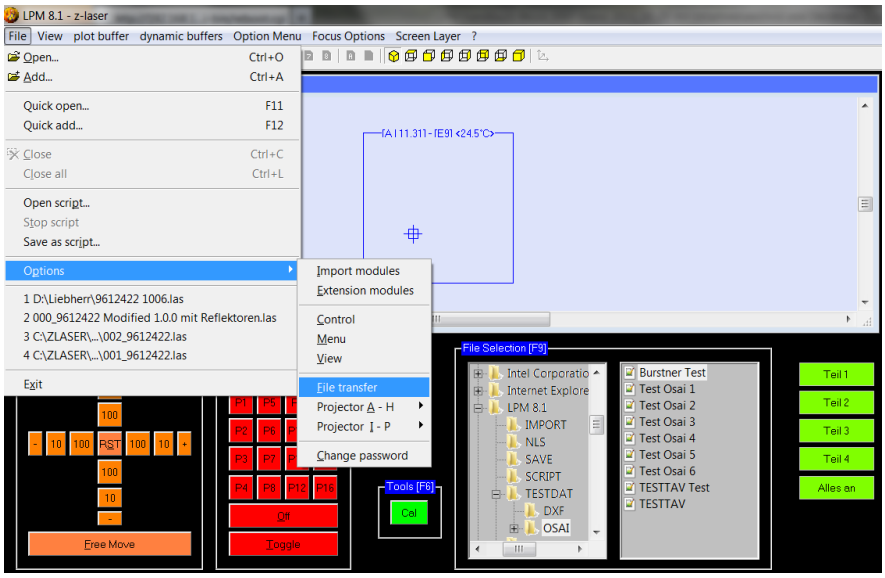
Example:

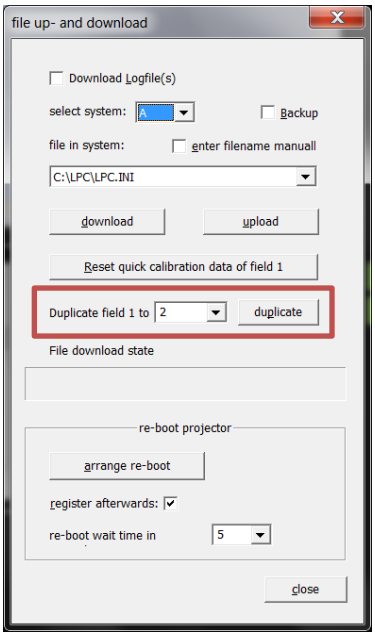
Working with two workbenches

While a work piece is being machined on workbench 1, a second work piece is placed on workbench 2, which is parallel to workbench 1. After the machine has finished with the first work piece on workbench 1, it moves on to workbench 2 to start the machining on the second work piece. This requires the user to create two different projection areas for the two workbenches.

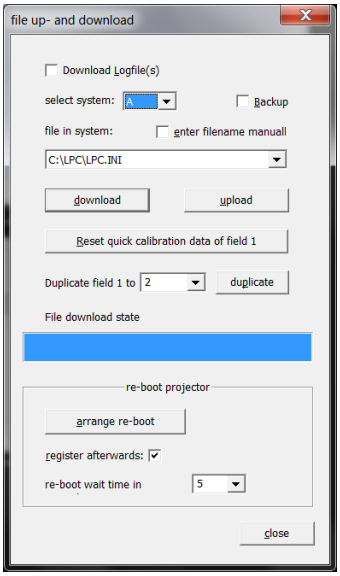
To create another projection area, proceed as follows:

1. Start LPM. Select File → Options → File Transfer. The file transfer window will open after entering the password cps.





2. To create a second field, select the button “Duplicate”. When the blue bar in the “File Download Status” progress bar is full, the duplicating process will be closed.

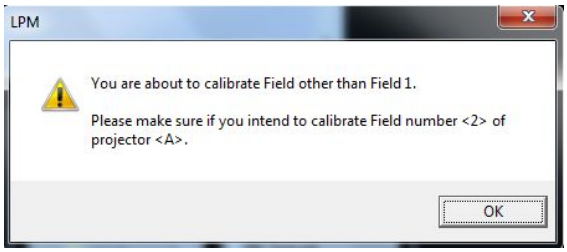


As Field 1 has now been duplicated into Field 2, the calibration data in both fields are now identical. Field 2 must therefore be set-up using different coordinates with the Z-Easy quick calibration.

3. In order to set-up Field 2 correctly, the current projector field must be switched to Field 2 using the command “/F 2”. Do not forget the space between the F and 2.



4. Now calibrate field 2 by entering the quick calibration by “File” → “Options” → “Projector A – H” → “Projector A”. After entering the password, you will see a dialog window open, warning you that you are calibrating field 2.

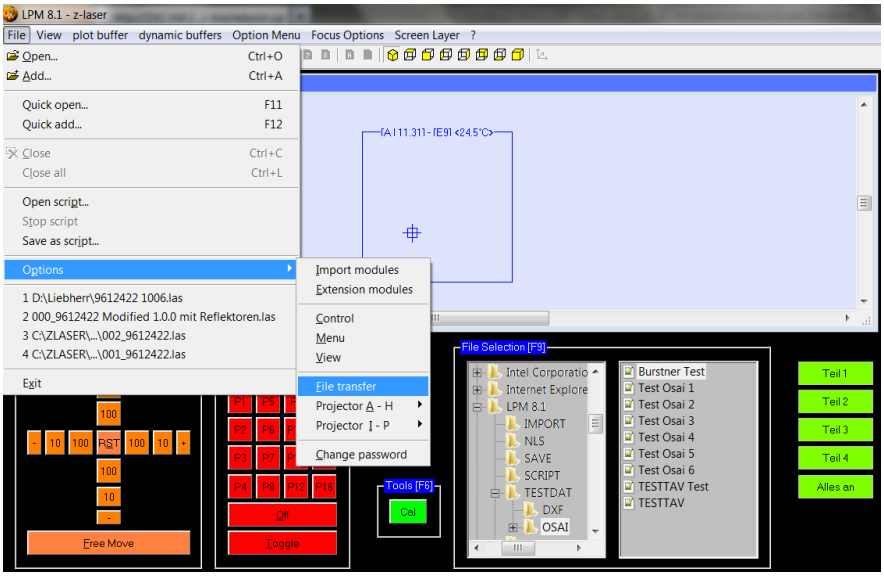


Confirm with "OK" and go through the process of the quick calibration of field 2. Field 2 is now calibrated. You can always switch from Field 1 to Field 2 by selecting "Options" > "Direct Command". The commands for this purpose are: /F 1, /F2, /F…

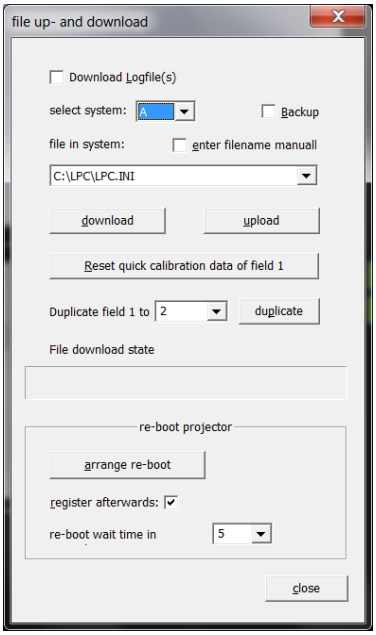
Data Backup

To avoid loss of calibration data that you set in the projector during the quick calibration process, you should make a data backup of all the files and fill in the installation protocol included in the appendix. Use the following steps:

1. Open LPM and go to **File** → **Options** → **File Transfer**. This will require entering the password to enter the file transfer menu. The default password is "cps".



2. You should now see the following window appear:

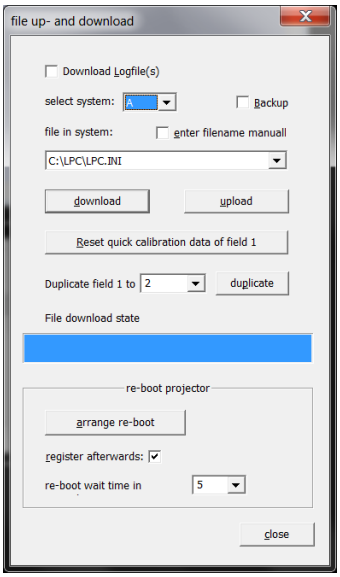


3. Select the system you would like backup, tick the "backup" box and select the “Download” button.

4. An Explorer window will now open in which you can select a path to save the backup to. It is recommended to create a new folder to which the files can be

saved. Press “OK” to start the download to the selected path.

5. You can view the progress of the backup in the status bar “File Download Status”.



6. An information window will appear, providing information about the status of the download.



The file ELC_F01.ACD will not be found if the quick calibration has not been made.

You should only make a backup of the projector files once a quick calibration has been made. A backup should also be repeated every time a change is made to the quick calibration data.

The files ELC_F01.REF and LPC_F01.REF will not be found, unless reference points have been set during the quick calibration.

Click “OK” to close the information window. Your projector's files have now been saved to the control computer and you may close the file transfer window by selecting the “Finish” button.

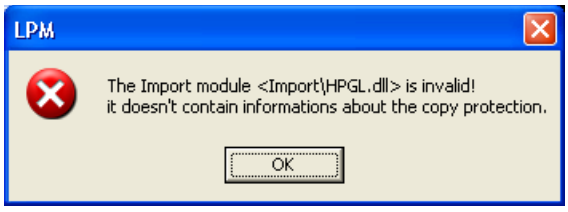
Store the files in a secure directory on your PC and burn it to a CD or DVD if necessary. You can send the files to Z-Laser and we can achieve them for you, but without guarantee.

If you damage your laser projector, send us your backup files immediately so that we may upload them to your exchange projector. However, in each case, a quick calibration of your device is essential.

Software Registration

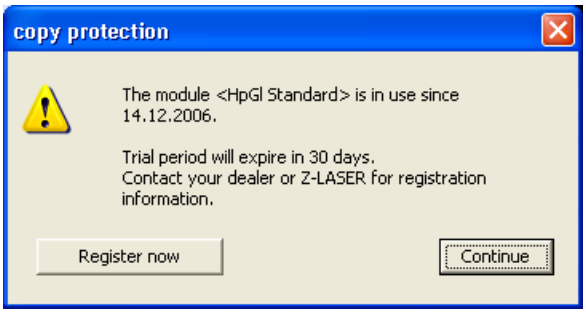
LPM consists of multiple additional modules and it is copy protected. You may test each of these modules with your laser projector for 30 days without charge.

LPM does not check if any of the modules are registered when it is in simulation mode. The 30 day limit does not begin until LPM connects with the projector for the first time. A warning message will be seen every time LPM starts up. LPM modules from previous versions will no longer work and you will see the following warning if you try to load an out of date module:

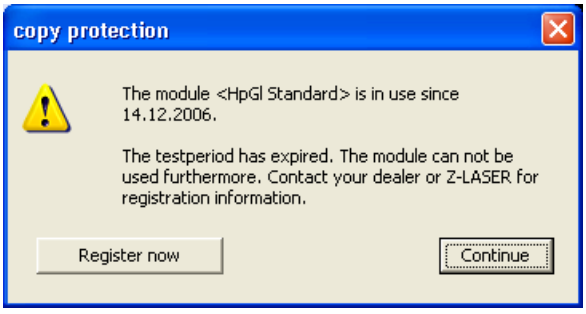


The basic modules, consisting of HPGL and DXF Standard, are always free to use but they must still registered.

During the test phase, when using a non-registered module, a message will appear for each module warning you that the module is not registered. For example, if you use the HPGL Standard import filter for the first time on the 25.07.2011, you will see the following warning message appear when you load a HPGL file:

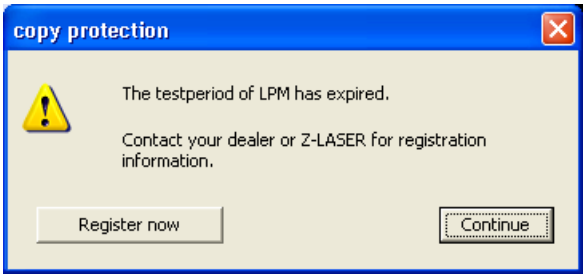


This message will appear after every restart of LPM or after every hundredth operation and it must be confirmed by selecting "Continue". The registration process may take place at any time after the installation is complete. During the test phase, all unregistered modules will operate as normal. After the 30 day test phase is finished, the unregistered modules will no longer work. In the example of the previous HPGL Standard import filter, then the following message will appear:



Please note that any manual date changing of the PC, the registry or the text file, LPM.INI, as well as simply copying the LPM directory will result in LPM stopping

working immediately. All modules that have been installed will not work and can only be unlocked again by registration. In this case, the following warning will appear:



Select “Register Now” to start the registration process. Select the “Next” button to return to LPM. You will only be able operate LPM in simulation mode after the 30 day period is finished.

Process of the Registration

Registration is exclusively controlled by Z-LASER. Once installed, the file, REGISTER.TXT, should be sent to Z-LASER. This can be done via e-mail or by fax. Even registration by telephone is possible. The registration number is calculated using this file and can be entered in directly, after which the selected modules are unlocked.

The following data is required for the registration of LPM:

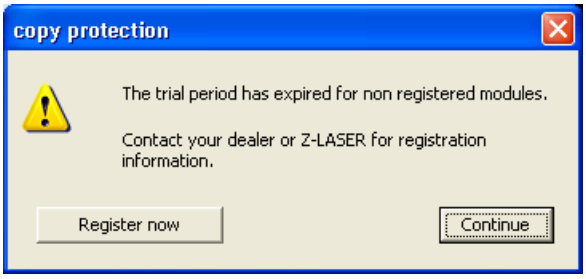
- The customer or dealer details
- The projector data, including the serial number
- The serial numbers of the modules to be registered.

This data is entered into the configuration file of LPM (LPM.INI). In addition, a new file, named REGISTER.TXT, is created in the Start menu. REGISTER.TXT can be found in the LPM folder and it can be sent directly to Z-Laser in this form.

Starting the Registration Process

There are two possibilities to start the registration process.

When LPM starts, a warning message appears informing you of the test phase. If one or more unregistered modules are found, you will see the following window:



If you know your registration number (Z-Number), you can register the module directly by selecting the "Register Now" button. If you have no registration number, or you would like to register your modules at a later date, select the "Next" button.

In the menu bar of LPM, under "?", you will find the menu item “Register LPM”. Select this to start the registration process.

If you would like information about your modules, select the menu item “Report about used modules”, also under “?”.

Please note that the “Register LPM” and “Report about used modules” items are not available in simulation mode.

The Registration Procedure

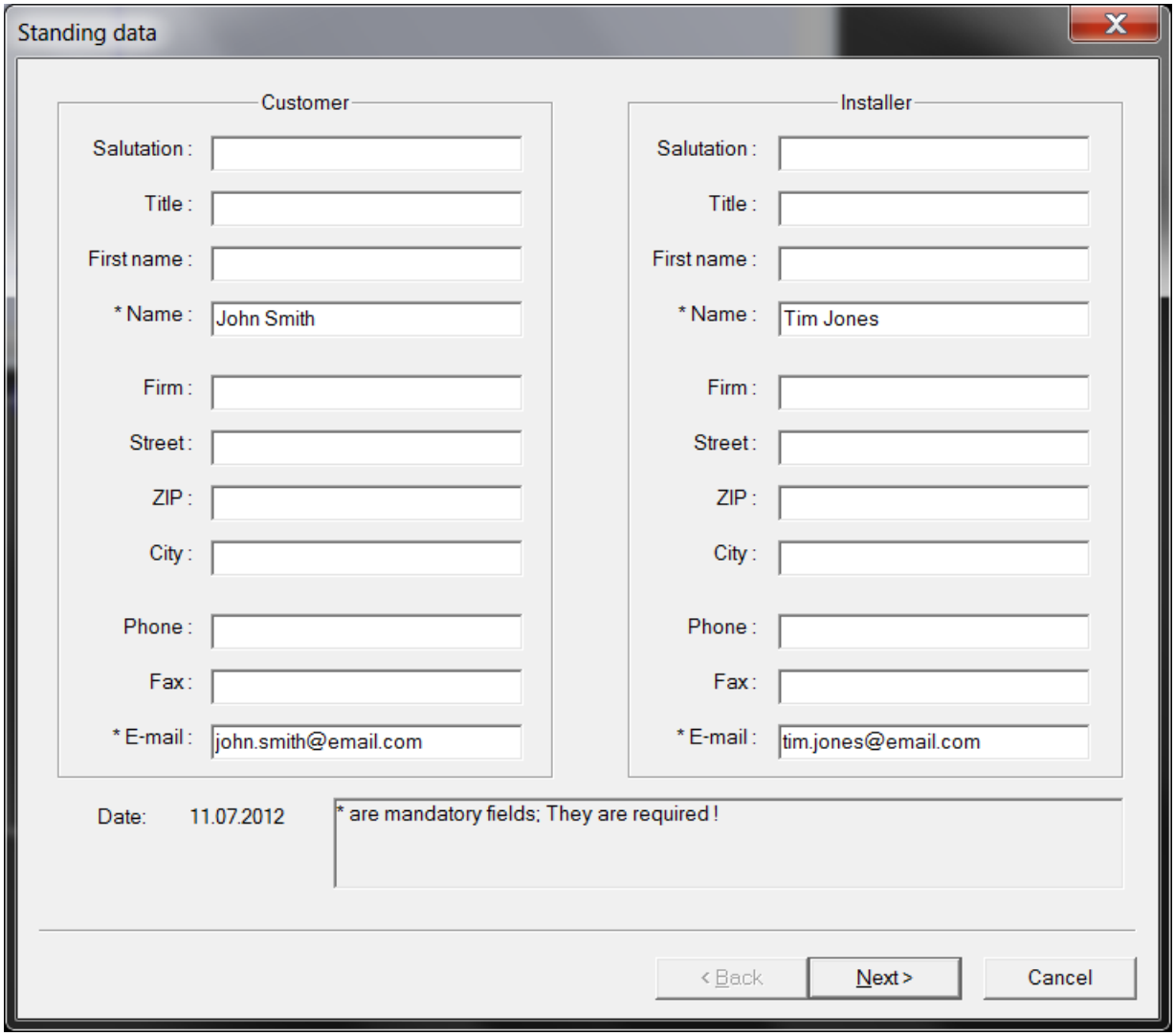
This procedure must be run twice. The first time is used to pass the required data to the REGISTER.TXT file and the second time is to enter the registration code which is supplied to you by Z-Laser.

The procedure consists of several stages and interactively guides you through the entire registration process. Use the "Next" and "Previous" buttons to switch between the next or previous pages.

Customer and Installer Details

This page contains information about the customer, the location of the laser system and the installer. Fields marked with an asterisk (*) which are mandatory and must be completed.

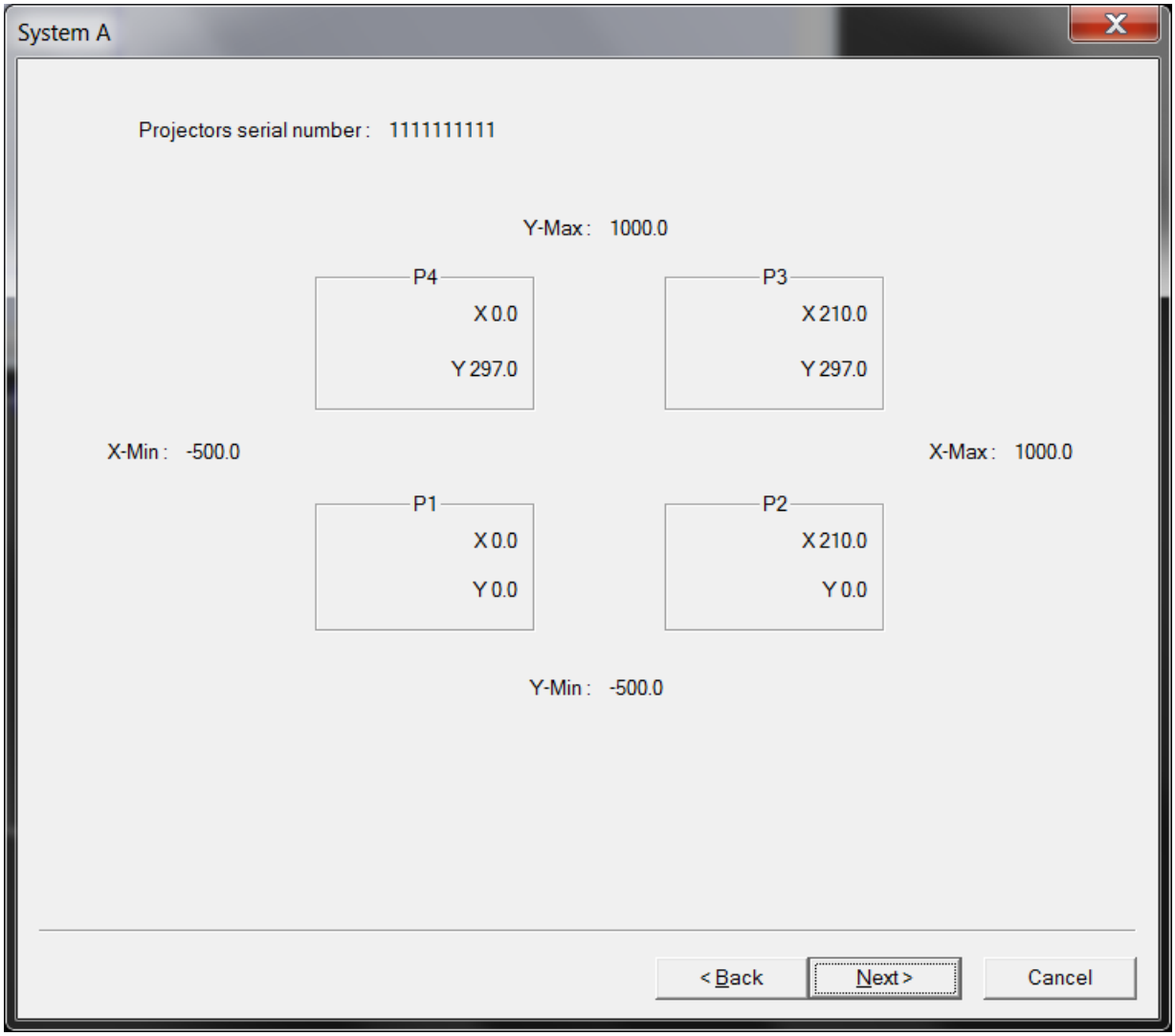
Before these fields have been filled, it is not possible to select the “Next” button. If these fields were filled in previously, then they will appear at the next registration walk-through.



Projector Geometry

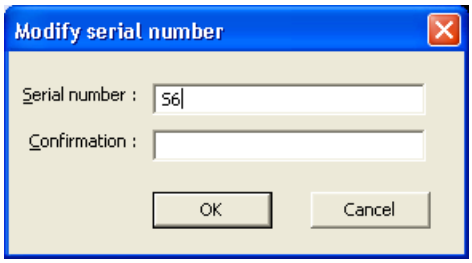
These pages contain information that gives an experienced installer the opportunity to check the coordinates of the projectors again. You cannot change the entries here.

Select “Next” to continue.

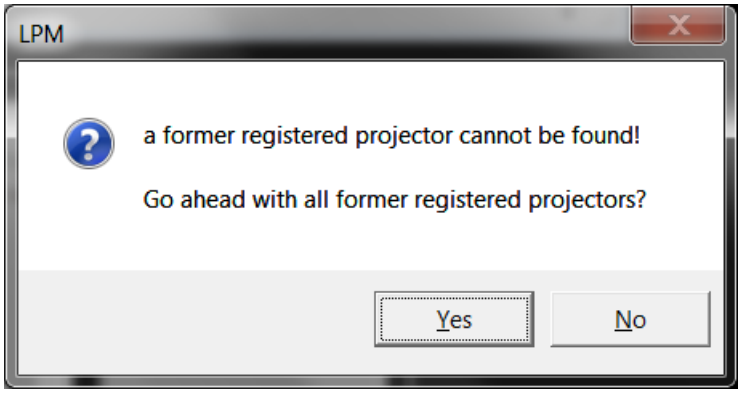


Serial Number

If, for any reason, the serial number is not found on the projector, then a dialogue box will appear and ask you for the 10 digit serial number of the laser projector. You can find this number on the silver plate mounted on the side of the housing of the projector. Cancelling this window will halt the registration procedure.



After entering the serial number, select the “OK” button. If you have previously registered modules, but you have changed your projector with a new one, you will get the following error message:

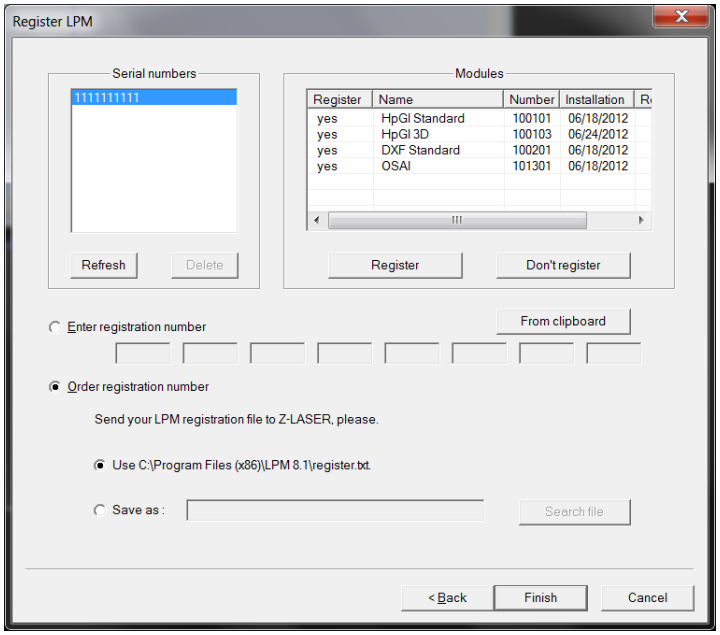


If you select “Yes”, then the original registration number for the serial number of the new projector will be accepted. This means that any previously registered module will now work with the new projector.

If you would like to register any additional modules, then select “No” to go to the next step to apply for a new registration code.

Module Registration

On the last page of the registration process, you can see a list of the serial numbers and the modules which will be activated upon receipt of the Z-Number. The following screen shot gives an overview of the layout of this window:



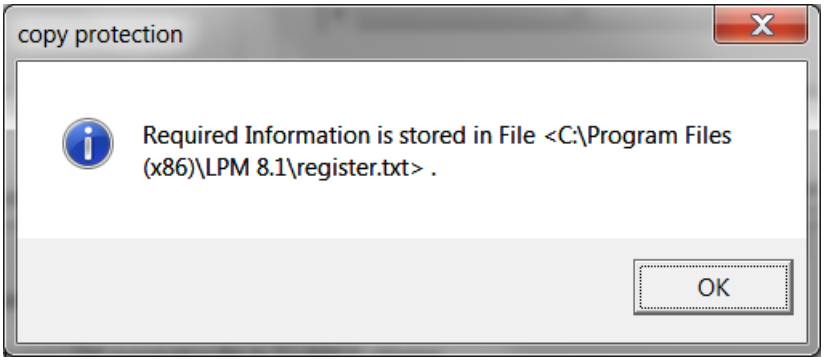
All serial numbers of the currently connected projectors are displayed under the heading “Serial Numbers”. If the serial number(s) remain the same, then the current Z-number will remain and a new one does not need to be ordered. However, if you wish to register additional modules, a new Z-number will be required by the registration process.

Click on the “Refresh” button for the serial number list to be updated (e.g. exchanging a projector) and a new Z-Laser must be ordered. Several modules may be listed although they were not explicitly selected in the installation. To generate a request for a new Z-Number, it is necessary to register the modules that are needed and that were integrated into LPM during the installation. At least one module must be selected. Click on the row in which the module is listed to select the appropriate module. By holding the Ctrl key, several modules can be selected. By selecting the “Register” button, the entry will change from “No” to “Yes”.

If you would like to register modifiers, such as LPM Selca, LPM Teach, LPM Steps or ULB Handler, then you must have used the respective module at least once otherwise you will be unable to register it.

If all modules to be used are selected and the entry under “Register” is set to “Yes”, you can order a new Z-number or registration code. You have several options in which to do this.

Tick the box next to the text C:\Program Files\LPM 8\REGISTER.TXT to save the registration data to the file REGISTER.TXT. Alternately, the file can be saved to a different folder or even given a new file name by selecting “Save As”. A dialogue box informs you of the saving process success or failure.



Select “OK” to continue.

Remember the path of this file well, as you will be required to send this file by email (or printed as a letter or a fax) to Z-Laser.

Select “Finish” completing the registration process. Again, a dialogue window will appear to inform you where and under which name the file has been saved to.

Usage of new or additional modules

If you want to use additional modules after the registration, you may test them up to 30 days after the first use (installation) of the module.

This means that the test period may already be finished by the first use of the module.

In this case, the module must be reinstalled. After the new installation, it may be tested for a further 30 days. Alternately, you can use it in simulation mode and use it as much as you wish.

Registration number on the delivery note

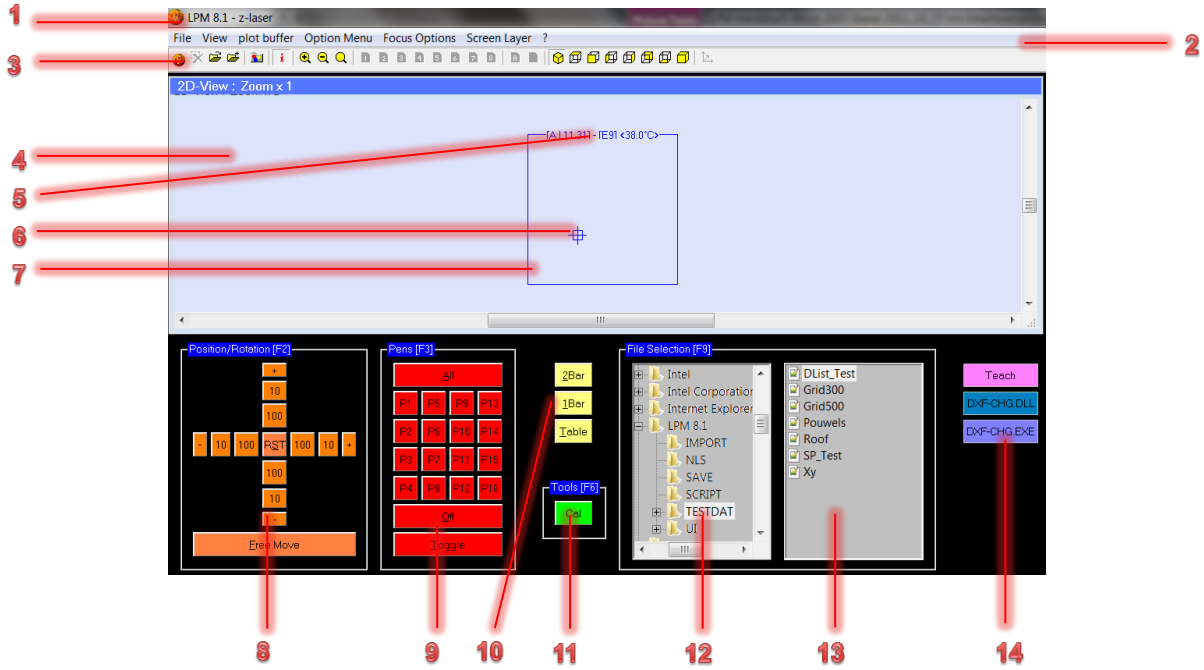
Depending on your application, there may be a registration number on your delivery note. This registration number can be inserted directly, as shown in the previous section **“Module Registration”**. In this case, it is not necessary to get the code from Z-Laser.

LPM user interface

LPM has a customisable interface. A standard interface has been created for the use. This standard is variable, to suit your requirements.

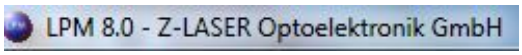
As an overview, the basic elements of the LPM interface are explained by means of the following diagram.

Once you have familiarised yourself with the various interface elements, you can carry out a few test projections.



1. Title bar
2. Menu bar
3. Toolbar
4. Graphical interface
5. Technical projector information
6. Field borders
7. Background image (default file 0M.PLT)
8. Buttons for moving and positioning
9. Buttons for pen selection
10. Buttons for height selection
11. Drift compensation search button
12. Folder selection (Windows Explorer)
13. File selection
14. Extensions

The title bar

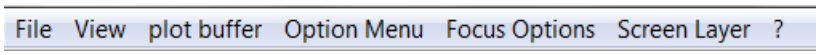


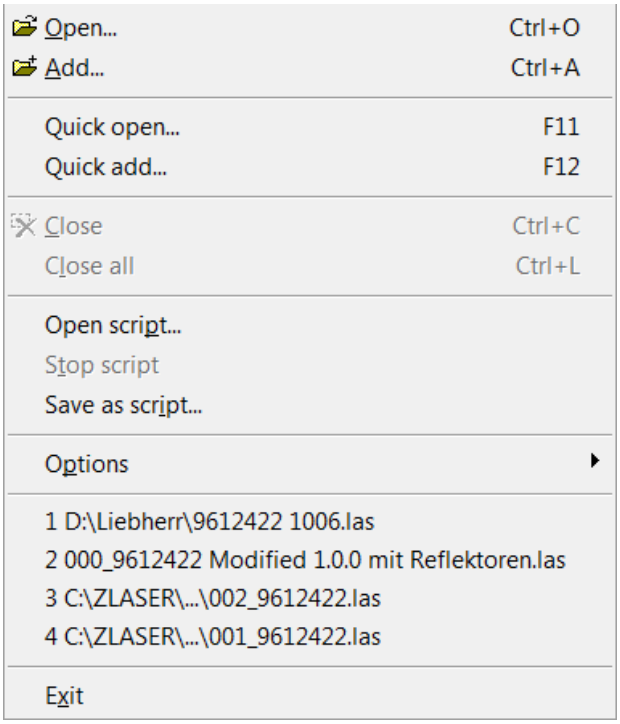
The **title bar** shows you (from left to right):

	The software logo
	Version number
	Your registered company name

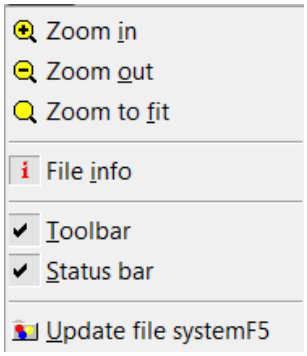
The menu bar

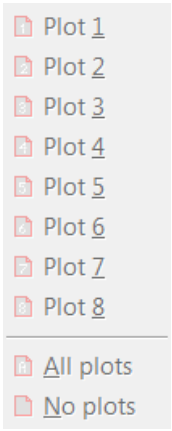
The menu bar comprises of various menus that you can open by clicking on them.

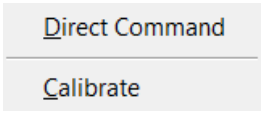


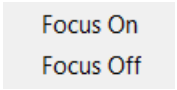
File	
	**"Open..."** opens a image file via the "open file" dialogue. All other images will be closed down.
	"Add..." adds a image file to the next available plot buffer via the "open file" dialogue.
	"Quick open..." calls up a pop-up window, where you can directly open a LPM.INI file format via a specified path, by entering the file and file name under "**File name"** and click "**OK**". Example: "**TESTDAT\Roof**" opens the "**Roof.PLT**" file in the "**TESTDAT**" folder. All other images will be closed down.
	"**Quick add...**" calls up a pop-up window, where you can directly open a LPM.INI file format via a specified path and add it to the next available plot buffer.
	"**Close**" closes all selected images in the activated plot buffers 1 – 8.
	"**Close all**" closes the images in all plot buffers.
	"**Open script**..." opens a script file and processes it.
	"**Close script...**" closes the current script and ends the processing of it.
	"**Save script...**" saves the current presentation, which can be reloaded again later.
	"**Settings**" enables the settings for "Import filters", "Extensions", "Controls", "Menu", "Views", "File transfer", Projector A - H", "Projector I - P" and "Change password"
	"**Exit**" ends LPM

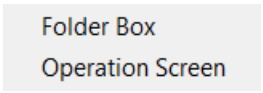
Views	
	"**Zoom in**" enlarges the image within the graphics area to a factor 2...32

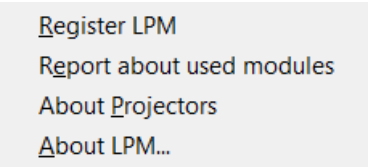
	"**Zoom out**" reduces the image within the graphics area to a factor 2...16
	"**Show all**" zooms in on the image to factor 1 and centralises it
	"**File Info**" activates or deactivates the file information.
	"**Toolbar**" shows the toolbar or hides it
	"**Status Bar**" shows the status bar or hides it
	"**Update file selection**" manages and updates the folder and file selections.

Plot Buffer	
	"**Buffer 1**" - "**Buffer 8**" activates/deactivates the respective image selected from the plot buffer
	"**All images**" selects all images in the plot buffers
	"**No Image**" deselects all images in the plot buffers

Option Menu	
	"**Direct command**" opens up the "direct command" window
	"**Calibrate**" enables the projector to perform a drift compensation search

Focus Options	
	"**Focus On**" switches on the focus
	"**Focus Off**" switches the focus off

Screen Display	
	"**Folder Box**" screen display for the directory structure
	"**Operation Screen**" screen display to execute technical operations and diverse LPM functions

?	
	"**Register LPM**" calls up the registration manager
	"**Info about used modules**" shows you which modules have been installed
	"**Info about projectors**" shows you which projectors are connected
	"**Info about LPM...**" shows you the version information for LPM

The toolbar

The toolbar enables you to gain instant access to actions (i.e. tools), by using various icons, that can be called up via the menu bar. You can hide the toolbar at any time.

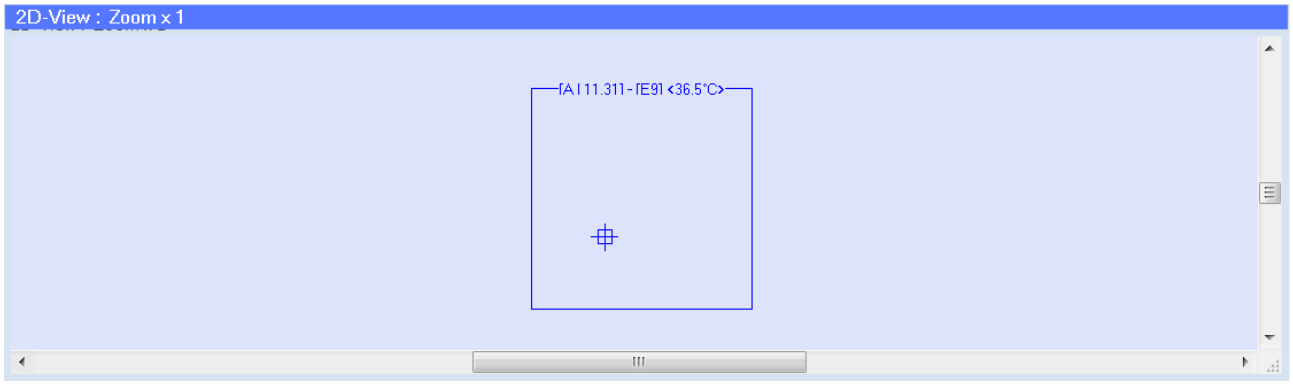


The toolbar shows the following (from left to right):	
	Software logo
	Closes selected image
	Opens image
	Attaches image
	Updates the file selection
	Displays the file information in the graphic window
	Enlarges the view by factor 2
	Reduces the view by factor 2
	Resets the zoom factor to 1
	Selects plot buffers 1 – 8
	Selects all plot buffers
	Deselects all plot buffers
	Shows the perspective 3-D view (only in 3-D mode)
	Shows the view from the top
	Shows the view from the front
	Shows the view from the left
	Shows the view from the right
	Shows the view from the back
	Shows the view from the bottom
	Shows the pseudo-perspective view
	Switches the 3-D orbit on/off

The graphical interface

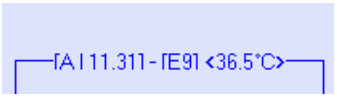
The laser working area, as well as your opened files, is displayed in the **graphical interface**.

2D-View : Zoom x 1	Shows which zoom factor is currently in use. This information is relevant only for the graphical display on the monitor. The laser image scaling will remain unaffected.



The technical information

The **technical information** is found at the top field border (clipping zone) and shows the following (from left to right):

	The respective projector A-P address
	The firmware's version number of the projector
	The current drift compensation status
	The current temperature of the projector

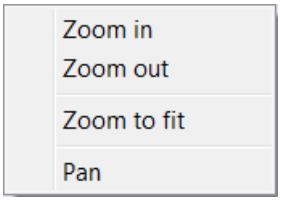
The drift compensation is defined by the following statuses:

[E9]	Drift compensation has not started
[~1]...[~4]	Drift compensation point 1....4 is currently being searched
[OK]	Drift compensation successfully completed
[E-1]...[E-4]	Error during drift compensation at point 1....4

The unit of temperature used for the operation temperature can be changed from °C to °F (Fahrenheit) in the configuration file LPM.INI.

Pop-up menu in the graphical interface

The following pop-up menus in the graphical interface can be activated using the right-click on the mouse pad:

	"**Zoom in**" enlarges the image by factor 2 and centres the selected point to the middle of the graphical interface
	"**Zoom out**" reduces the image size by factor 2 and centres the selected point to the middle of the graphical interface
	"**Show all**" zooms in on the image by factor 1 and centres the selected point to the middles of the graphical interface
	"**Centre**" centres the selected point to the middle of the graphical interface

Pop-up menu in the working area

A customised pop-up menu in the working area can be activated by using the right-click on the mouse pad:

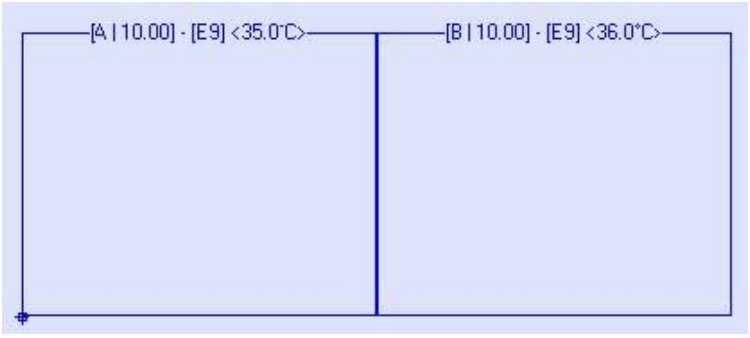
	"**Direct command**" opens the "direct command input" window
	"**Calibrate**" enables the projector to conduct a drift compensation search

The field borders

The field borders (clipping zone) indicate your projector's maximum working area.



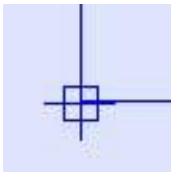
This defines each registered projector, specific to its own field borders.



Background image

With LPM, you have the option to create a background image, which should be displayed in the graphical interface on the user interface. This is useful, for example, when you are highlighting several operating areas with your laser projector at the same time, and you need to align the work piece contours onto your working area directly on the screen.

As a default, a small square with a cross at the centre of the coordinates, will be displayed.



Read up in the chapter **"Creating a background image"**, on how to create your own background image.

The status Bar

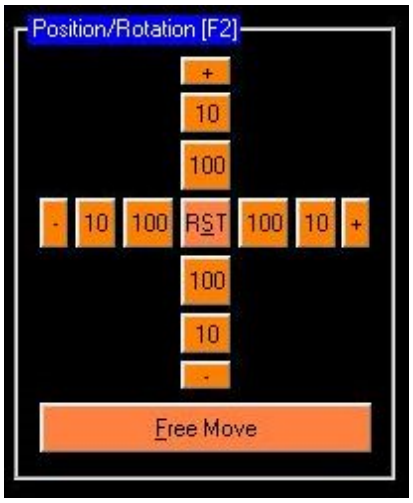


The status bar shows you the following information (from left to right):

1. A summary for the selected menu points and icons
2. Communication status
3. Plot buffer status
4. Last command sent
5. Date and time

Buttons for shifting and position changes

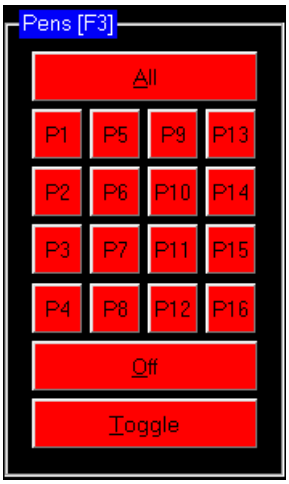
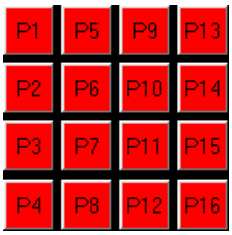
The presentation can be shifted by any required value, using the buttons for shifting/position changes.

		shifts the drawing up/right by 1mm
		shifts the drawing down/left by 1mm
		shifts the drawing up/down/right/left by 10mm
		shifts the drawing up/down/left/right by 100mm
		shifts the drawing back to its original position
		calls up a pop-up window that allows you to move the drawing in an x-/y- axis. The unit of measurement here is in metres!

You can change the position of the display by using the mouse or using the keyboard. Please note: If the display moves outside of the field borders, you might still be able to see them on your computer screen, however, the laser will only project the part of the display that lies within the field borders.

Buttons for pen selection

Using the buttons for the pen selection, you can select or deselect specific elements of single polygons in order to get a better overview when projecting several objects at the same time.

		Switches on all pens
		Adds the respective pen colour/layers. You can define up to 16 colours/layers.
		Switches off all pens
		Adds all pen colours/layers in succession

Depending on the chosen import module, the pen numbers define:

– Pen colours (HPGL)

– Colours or layers (DXF)

– Tool numbers (CNC)

Choose the corresponding attachment for the respective single import modules.

Buttons for height offset

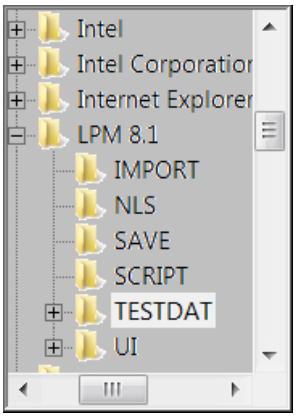
If you want to project on a different height, rather than on your calibrated workspace, you can communicate your requirement to the projector by using the height offset buttons. The buttons can be adapted to suit your requirements and can be configured to standard projection heights.

	"2Bar" raises the projection to 38,1 mm. "1Bar" raises the projection to 19 mm. "Table" resets the projection back to the original height of your calibrated workspace.

Calibration button

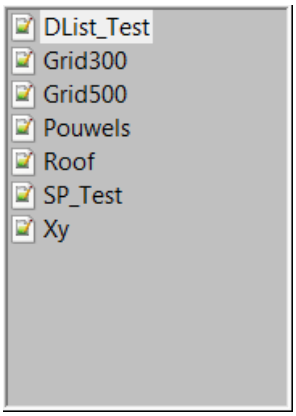
	The calibration button activates the drift compensation search.

Folder selection (The Explorer Window)



Like with Windows Explorer, you can navigate through the folder selection (Explorer Window) to access all folders and files on your computer.

File selection



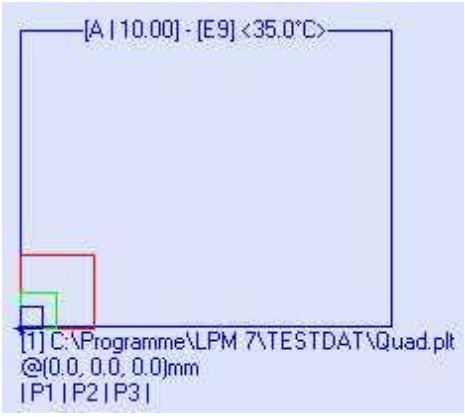
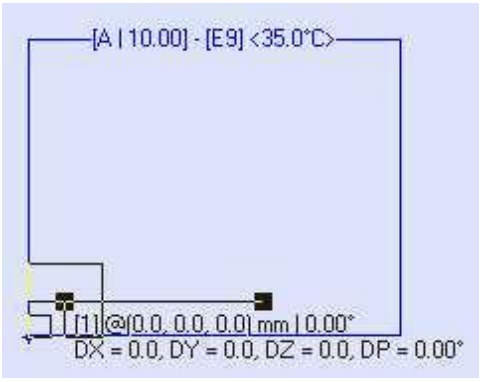
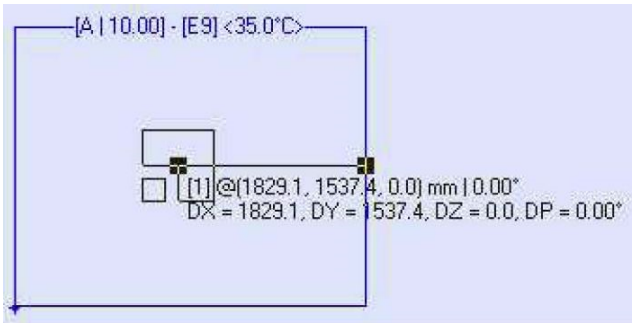
The file selection box displays the respective files for selected folders. The image files can be opened by double clicking on the mouse, or by selecting the cursor keys ↑ and ↓, and then pressing the Enter or Space bar button.

Test Projection

This chapter provides the user with an insight into the range of applications of LPM. Here, you can make your first steps in laser projection. Turn on your projector and then start LPM.

Open the file QUAD.PLT which you will find in the directory in the path C:\PROGRAM FILES\LPM\TESTDAT\.

Pay attention to the actions you perform while using your PC, as to the simultaneous changes of the laser projection.

	You should now see 3 nested squares on your screen as well as on your projection surface.
	Click on the drawing. The contour of the object appears black. In addition, 2 black boxes appear, which can be selected with the mouse. Now move the object by clicking with the left mouse button on the box in the centre of the object and drag the object with the left mouse button to a different location.
	Now click on the box at the end of the line and rotate the object.

	The object is now moved and rotated.

If you click on the  button in the middle of the position buttons, the object will return to its original starting position.

As already mentioned, you can change the position of the object within a defined framework with the help of the position buttons.

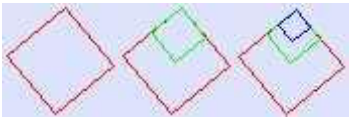
To change the position of the projection, you can use the following keyboard combinations:

[Cursor]	1mm offset
[Ctrl + Cursor]	10mm offset
[Shift + Cursor]	100mm offset
[Shift + Ctrl + Cursor]	1mm offset

To change the rotation of the projection, you can use the following keyboard combinations:

[Page Up/Down]	1° rotation
[Ctrl + Page Up/Down	10° rotation
[Shift + Page Up/Down]	90° rotation
[Shift + Ctrl + Page Up/Down]	0,1° rotation

You are able to turn the individual squares on and off with the use of the pen selection buttons:

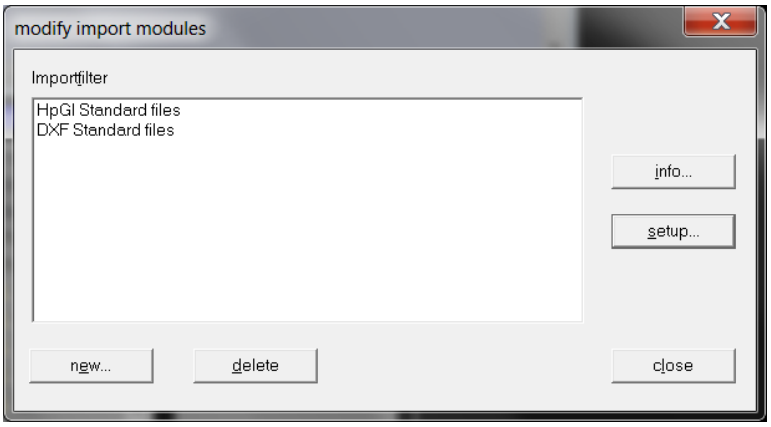
By selecting the OFF pen selection button, you will be able to make the contours of the squares invisible but the file information will remain visible.	
By selecting the buttons P1, P2 and P3, all contours will become individually become visible again.	

Set up of import filters

The following sections will help you set up the standard filters for the import of HPGL standard files as well as for DXF standard files.

HPGL Import filter

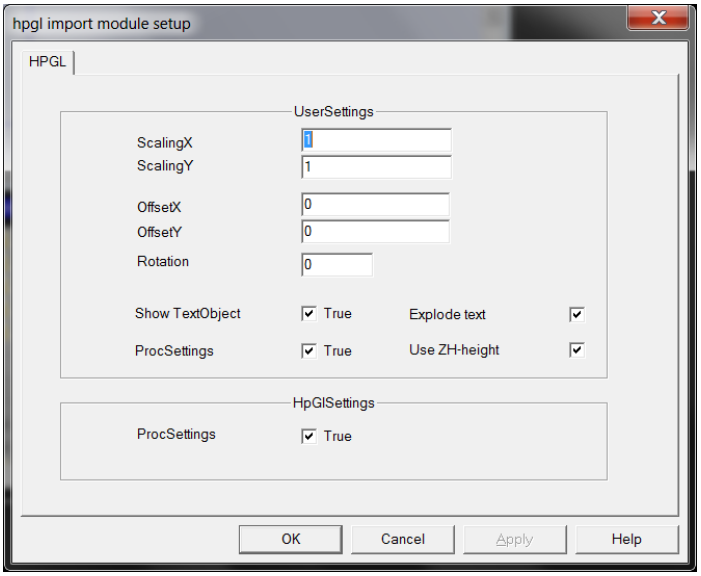
To set up the HPGL import filter, go to the menu bar and open “File” → “Options” → “Import filters”. The window “Modify import filter” should open. All import filters that have been selected and installed during the installation of LPM 8 are listed here.



Select “HPGL standard files” and click on “Set-up...”.

The window “HPDL standard files” should appear.

From here, you can set up different parameters, which are explained as follows:



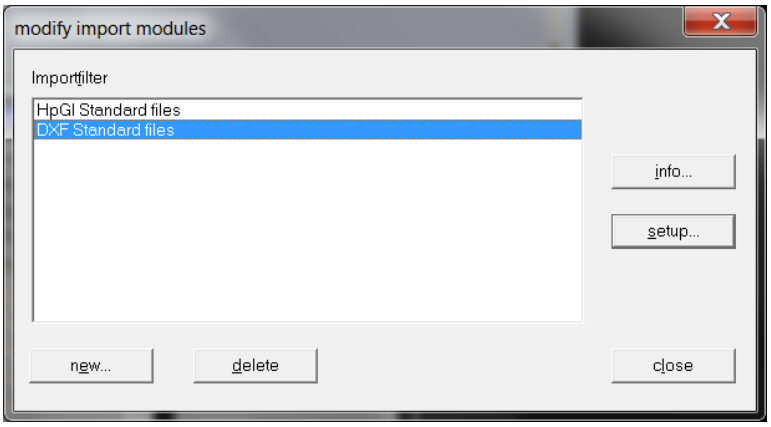
We have provided an overview for all HPGL commands, as well as additional commands developed by us to support your laser projector (Z-commands). These can be found in the following appendix:

ScalingX:	Factor, to which the HPGL file is to be scaled in the x-axis
ScalingY:	Factor, to which the HPGL file is to be

	scaled in the y-axis Please note that an increase in the scaling factor could lead to discrepancies and inaccuracies.
OffsetX:	Shifting of the HPGL file as it relates to the projection point of origin in the x-axis in (mm)
OffsetY:	Shifting of the HPGL file as it relates to the projection point of origin in the y-axis in (mm)
Rotation:	The rotation of the HPGL file around the programmed point of origin, measured in degrees
ShowTextObject:	Enables the fading in- and out of a text in the HPGL file
ProcSettings (UserSettings):	Once the check box under "**User Settings**" is ticked, any new settings that have been made in Scaling, Offset and Rotation will be taken into consideration. Otherwise, the standard values apply as before: Scaling =1, Offset = 0 and Rotation =0.
Explode Text:	When activating the check box, the edges of the text will dissolve – this usually occurs with older LP models. On the other hand, when deactivating the check box, the HPGL Text Object gets used instead.
Use ZH-height:	When the command "ZH" occurs in the HPGL file, only the last "ZH" command is forwarded to LPM as a height offset command "/HA" (please refer to chapter HPGL- and Z-commands)
ProcSettings (HPGLSettings)	Once the check box in HPGL Settings is activated, HPGL commands, such as R0, are interpreted (also see chapter Projector Commands), corresponding with the set-up of drawing files.

DXF Import modules

To set up the DXF import filter, go to the menu bar and open the menu entry: "**File → Options → Import → filter**". The "**Modify Import modules**" window should open.



Select the entry "**DXF Standard Files**" and click on "**Set-up...**"

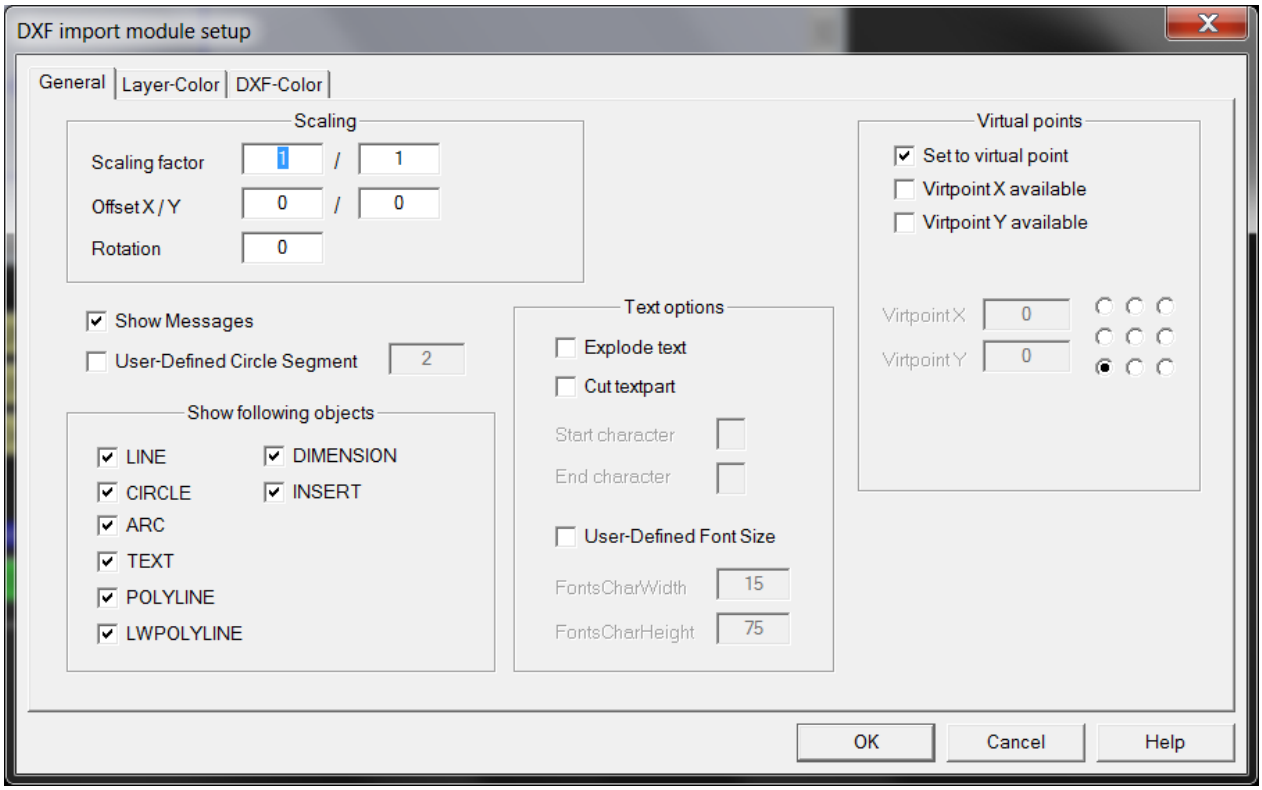
The "**DXF import module set-up**" window should open, which, in turn, is divided into 5 registers: **"General", "Layer Buffer", "Layer Colour", "Object Colour", and "DXF colour".**

During the installation of the LPM, you can install various DXF import modules. Depending on the installed module, different registers are available in the DXF import set up:

1. DXF Standard: General, Layer Colour, DXF Colour
2. DXF Spline: General, Layer Colour, DXF Colour
3. DXF Objects: General, Layer Colour, DXF Colour
4. DXF Buffer: General, Layer Buffer, Layer Colour, DXF Colour
5. DXF All: General, Layer Buffer, Layer Colour, Object Colour, DXF Colour.

You can set up different parameters, which are explained as follows:

Register "General":



Scaling:

Scaling Factor:	The factor by which you can scale (maximise/minimise) the DXF file.
Offset X/Y/Z:	The file is shifted by the value entered, while considering the scaling
Rotation:	The rotation of an object

Show messages:	When the check box is activated, an error message will appear should the DXF file be defective.
User defined circle segment:	Optimising the single lines on which the circle shape is built. The higher the value entered, the lower the resolution or interpolation of the circle. This indicates that primarily longer lines are taken into consideration. The circle becomes angular.

Show following objects:

You will see a list of supported CAD objects. Select the objects you want to project with your DXF file. Please note that the CAD object "Text" only allows you to project multi-line

texts, no single line texts.

Virtual Point:

The virtual point function searches the DXF object using the selected method to find the coordinates, which lie closest to the virtual point of the object. Subsequently, the object gets moved to the virtual point.

Set to virtual point:	Here you can select a virtual point from the DXF files and move it to an adjustable coordinate.
VirtpointX available:	Activate the check box if you want to determine the x-coordinate of the virtual point yourself.
VirtpointY available:	Activate the check box if you want to determine the y-coordinate of the virtual point yourself.
VirtpointX:	Enter the x-coordinate of the virtual point here.
VirtpointY:	Enter the y-coordinate of the virtual point here.
VirtpointZ:	Enter the z-coordinate of the virtual point here.
	Grip point for the virtual point. The DXF import allocates 9 positions, that can be used for relocation. For example: Depending on the Grip Point, taken from the drawing's overall data, the bottom left corner gets moved to the entered point zero. Likewise, the top middle point will be moved to a fixed point.

Text Options:

Explode Text:	Resolves the text in line segments. This is implemented when using an external document. This typeset is found in the Import.VF file and can be modified. The parameter FontFile is the reference to the

	font file in LPM.INI. Should Explode Text not be set, the text elements will be transferred directly to the projector, and will be projected from the internal font. NOTE: Not every text that has been admitted in DXF can be shown.
Cut Textpart:	Cut Textpart can crop entries made in the text field. This includes the parameter Start and End characters. When a start character is defined, only the selected text will be shown from that point, up until the end character.
Startcharacter:	Enter the character to mark the point, from which the text object can be cropped. Please note upper and lower case spelling.
Endcharacter:	Enter the character to mark the point, from which the text object can be re-displayed. Please note upper and lower case spelling.
FontsChartWidth:	Defines the width of the character.
FontsChartsHeight:	Defines the height of the character when scaling the set up co-ordinations.

Filter Settings:

Various filters can be set here. These are “Layer to Buffer”, “Layer to Colour” or “Object to Colour”. The filter settings determine whether the single DXF layers are to be stored on image memory (Buffer) or highlighted with the use of different pen colours.

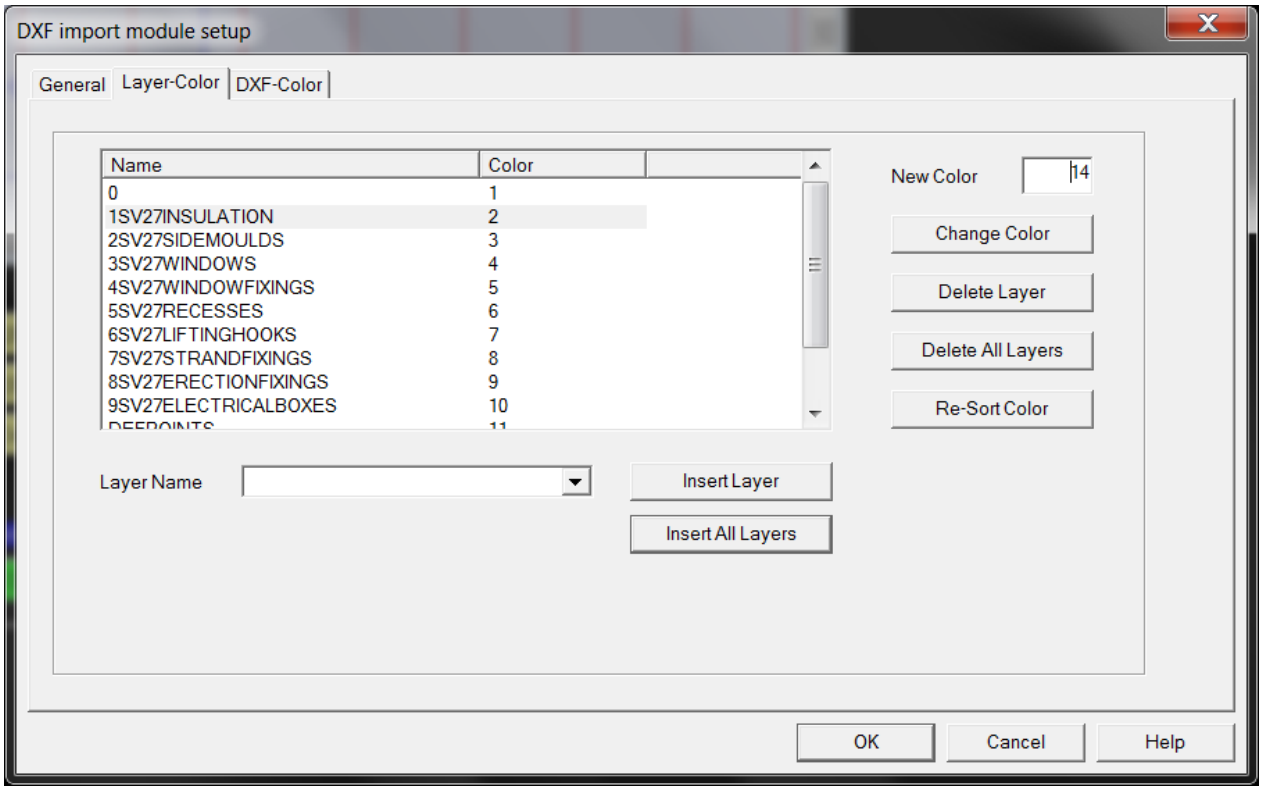
Should, for example, contours that have been saved under separate DXF layers be moved/rotated individually, the program will advise the Layer to Buffer setting.

Layer to Colour is used when the individual DXF layers need to be allocated to specific pen colours (Colour). Layer to Colour mode is used when working with complex files that have already been divided up into different layers, and that can be either projected or turned off by activating the particular pen colours.

Rotate Object around Axis:

Rotation of the axis X, Y, and Z

Register „Layer-Buffer“:



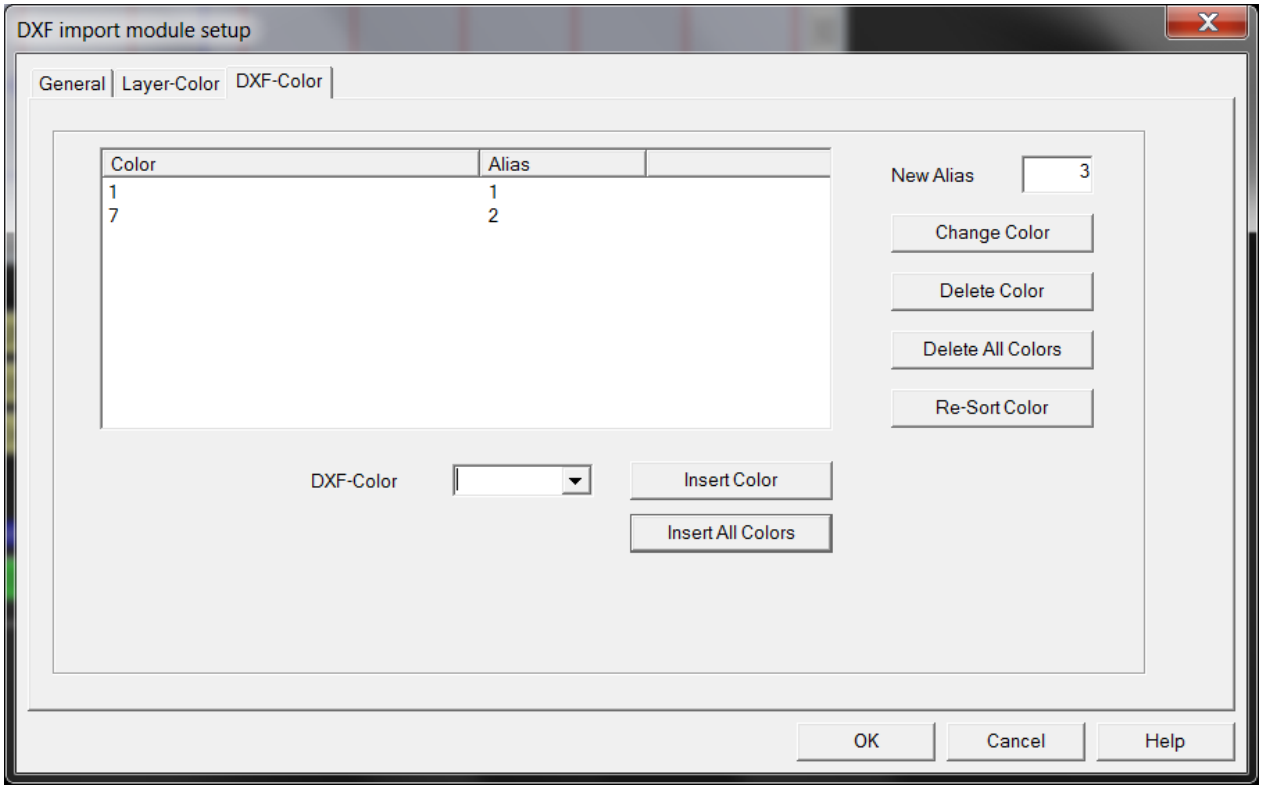
Every layer can be downloaded selectively onto any image memory. This makes it possible to manipulate individual elements in a drawing.

If the extension module DXF Changer is used alongside the DXF Import, a new, modified, but similar, file can be written after the data has been successfully manipulated.

Layer name:	Here you can choose the layer/s from the LPM 8 DXF file and re-assign its pen colour.
Insert Layer:	click on this button to select the layer in “Layer name”. The selection will be visible in the action window.
Insert All Layers:	Inserts all layers currently active in the DXF file to the action window.
Change Buffer:	Changes the buffer setting
Delete Layer:	Deletes a layer selected from the action window
Delete All Layers:	Deletes all inserted layers from the action window
Re-sort Buffer:	Resets the numbering of the layers to their original settings. It is listed in ascending order, starting at 1.

Register "Layer-Colour":

In the register “Layer-Colour” you can reassign the pen colour of a particular layer. These settings will subsequently be used for every layer with the same name, irrespective of the DXF file.



Please note:

The colour choice will be determined by the DXF file, should no colour be allocated specifically.

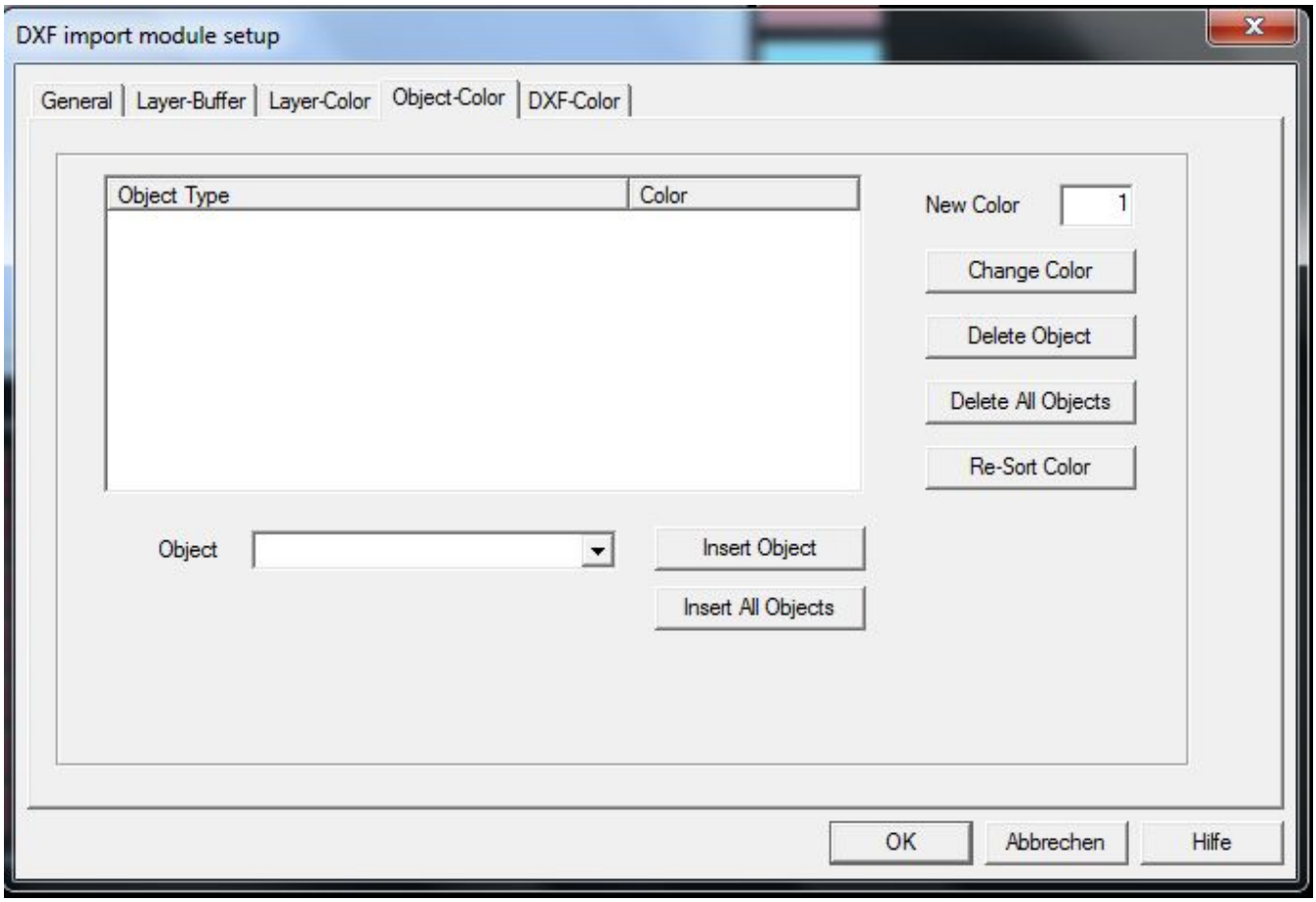
Layer name:	Here you can select the particular layer from the LPM 8 DXF file, and choose its designated pen colour.
Insert Layer:	Click on this button to ultimately select a layer from “Layer name”. The options are shown in the action window.
Insert All Layers:	Adds all layers from the DXF file that are currently active to the action window.
Change Colour:	The pen colour can be changed under “Change Colour”, once a colour has been selected in “New Colour”.
Delete Layer:	Deletes a layer from the action window

Delete All Layers:	Deletes all inserted layers from the action window.
Re-sort Colour:	Resets the numbering of the layers to their original settings. It is listed in ascending order, starting at 1.

Under field selection “**Pen Colours**”, you can select a pen colour that you want to use for one or more layers from your DXF file.

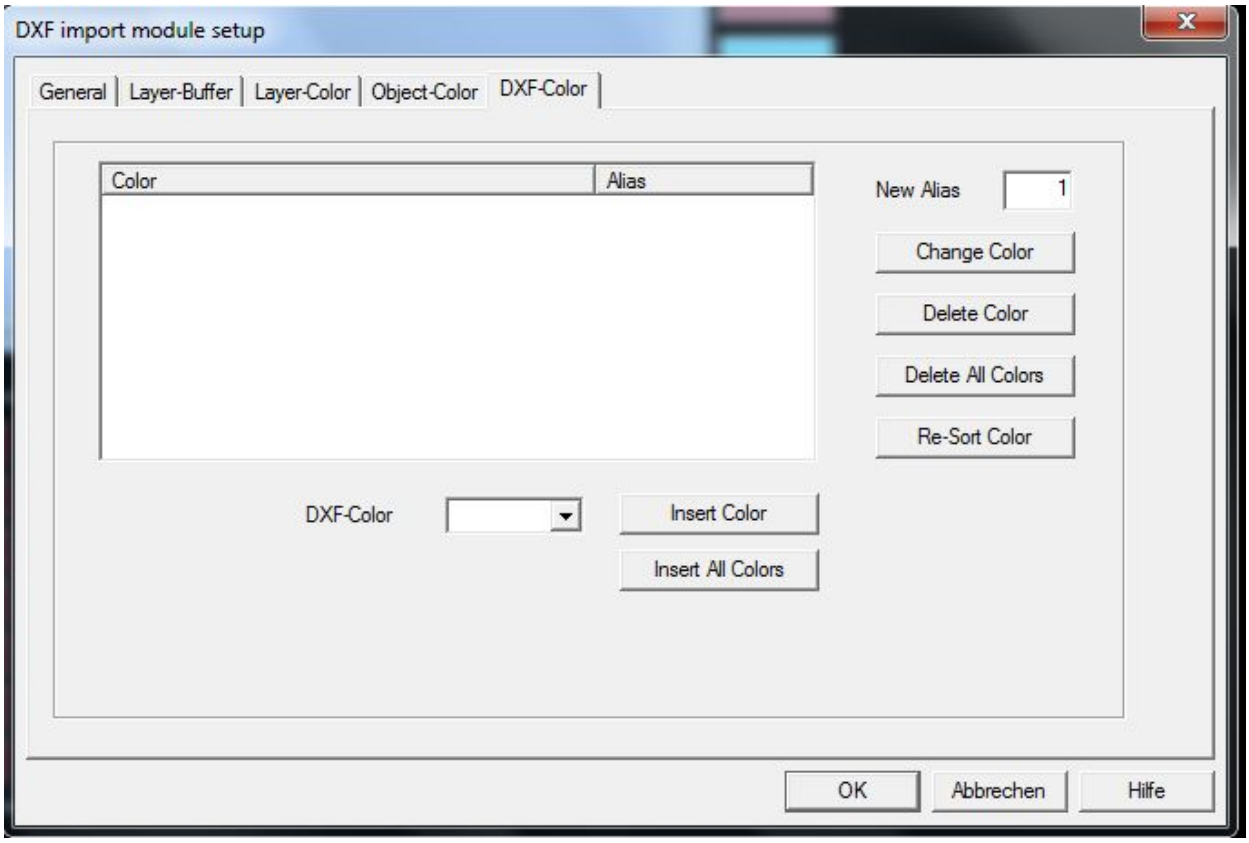
Change Colour:	Click on this button to select the pen colour, previously specified from “New Colour”, to be used for a particular layer. You can select more than one layer simultaneously, by holding the CTRL key, clicking on your chosen layer on the action window, using the left click on your mouse.

Register „Object Colour“:



Here you can assign individual objects to pen colours. Objects are standard elements from the DXF definition line, circle, spline curve etc.

Register „DXF-Colour“:



In the register “DXF-Colour” you can reassign the pen colour of a particular object, irrespective of the layer. The clear assignment of a pen colour enables an additional differentiation between objects found on the same layer.

In this way, you can change a pre-set pen colour in the CAD program by matching a new pen colour to it for LPM. The pen colour amendment will be changed on every loaded DXF file. Every pen colour will be defined by an ACI number in the CAD program, which is a number between 1 and 255.

However, standard names for pen colours are only used for the numbers 1 to 7.

The numbers have the following meanings:

1. Red
2. Yellow
3. Green
4. Turquoise
5. Blue
6. Magenta
7. White/Black

The allocation can be altered in the LPM.INI configurations file:

DXF-Colour:	Enter the pen colour number, as designated from the CAD system, to change its colour allocation.
Insert Item:	Click on this button, to allocate the pen colour number given in “DXF Colour”, to a pen colour selected under “Pen Colour” and insert into the action window.
Delete Item:	Deletes the pen colour number from the CAD system.
Change Colour:	Matches the pen colour set up in “Pen Colour” with the colour selected in the action window.
PenColour:	
Off:	Allocates no pen colour
1.....16:	Allocates the pen colours to 1....16.

Scripts in LPM

You can save image arrangements in LPM as scripts in order to open them at a later stage. In addition to that, you can animate scripts so that different objects can be faded in or out or rotated one after the other. You will learn how to generate and modify a script in the following chapter.

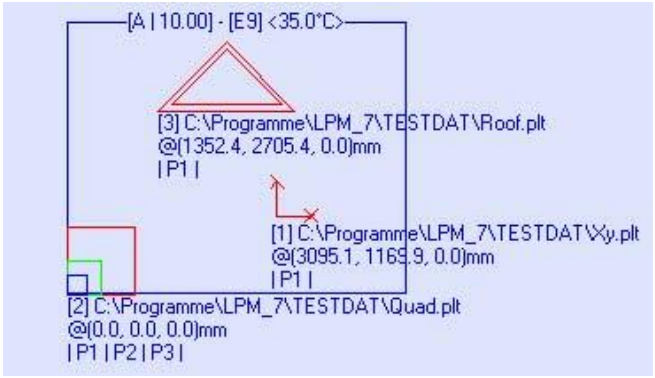
Generating a Script

In this paragraph, you will be able to see an example on how to generate a script.

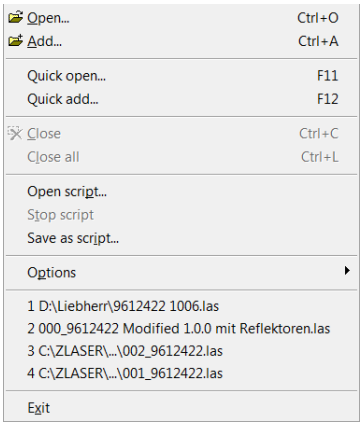
1. Open LPM 8

2. Load the test files QUAD.PLT, ROOF.PLT and XY.PLT, which you will find in the path C:\Program Files\LPM 8\TESTDAT.

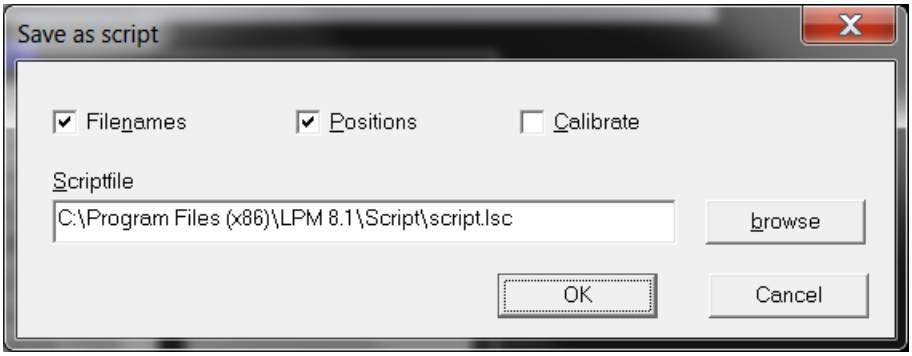
3. Now place the three images in such a way that they do not lie on top of each other and that the file information is visible.



4. Open the file menu in the menu bar and select “Save as Script...”



5. The following dialogue window appears as “Save as Script”



Make sure that the check boxes “Filenames” and “Positions” are ticked.

Select the button “Browse” to name the path under which the script should be saved.

Name the script. In this example, we named it “Testscript”. Select the “OK” button.

The script is now saved.

Close all images that have been previously loaded.

You can open the script again by opening the file menu in the menu bar and select “Open”. You will now see the images displayed as you had saved them.

In the next part of the chapter, you will learn how to “animate” your script.

Understanding the commands

In this section, you learn the basics of script animation, with which you can fade in or out, rotate, or shift images one after the other.

This section will help you to slowly familiarise yourself with the script commands you need to generate a script outside of LPM 8 with any type of test editor e.g. Notepad by Windows.

In order for the file to be loaded by LPM 8, the file must have the suffix .LSC.

To take your first step, open the file “Testscript“ with Windows Notepad.

You will see the following entry which may look different, depending on the order in which you loaded the images and their position prior to saving the script:

```
REM ***************** SCRIPT START *****************
REM Created 19.10.2006 10:53:42 by LPM

:START

CLOSEALL

LOADBUF 1 C:\Programme\LPM 8\TESTDAT\Xy.plt

EXEC /P# 1 /MA 38496 13000 /RA 0 /RZA 0 /HA 0 /MV0 /MH0

EXEC /P# 1 /GE 1 /GE 2 /GE 3 /GE 4 /GE 5 /GE 6 /GE 7 /GE 8 /GE 9 /GE 10 /GE 11 /GE 12 /GE 13 /GE 14 /GE 15 /GE
16

LOADBUF 2 C:\Programme\LPM 8\TESTDAT\Quad.plt

EXEC /P# 2 /MA 0 0 /RA 0 /RZA 0 /HA 0 /MV0 /MH0

EXEC /P#2 /GE 1 /GE 2 /GE 3 /GE 4 /GE 5 /GE 6 /GE 7 /GE 8 /GE 9 /GE 10 /GE 11 /GE 12 /GE 13 /GE 14 /GE 15 /GE
16

LOADBUF 3 C:\Programme\LPM 8\TESTDAT\Roof.plt EXEC /P# 3 /MA 4158 24968 /RA 0 /RZA 0 /HA 0 /MV0 /MH0

EXEC /P# 3 /GE 1 /GE 2 /GE 3 /GE 4 /GE 5 /GE 6 /GE 7 /GE 8 /GE 9 /GE 10 /GE 11 /GE 12 /GE 13 /GE 14 /GE 15 /GE
16

END REM ***************** END OF SCRIPT *****************
```

The script should be modified later in such a way that the objects are faded in and out one after the other repeatedly in the order XY.PLT, QUAD.PLT and ROOF.PLT.

At the same time, the single pen colours of the QUAD.PLT file should also be displayed one after the other.

You will also see an explanation of the used commands. For an overview of the **script commands**, look here.

```
REM ***************** SCRIPT START *****************
REM Created 20.05.2007 14:09:03 by LPM
```

This is the header of your script. It contains, for example, the date and time when the script has been generated.

:START	signals the beginning of the script.

CLOSEALL	closes all files currently loaded in the plot buffers.
LOADBUF 1 C:\Program Files\LPM 8\TESTDAT\Xy.plt	loads the file Xy.plt from the given folder into the plot buffer 1.

EXEC /P# 1 /MA 34000 11750 /RA 0 /RZA 0 /HA 0 /MV0 /MH0	
<EXEC>	The file Xy.plt is executed. This command is an identifier in the script language which is followed by one or more commands for LPM 7.
<P# 1>	The file is loaded into the the plot buffer 1 and moved by the amount x 1/10 mm and y 1/10 mm <MA 34000 11750>.
<RA 0>	The file is not rotated around the centre or a already made rotation around the balance point is set back to 0.
<RZA 0>	The file is not rotated around the point of origin or a already made rotation around the point of origin is set back to 0.
<HA 0>	The height offset is set back to the calibration level.
<MV0>	The file is not mirrored vertically at the centre point or a mirroring around the balance point of the object will be reset.
<MH0>	The file is not mirrored horizontally at the centre point or a mirroring around the balance point of the object will be reset.

EXEC /P# 1 /GE 1 /GE 2 /GE 3 /GE 4 /GE 5 /GE 6 /GE 7 /GE 8 /GE 9 /GE 10 /GE 11 /GE 12 /GE 13 /GE 14 /GE 15 /GE 16	
<EXEC /P# 1>	The object in plot buffer 1 is executed.
<GE 1 /GE 2 /GE 3 /GE 4 /GE 5 /GE 6 /GE 7 /GE 8 /GE 9 /GE 10 /GE 11 /GE 12 /GE 13 /GE 14 /GE 15 /GE 16>	The pen colours 1 – 16 are switched one after the other.

The listed commands are also used for the other two images, QUAD.PLT and ROOF.PLT.

Modifying Scripts

As you know understand the meaning of the script commands in the previous section, you will now learn how to configure the script file.

Open your script file in Notepad.

Rewrite the file as follows and save it afterwards:

REM ********** SCRIPT START **********

REM Created 20.05.2007 14:09:03 by LPM

Do not change any entry in the headline.

CDLY 1500	Command delay after executing a command in [ms]. The command delay here is 1,5 s.
CLOSEALL	closes all files currently loaded in the plot buffers.
LOADBUF 1 C:\Program Files\LPM 8\TESTDAT\Xy.plt	loads the file Xy.plt from the given folder into the plot buffer 1.
EXEC /P# 1 /MA 38496 13000 /RA 0 /RZA 0 /HA 0 /MV0 /MH0	executes the listed commands in plot buffer 1. The file is moved to the default position. Rotation, height offset and mirroring will be reset.
LOADBUF 2 C:\Program Files\LPM 8\TESTDAT\Quad.plt	loads the file Quad.plt from the given folder into the plot buffer 2.
EXEC /P# 2 /MA 0 0 /RA 3500 /RZA 0 /HA 0 /MV0 /MH0	executes the listed commands in plot buffer 2. The file is moved to the default position. The file is rotated counter-clockwise in 3500 angle steps around the centre point.
LOADBUF 3 C:\Program Files\LPM 8\TESTDAT\Roof.plt	loads the file Roof.plt from the given folder into the plot buffer 3.
EXEC /P# 3 /MA 4158 24968 /RA 0 /RZA -1800 /HA 0 /MV0 /MH0	executes the listed commands in plot buffer 3. The file is moved to the default position. The file is rotated clockwise in 1800 angle steps around the point of origin.
:START	Jump mark
EXEC /P# 1 /G9	activates plot buffer 1, all pen colours are

	switched on.
EXEC /P# 1 /G0	activates plot buffer 1, all pen colours are switched off.
EXEC /P# 2 /GS 1	activates plot buffer 2, only pen colour 1 is switched on. All other pen colours are switched off.
EXEC /P# 2 /GS 2	activates plot buffer 2, only pen colour 2 is switched on. All other pen colours are switched off.
EXEC /P# 2 /GS 3	activates plot buffer 2, only pen colour 3 is switched on. All other pen colours are switched off.
EXEC /P# 2 /G9	activates plot buffer 2, all pen colours are switched on.
EXEC /P# 2 /G0	activates plot buffer 2, all pen colours are switched off.
EXEC /P# 3 /G9	activates plot buffer 3, all pen colours are switched on.
EXEC /P& 11111111 /G0	activates all plot buffers, the pen colours of all loaded files are switched off.
GOTO START	starts an action loop by jumping back to the jump mark; here: infinite loop.
END	End of script; no further commands follow.

REM ********** END OF SCRIPT **********

Script Commands

For the script generation you have a number of commands.

The most important are listed in the following:

AD X:\directory\file name.ext or ADD X:\directory\file name.ext	Opens the plot file of a special extension out of the path and adds it to the next free plot buffer.
PAD file name.ext or PADD file name.ext	Loads an plot file in the next free plot buffer from the given path under the command "PATH".
CDLY n or CMDDELAY n	Command delay after executing a command. n = 0...60000 ms
CL or CLOSE	Closes the plot file in the current plot buffer.
CLA or CLOSEALL	Closes the plot files in all plot buffers.
CLB n or CLOSEBUF n	Closes one defined plot buffer. n = 1...8
END	Terminates the end of a script.
EXEC	Executes an external Program or command.
GOTO x	Jump to another command x.
P# n	Activates the defined plot buffer. n = 1...8
P& 00000000	Deactivates all 8 plot buffers.
P& 11111111	Activates all 8 plot buffers.
REP n or REPEAT n	Opens a loop of commands. n = repeats (1...8)
ENDREP or ENDREPEAT	Ends a loop of commands.
REM or REMARK	Enables inserting a comment. The comment only is visible in the *.lsc Script file.
QTH n or QUEUETHRESHOLD n	Controls the amount of commands enqueued. If there are more than n commands the

	script execution will be delayed. n = 0...1000
GOSUB x	Opening of a sub-program called "x". A maximum of 8 sub-programs can be opened.
RET or RETURN	Jump back from a sub-program.
PATH X:\path\...\directory	Sets the path for LOAD and ADD commands beginning with the letter "P", e.g. PLOAD or PLOADBUF.
LD X:\path\...\file name.ext or LOAD X:\path\...\file name.ext	Loads the selected plot file.
LD file name.ext or LOAD file name.ext	Loads a plot file with the given extension.
PLD file name.ext or PLOAD file name.ext	Loads a plot file from the path inserted under the command "PATH".
LDB n X:\path\...\filename.ext or LOADBUF n X:\path\...\filename.ext	Loads the selected plot file in the chosen plot buffer. n = 1...8
PLDB n filename.ext or PLOADBUF n filename.ext	Loads a plot file from the path inserted under the command "PATH" in the chosen plot buffer. n = 1...8
LDX p X:\path\...\filename.ext or LOADX p X:\path\...\filename.ext	Loads the selected plot file by naming the part. p = 1.....n
PLDX p file name.ext or PLOADX p file name.ext	Loads a plot file from the path inserted under the command "PATH" by naming the part. p = 1...n
LDBX n X:\path\...\filename.ext or LOADBUFX n X:\path\...\filename.ext	Loads the selected plot file in the chosen plot buffer by naming the part. n = 1...8 p = 1...n
PLDBX n p file name.ext or PLOADBUFX n p file name.ext	Loads a plot file from the path inserted under the command "PATH" in the chosen plot buffer by naming the part. n = 1...8 p = 1...n
ADX p X:\path\...\filename.ext	Adds the selected plot file to the next free

or ADDX p X:\path\...\filename.ext	plot buffer by naming the part. p = 1...n
PADX p file name.ext or PADDX p file name.ext	Loads a plot file from the path inserted under the command "PATH" in the next free plot buffer by naming the part. p = 1...n

Script Error Codes

If any errors occur while executing a script, the errors will be indicated by an error code and the corresponding line number.

The following are 8 different code numbers and their meanings:

1. unknown error
2. Parameter is missing
3. Parameter outside a valid range
4. Label not found
5. GOSUB-encapsulating larger than 8
6. RETURN without GOSUB
7. REPEAT-encapsulating larger than 8
8. ENDREPEAT without REPEAT

Faulty Scripts are ended automatically before being completed.

LPM Configuration

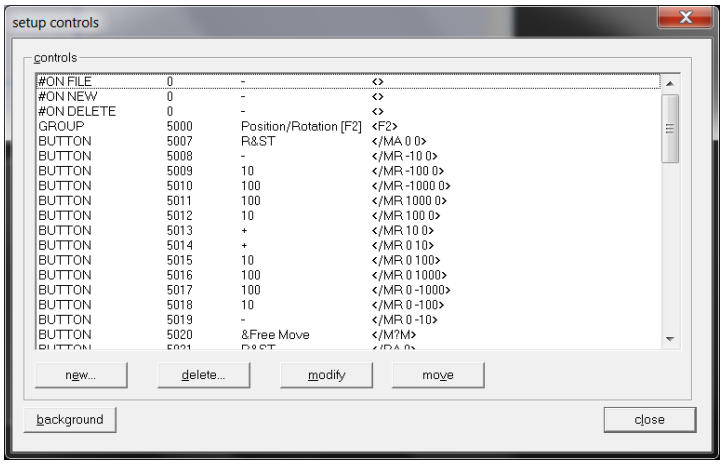
Configuration of the user surface

You can design the user interface for LPM any way you chose. Dialogue windows and control buttons can be added, changed or removed entirely. When installing LPM, you have four sample layouts available to choose from for your desktop. You can change these later, should you chose, in the configuration file LPM.INI.

For this, go to the LPM.INI – paragraph [**General**] to the entry file **CtrlFile=UI\LPM1.CTL.**

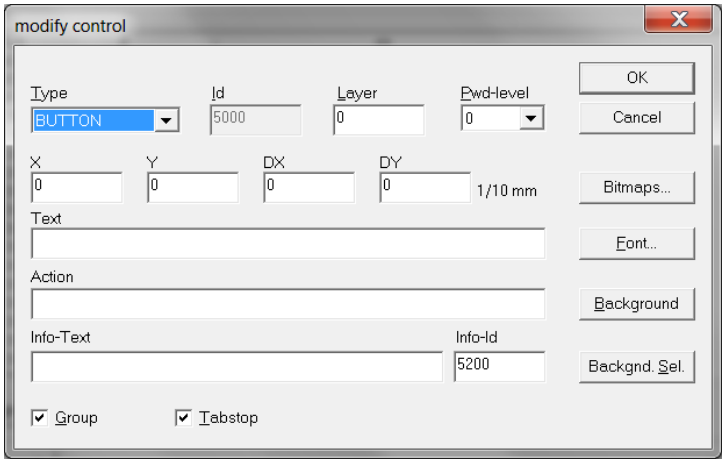
If you chose to use interface 4 instead of interface 1, then change the entry as follows: CtrlFile=UI\LPM4.CTL.

You can make any changes following the installation, or during the operation of LPM, by going to “Set-up controls” in LPM, via File → Options → Control.

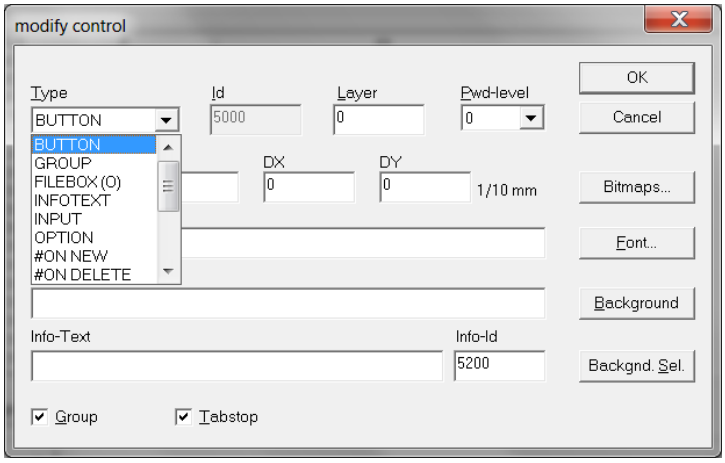


If you want to add a new control element or something similar to your user surface, click the “New...” button.

The “Modify control” window should appear:



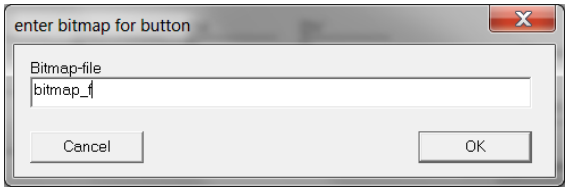
Under “Type”, you can determine the different control elements:



Type: “Switch”	Adds a control button which sets off a certain action.
Layer:	Determines the start-up layer on which the control button should be displayed. You can generate up to 32 different start-up layers with different control buttons for each configuration. Control elements found on the start up layer 0 are displayed on all layers.
X:	Enter the x-coordinates here, where the upper left corner of your control button should be.
Y:	Enter the y-coordinates here, where the upper left corner of your control button should be.
DX:	Enter the width of the control button, in pixels
DY:	Enter the length of the control button, in pixels
Text:	Enter the description which you want displayed on the control button.
Action:	Enter the command that should be executed when pushing the control button. An overview of possible control demands can be found in chapter “Projector commands”.
Info-text:	Here you can enter personal information on the control button.
Info-ID:	Enter an ID number for an info-text element here, that will display information in a specially generated text window, when pushing your customised control button.

Activate the check box “**Group”**, if you want to implement the control button via a quick-start key, like F2....F9. Please note, the type “**Group”** has to be determined and allocated first. Activate the check box “Tab stop”, for accessing the control button or tab stop key, on your keyboard.

Bitmaps: Allows the allocation of a graphic to a control button. The “Enter bitmap for button” window is opened.



Enter the name of the bitmap image here and save it under **UiBmpPath=C:\Program Files\LPM 7\UI\BMP** . Click “OK”. Check the LPM.INI paragraph [**General**] as well as the entry “**UiBmpPath”.**

You will find 6 sample files in the BMP folder. Three of them illustrate an arrow pointing left and the other three shows a white square. You will notice that each bitmap image has a slightly different name: i.e. Name suffixes _d, _f and _u. When you create your own bitmap, **please remember to save it three times** with its respective name supplements. Marking each bitmap with the respective _d, _f and _u is essential for correctly allocating them to the control button.

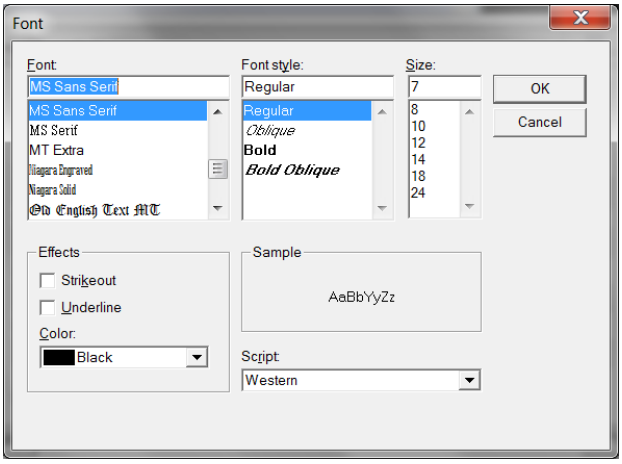
The name supplements/markers mean the following:

_d	Down This bitmap is used when pushing the control button.
_f	Focus This bitmap is used when the focus is heightened with the control button
_u	Up This bitmap is used when neither the control button nor the focus is in use.

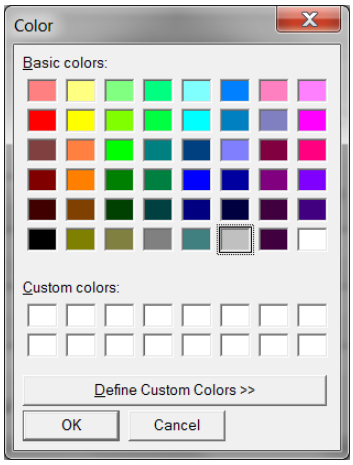
Example: If you want to name your bitmap “Socket”, then save the bitmap in the BMP folder, three times: “**Socket_d”**, “**Socket_f”** and “**Socket_u”**. To consign your “Socket” image to the control button generated by you, enter “**Socket_”** in the “Enter Bitmap for

Button" window. Please do not forget the underscoring line "_".

Font...: Click on this button to change the font type and/or size or any other attribute. The "**Font"** window will open.



Background: Click here to set or determine the background colour of your control button. The "Colour" window is opened.



Background Sel: This defines the background colour of a control button, once pushed.

Type Group: This includes a number of control buttons which mutually set each other off. A group is displayed as a frame on the LPM interface. Above the frame, you will find the corresponding group name. You can select each respective control button within a group by using the cursor key on your keyboard to navigate, and pushing space bar tab afterwards.

Layer:	Insert the number of the layer, which will be used for the group function, here. Up to 32 different layers can be used. Enter the value 0, so that an allocated group function is available for all start up layers.
X:	Enter the x-coordinates here, where the upper left corner of your control button

	should be.
Y:	Enter the y-coordinates here, where the upper left corner of your control button should be.
DX:	Enter the width of the control element “Group”, in 1/10mm
DY:	Enter the length of the control element “Group”, in 1/10mm.
Text:	Enter a description for the control element “Group”.
Selection (F2....F9)	Enter a quick-find function key that will enable you to select the first element in the group quickly. Please note, quick-find function keys such as F1, F5, F11 and F12 are already assigned elsewhere.
Font:	Click on this button to change the font type and/or size or any other attribute. The “**Font”** window will open.
Background:	Here you can highlight the text you have allocated to your “Group” control element.
Type “File box (O)”:	This is an alternative to the File-Open-Dialogue, for when loading or adding an image file. Unlike the type “Filebox”, this is a separate, independent Explorer-window, where you can select paths, files and folders directly.
Type “Infotext”:	Create a field and enter a text that will be displayed within this field once the respective control button has been pushed. Please note, the “ID” of the info text that has been pre-allocated should correspond with the “Info-ID” of your control button.
Type “Input”:	Creates an input button. Windows created via this input should always be grouped. Within this group, you can alternate from one button to the next, which causes the previous control button to automatically deactivate. (Toggle effect)
Type “#ON NEW”:	Determine the control command that should be carried out when opening an image file.
Type “#ON DELETE”:	Determine the control command that should be carried out when deleting an image file.

Type "ON FILE":	Determine the control command that should be carries out when loading an image file.
Type "~REPORT":	Assign a specific report value to a control element in order to activate a function of the laser menu that has been pre-determined by you. Refer to the chapter **"**Laser Menu**"**.
Type "INFOFIELD":	Create an info field on the LPM screen that will display information on the current image data. The "ID" for this field corresponds with the image buffer for which the information is displayed. Under „Text", you can enter the following command: FILENAME shows the name of the file (including the path) that is found in the selected image buffer 1...8. FILEINFO shows only the name of the file that is found in the selected image buffer 1...8 (dependent on the import module) X indicates the x-position of the image file. Y indicates the y-position of the image file. Z indicates the current height offset of the image file. R indicates the angle of the current rotation around the centre RZ indicates the angle of the current rotation around the point of origin. Upper and lower case letters are irrelevant.

Creating a background image

LPM gives you the option of creating a background wallpaper, which will be shown on the graphic interface. This is useful, particularly when you want to light up several working surfaces with your laser projector simultaneously, as well as align your work piece contours directly on your screen on your working space.

By default, the coordinate origin is indicated by a small square with a cross.

There are two ways for you to create your own background wallpaper:

1. Change the file of the standardised background wallpaper. To do this, you open the file **0M.plt** that is found under C:\Programme\LPM 8\UI. Replace the HPGL commands with your own and then save the file.
2. Create a background wallpaper in HPLG code, using Windows Text Editor or Notepad. Save the file as a .PLT file, choosing a random name, under C:\Programme\LPM \UI. Open the LPM configuration file LPM.INI, found under C:\Programme\LPM. In the section [View], you will find the entry **BkFileName=UI\0M.PLT.** Replace the “0M” with the file name you have chosen. Save your settings. Regardless of how you set it up, when you open LPM you should see your new background wallpaper.

Practice:

Please note that the following exercise only gives a small insight on how to create a background wallpaper. The values used are hypothetical.

Theory:

Let us assume that you have 4 working areas onto which you want to project. Two of those surfaces have the measurements 5000mm x 900mm and are parallel to each other. They are spaced apart 400mm. You would like to highlight the one surface in turquoise, the other one in green. Parallel to the two large working surfaces, there are two smaller ones next to each other with the measurements 2000mm x 1500mm. The smaller surfaces are spaced apart 300mm from the bigger surfaces and 1000mm from each other. The two smaller surfaces should be displayed in black and be shaded. The small default square with the cross should be displayed in red.

Open the Windows Editor for this exercise. Type in the following HPGL commands:

PU;

SP1;

PA -100, 0;

PD;

PA 100, 0;

```
PU;
PA 0, -100;
PD;
PA 0, 100;
PU;
PA -50, 0;
PD;
PA -50, 50;
PA 50, 50;
PA 50, -50;
PA -50, -50;
PA -50, 0;
PU;
SP6;
PA -0, -0;
PD;
PA 5000, -0;
PA 5000, 900;

PA -0, 900;
PA -0, -0;
PU;
SP2;
PA -0, 1300;
PD;
PA 5000, 1300;
PA 5000, 2200
PA 0, 2200;
PA 0, 1300;
PU;
SP8;
PA 0, 2500;
PD;
PA 2000, 2500;
PA 2000, 4000;
PA 0, 4000;
```

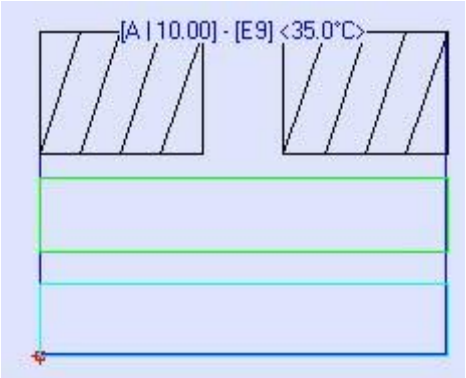
PA 0, 2500;
PA 500, 4000
PA 1000, 4000
PA 500, 2500
PA 1000, 2500
PA 1500, 4000
PA 2000, 4000
PA 1500, 2500
PU;
SP8;
PA 3000, 2500;
PD;
PA 5000, 2500;
PA 5000, 4000;
PA 3000, 4000;
PA 3000, 2500;
PA 3500, 4000

PA 4000, 4000
PA 3500, 2500
PA 4000, 2500
PA 4500, 4000
PA 5000, 4000
PA 4500, 2500
PU;

You will find an overview of the HPGL commands and their relevance, in the HPGL and Z Commands chapter.

- Save the exercise file in .PLT format under C:\Programme\LPM \UI.
- Open the configuration file LPM.INI.
- Go to paragraph **[View].**
- Go to the entry **BkFileName=UI\0M.PLT**
- Rename this entry as **BkFileName=UI\Exercise.PLT.** We have named the newly created file **Exercise**. Instead of 0M or Exercise, you can enter the name you have chosen for the file.
- Open LPM.

The new backdrop wallpaper should look as follows:

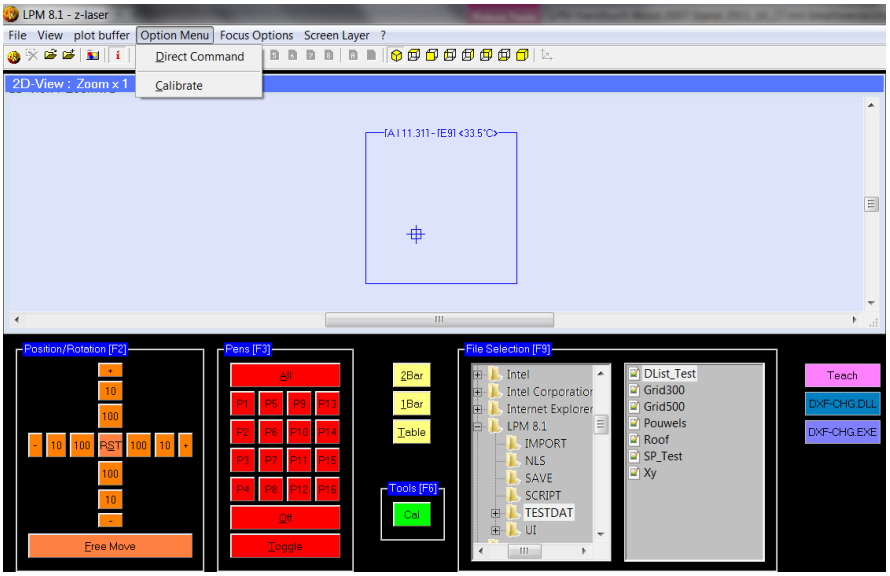


Please note there might be some deviation regarding the field boundaries.

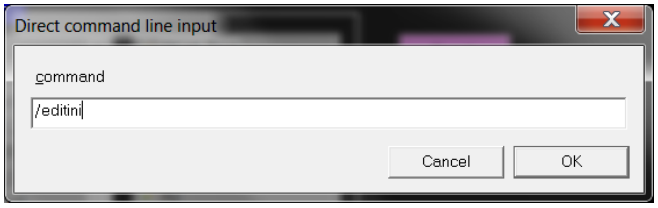
LPM Configuration by LPM.INI

You have the option to carry out various settings directly via the configuration file LPM.INI. You can edit the configuration file LPM.INI as follows:

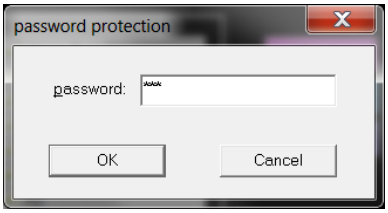
1. Via the Menu Option, go to Direct Command.



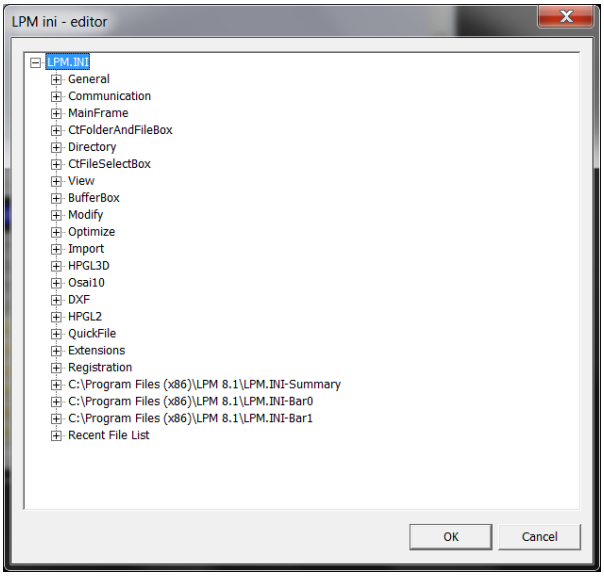
2. The window „Direct Command input" will now open. In the field Command, enter "/editini" and click "OK".



3. You will now be asked to enter your password. By default, this is "cps". Click "OK".



4. The configuration file LPM.INI will now be open. At this point you can change entries, like for example, the company name. You will find the entry under "Main Frame". After entering the changes, click "OK". The configuration file LPM.INI will now close.



In the following, you will find all LPM.INI entries, together with their respective meanings and examples (illustrated in italics). All entries are divided into sections which are written in square brackets […]. This makes finding specific entries easier. Please note, any manual modifications made in the LPM.INI file will not be considered in a repeated installation routine.

[General]	
MenuFile=UI\LPM.MNU	Directory and name of the menu file. The user-defined custom menus, which can be directly configured via LPM, are saved here. For this, open the “Set-up menu” window via File → Options → Menu. Please note: this action requires your password.
CtrlFile=UI\LPM1.CTL	Directory and Name of the Control file. All settings for the user interface are saved here. This is the same LPM user interface that can be selected during the set up. There are four user interfaces. If you want to use a different user interface, change the entry as CtrlFile=UI\LPM4.CTL and select the user interface 4.
NlsFile=NLS\LPM049.NLS	Directory and name of the language-specific resource file. Choose LPM049 for German, LPM001 for English, LPM033 for French and LPM039 for Italian.
FontFile=UI\import.vf	Directory and name of the font file, which is used by LPM.
UiBmpPath=C:\Program Files\LPM 7\UI\BMP\	Directory for program specific bitmaps and animations.
Password=PHB	Password code. Your password will be

	shown in cryptic code. This code may not be changed! Any changes will lead to limited access to various LPM actions.
DateFormat= DD.MM.YYYY	Shows the date format.

[Communication]	
Simulation=1	You can operate LPM in the simulation mode, meaning that communication to a laser projector is not required. 1 = Simulation on 0 = Simulation off
UseCommunication=0	0 = serial communication mode on 1 = network mode via NetBIOS / IPX/SPX - protocol on 2 = network mode via TCP/IP – protocol on
StaticIPAddresses=0	Displays the port number via which the projector is communicated with. When StaticIPAddresses = 1, a connection is established only to those projectors, whose IP addresses have been specified in the entry “IPAddresses”. When StaticIPAddresses= 0, all projectors featured in the net are searched.
IPPort=50000	Specifies the port via which the projector should be addressed by under IP protocol. Valid values are 50000 - 60000.
IPAddresses=192.168.10.10	Shows the IP address via which a connection to the projectors should be established. More than one IP address should be separated by commas.
NetbiosLana=99	Network address for network communication via NetBIOS/ IPX/SPX Protocol. 99 = automatic determination of the network address
NetbiosPlantId=0	The address for a group of projectors which should be used independently from each other within one network.
Port=0	Serial communication interface. 0 = automatic COM port search
Baud=0	Transfer speed to serial interface (bit rate). 0 = automatic determination of the bit rate.
CommandAtProgEnd = /P& 11111111 /BS	Defines the final commands once the program ends. Stops the image projection in all 8 image buffers.
ShowTxErrorInt=1	1 = activated error message in incoming

	data with communication problems. 0 = error message deactivated.
ShowTxErrorExt=1	1 = activated error message in outgoing data with communication problems. 0= error message deactivated.
Slaves=1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0	For any connected laser projector the number "1" represents a projector that is ready to be registered. Up to 16 projectors can be connected, resulting in a sequence of numbers A...P. Slaves=1,0,0,1 means that only those laser projectors with client reference A and D are searched and, where required, registered.
CREFSequenz=A1B2	Uses projector A, with measured reference point 1, and projector B, with measured reference point 2, to reference with the DirectCommand *CREF. The reference points should have been measured up whilst setting up the projectors. Common values for the projectors are A...P, and the reference points 1 or 2.
IdlePollingEnable=1	1 = laser menu inquiry is activated. 0 = laser menu inquiry is deactivated.
ExpertMode=0	Never change this value, or you risk deleting all manufacturer's calibration
; Slave_A ... Slave_P=ClippingArea: Xmin,Ymin,Xmax,Ymax for all Slaves set to 1	This entry is commented out by the preceding semicolon (;) and is therefore not considered as a command. It is merely information on the field boundaries in simulation modus.
Slave_A=0,0,50000,40000	Indicates the coordinates of the laser operating range of projector A in the simulation mode: the corners of the rectangular are at (0/0), (50000/0), (50000/40000) and (0/40000)
Slave_B=50000,0,100000,40000	Indicates the coordinates of the laser operating range of projector B in the simulation mode: the corners of the rectangular are at (50000/0), (100000/0), (100000/40000) and (50000/40000).

[MainFrame]	
rect=0168 0052 0949 0652	Last position and size of the screen window in normal display status.

GrRect=0000 0032 0926 0309	Size and position of the graphic interface.
BkColor=15859736	Colour of the LPM user interface. The RGB-code is calculated as follows: R * 20 + G * 28 +B * 216 Example: You require a grey LPM user interface. In the RGB colour range, grey has the values 126, 126, 126. Thus, the following RGB code can be calculated: 126 * 20+ 126 * 28 +126 * 216 = 8289918
icon=0	This entry corresponds with the entry "max". icon = 1 shows the LPM window as an icon in the status bar. To that end, the entry “max” needs to be set to 0.
max=1	Maximises the LPM window (full screen). max = 0 minimizes the LPM window. When icon=1 and max=0, then the LPM window is minimised and shown as an icon in the status bar.
LPM=LPM 8.0	Specifies the version number of LPM.
CompanyInfo=Z-Laser Optoelktronik GmbH	Displays the name of your company as specified in the set up. The name can be changed manually in the LPM.INI at any time.
StartUpLayer=6	Here you can determine the start-up layer of LPM. In total, there are 32 different layers available.
GetTempEnable=1	Switches on the automatic temperature check. GetTempEnable=0 switches off the automatic temperature check.

[CtFolderAndFileBox]	
LastPath=C:\Program Files\LPM\TESTDAT	Shows the path of the last opened directory. This entry can vary, depending on the selected installation site of LPM and the last opened directory.
FileMask=*	File filter with which you can set up which file formats should be visible in the file selection. * = all file formats plt = all HPGL files dxf = all DXF files cnc = all CNC files etc.

[Directory]	
InitialDir=C:\Program Files\LPM\TESTDAT\	The last directory used in the file box, which was configured in the control area, is saved here.
InitialFltrNdx=9	The last file filter used in the file open dialogue is saved here. The number of the file filter is dependent on the sequence of the modules (filters) previously installed.

[CtFileSelectBox]	
InitDir =C:\Program Files\LPM\Testdat\	This entry can only be changed when the control element "File box(0)" in LPM under File → Options → Control has already been created. The directory last used in the file box is saved.
InitMsk=*.*	File filters which display files in the file box (0), either with a certain file extension or a certain file directly.

[View]	
RotateTrackAroundZero = 0	The centre of gravity can be moved to the point of origin when rotating an object if RotateTrackAroundZero = 1.
UseLPMKlick =1	Using inactivated LPM click, you can select single lines from an object to project them.
ShowLPMKlickAtOnce = 1	Displays the selected line(s) on the screen, which are directly projected by the laser projector. ShowLPMKlickAtOnce = 0 displays only the selected line(s) on the monitor.
TemperatureUnit=0	You can change the temperature unit from °C to °F. 0 = °C 1 = °F
LengthUnit=0	You can change the unit of measurement from mm to dec. inch. 0 = mm 1 = dec. inch
BkTransparent = 0	You can make the clipping zone/field borders fade in or out, i.e. switch to transparent, so that they are no longer visible in the graphic interface.

	0 = field borders on 1 = field borders transparent
FileInfo=1	1 = switches on the file information for every loaded object. 0 = switches off the file information for every loaded object.
ClientInfo=1	1 = switches on the technical information of the projector. 0 = switches off the technical information of the projector.
ExecSubstSign = 0	This entry should not be changed.
CriticalSystemTemperature=40	When exceeding the given temperature in °C, a warning message will appear on the screen.
BkColor=221,227,250	RGB background colour for the graphic interface. Here: light blue.
AreaBrushColor=221,227,250	RGB background colour for the projection interface (interface within field borders/ clipping zone). Here: light blue.
AreaPenColor=0,0,255	RGB colour of the field borders/clipping zone Here: dark blue.
AVIRegister=UI\REG.AVI	Path and file name of the virtual animation monitor during the registration of the projector.
AVIInitConnection=UI\INIT.AVI	Path and file name of the virtual animation monitor when setting up the connection and registration the projector.
InitialScriptDir=C:\Program Files\LPM\SCRIPT\	The last used open-script directory is saved here.
InitialScriptFltrNdx=1	The last used open-script filter index is saved here.
UpdateLaserOnTrack=1	1 = the relocation of an object within the graphic interface is directly displayed bythe laser. 0 = Any change in position is forwarded to the laser only after the object has been relocated.
BkFileName=UI\0M.PLT	Name and path of a standard background object in HPGL format.
BkScaleX=1.0	Scaling factor of the background object with reference to the x-coordinates in [mm].
BkScaleY=1.0	Scaling factor of the background object with reference to the y-coordinates in [mm].

BkOffX=0.0	Offset for the background object in [mm] determined by the entry in "BkFrameShowComplete": “1” in this entry moves the background object relative to the configured background frame in "BkFrame". “0” in this entry moves the background object relative to the smallest X-coordinate.
BkOffY=0.0	Offset for the background object in [mm] determined by the entry in "BkFrameShowComplete": “1” in this entry moves the background object relative to the configured background-frame in "BkFrame". “0” in this entry moves the background object relative to the smallest Y-coordinate.
BkFrame=0,0,5500,4000	Size of the background frame in [mm].
BkFrameShowComplete=0	Scales the background frame so that large drawings which lie outside of the background frame, can also be displayed entirely on the graphic interface. See also BkOffX / BkOffY.
PlotPenColor_1=255,0,0	pen colour 1 = red
PlotPenColor_2=0,255,0	pen colour 2 = green
PlotPenColor_3=0,0,255	pen colour 3 = blue
PlotPenColor_4=255,255,0	pen colour 4 = yellow
PlotPenColor_5=255,0,255	pen colour 5 = magenta
PlotPenColor_6=0,255,255	pen colour 6 = turquoise
PlotPenColor_7=192,192,192	pen colour 7 = grey
PlotPenColor_8=0,0,0	pen colour 8 = black
PlotPenColor_9...16=255,0,0	pen colour 9...16 = red
PlotPenAlias_1...16=P1...16	You can appoint names for each respective pen numbers here. Instead of using current names P1...16, you can also T (for tool), S (for socket), etc...

[Import]	
DefaultExtension= .plt	This file extension is accepted for all file types which are not listed under register.
register=.plt,.dxf	All import modules that were installed during the set up are listed here and are connected with the following corresponding .DLL-file (import filter):

	.plt= HPGL files,Import\HPGL.dll .dxf= DXF files,Import\DXF.dll
Export=export.plt	With this parameter, the data that has been read by the import filter and has been edited for the projector is saved in a file called export.plt. This file contains the exact commands (HPGL) that are sent from LPM to the projector after import. By default, this file resides in the LPM directory. Should you require a different file location, a new path can be prefixed for this parameter.

Projector Commands

For the direct communication with your projector via LPM 8, there is a number of commands which you can insert in the input window "Direct Command". The input window can be opened either with the menu entry “Option Menu” → “Direct Command” or with the Pop-up window that is opened when clicking the right mouse key within the LPM 8-user interface.

The commands are not case sensitive and can be upper or lower-case.

*ABORTSRCCAL	Abort green driver operation point search in progress
*ADR a...p or z a = A...p = P z = all projectors	Address a specific projector for the next command. This is only a valid command when it proceeds the following commands: *CAL, *REF, /REF, /K, /L and *SETUP e.g. Start DCP search for projector K *ADR K *CAL
*CAL	Start search for drift compensation points for all projectors connected to LPM 8, with monitoring. Use the *ADR command to individually search for DCPs.
*CREF	Synchronous referencing of two reference points. LPM waits for information from the projectors. It requires the "CREFSequence=A1A2" entry in the LPM.INI file, under the heading [Communication], where the reference points must be defined. The entry A1A2 refers to Projector A only. When operating multiple projectors, e.g. 3 projectors, Projector A should supply the first reference point and Projector C should supply the second point. Use the command "CREFSequence=A1C2".
*DREF dx1 dy1 dx2 dy2	Relative movement of the 2 reference points (in 1/10 mm).

	The search for reference points only takes place if not previously referenced, or the command *CREF was invoked.
*DELAY n n = 0....∞ ms	Delay in ms. During this delay, no further commands can be sent to the projector.
*DIAG	Used to diagnose the Client
*DVIEW	Activate LPM frame full-screen mode
*GETFRATE	Recall client frame rate
*GETSRCSTAT	Recall green driver status
*GETTEMP	Get the system temperature (automatically every 60 seconds)
*GETTIME	Client runtime inquiry
*NEXTPLOT	Get next file in directory
*PREVPLOT	Get previous file in directory
*REF	Starts referencing for the selected Projector in LPM 8. The command *REF must be preceded by the *ADR command.
*REGISTER	Register all connected projectors listed in the LPM.INI
*SETUP	Start set-up - internal only
*SRCSCHEDULE	Green driver operation point search scheduling
*SRCCAL	Start green driver operation point search
*TRANSPLOT	Upload plot file
*X	XI->XD Test function
/AAX	Same as /AAX 0
/AAX y	Auto align to x-axis. The first edge of the image with the smallest angle to the x-axis is rotated to match the x-axis. This edge is placed at the y-position y (in 1/10 mm), the x position is that of the edge.
/AAX x y	Auto align to x-axis. The edge is placed at x,y (in 1/10 mm)
/AAY	Same as /AAY 0
/AAY x	Auto align to y-axis. The first edge of the image with the smallest angle to the y-axis is rotated to match the y-axis. This edge is placed at the x-position x (in 1/10 mm), the y-position is that of the edge.
/AAY x y	Auto align to y-axis. The edge is placed at x,y (in 1/10 mm)

/BA	Halt projection. The object will remain loaded in the plot buffer.
/BE	Continue halted projection. Reset motion detector (PIR)
/BS	Stop projection. The object will be removed from the plot buffer.
/CAL	Start search for drift compensation points for all projectors connected to LPM 8, without monitoring.
/CANCELTRACK	Rejects the active position in tracking mode and ends tracking mode. The picture moves to position before tracking started.
/CLICKSHOWON	For LPM-click: the new selected parts are shown at once in the projection
/CLICKSHOWOFF	For LPM-click: the click-selection is not shown at once in the projection
/CLICKSHOWTOGGLE	Toggles between the 2 commands from above
/CLICKSEND	The LPM-click selection is shown in the projection
/CLICKOFF	Returns to the normal shown picture and switches LPM-click off
CLIENTINFO n n = 0 or n = 1	n=0: the information from the clients are not shown in the view-window . n=1: the information from the clients are shown in the view-window. Without the parameter: toggles the view This information will be stored in the LPM.INI file
/DCMD?	Opens direct command input via dialogue window
/DELFILE n n = Buffer Number 1...8	Delete File loaded in Plot Buffer 1...8
/DGLIST P1...Pn Pn = Pen Colour 1...16	Creates a dynamic list of pen colours. This list is generated from various import filters, depending on the source file. LPM offers 16 available pen colours and you can arrange them in any order in the list. Example: /DGLIST 1 3 5 12 13 creates a list with the pen colour sequence

	1 3 5 12 and 13.
/DGLIST	Toggles through the list. If the last one is reached, it will restart.
/DGLIST "[ext.cmd]P1 ... Pn"	/DGLIST with extra Commands to Extension. Example: "/DGLIST [$c55.bnk Ab5]1 2 [@c55.clr ChFF]3" creates a list with commands to C55 extension Module. If P1 is selected, command "$c55.bnk Ab5" will also be sent. If P2 is selected, there is no extra command to be sent. If P3 is selected, command "@c55.clr ChFF" will be sent.
/DGLIST "[Bn]"	/DGLIST [Bn] toggles Pen only for specified Picture Buffer. And the chosen Buffer will be activated. NOTE: Buffer number has to be written in square brackets
/DHLIST H1...Hn	Creates a dynamic list of pen colours. This list is generated from various import filters, depending on the source file. The single heights are stored in H1 ... Hn (in 1/10 mm). The first height will be set directly.
/DHLIST	/DHLIST toggles through the above list. If the last one is reached, it will restart.
/DHLIST "[ext.cmd]P1 ... Pn"	/DHLIST with extra Commands to Extension. Example: "/DHLIST [$c55.bnk Ab5]1 2 [@c55.clr ChFF]3" create a list with commands to C55 extension Module. If P1 is selected, command "$c55.bnk Ab5" will also be sent. If P2 is selected, there is no extra command to be sent. If P3 is selected,command "@c55.clr ChFF" will be sent.
/DHLIST "[Bn]"	Toggles Height only for specified Picture Buffer. And the chosen Buffer will be activated. NOTE: Buffer number has to be written in square brackets
/EXEC "<X:\..\Program.EXE> ..." or	Executes an external program (file extension .EXE or .BAT), and overwrites the

/EXEC "<X:\..\Program.BAT> ..."	following parameters: n = Picture Buffer @@Xn@@ current x-coordinate of the zero point @@Y@@ current y-coordinate of the zero point @@FILEn@@ full filename @@NAMEn@@ file name without path and extension @@Rn@@ current rotation around the centre point @@RZn@@ current rotation around the zero point @@DX@@ displacement in the x-axis direction with the mouse or keyboard @@DY@@ Displacement in the y-axis direction with the mouse or keyboard @@DR@@ rotation with the mouse or keyboard @@INI@@ is replaced by the name including the path to the configuration file LPM.INI @@HWND@@ is replaced by the Windows handle to LPM Example: /EXEC "<C:\Programme\INFO.EXE> @@X1@@ @@Y1@@ @@NAME1@@" There are one or more output-created text files containing the new parameters. A prerequisite for the execution of this command is that you have administrator rights on your PC.
/EXECBUFFER "<X:\..\Program.exe> ..."	Execute program and merge parameters nearly as above, but the parameters will be replaced for each active plot buffer. The parameters in detail: @@X@@ @@Y@@ @@R@@ @@RZ@@ @@FILE@@ @@NAME@@ @@INI@@ @@HWND@@ The significance of the parameters corresponds to the command / EXEC. The parameters will be applied for each active frame buffer, which will open a DOS

	window. Example: In the picture buffer 1 is the file: "C: \ WINDOWS \ LPM 8 \ TEST.DXF" at coordinates (10.2/20.1). In the picture buffer 2 is the file: "C: \ WINDOWS \ LPM 8 \ QUAD.PLT" at (0/0). Enter the command with the following parameters: / EXECBUFFER "<X:\..\Program.exe> @ @ file@ @@ @ x @ @ @ @ INI @ @ @ @file @ @ " The input parameter @ @ file@ @ @ @ x @ @ be replaced by: "C: \ WINDOWS \ LPM 8 \ TEST.DXF" 10.2 "C: \ WINDOWS \ LPM 8 \ QUAD.PLT" 0.0 The input parameter @ @ file@ @ @ @ @ INI @ @ will be replaced by: "C: \ WINDOWS \ LPM 8 \ TEST.DXF" with "C: \ WINDOWS \ LPM 8 \ LPM.INI" "C: \ WINDOWS \ LPM 8 \ QUAD.PLT" with "C: \ WINDOWS \ LPM 8 \ LPM.INI"
/EXIT n	Exit client with exit code n n = 1 -> start client in network (NETBIOS/TCP) communication mode n = 99 -> start client in serial communication mode 49999 < n < 60001 -> (TCP/IP mode only) start client with specified port, e.g. "/EXIT 50000" -> start client with TCP port 50000
/EXTERNLOAD	The current picture of the active picture buffer is passed to the modifier, if one is active.
/EXTERNMODIFY n	If a modifier is loaded, it is now active for each frame buffer for the mode n
/F n n = 1...16	Select field n (1 ... 16)
/FA "X:\...\Filename.EXT"	Opens a new file for the next free plot buffer. Path and file name with suffix are required.
/FB n "X:\...\Filename.EXT" n = 1...8	Opens a new file for the plot buffer n (1 ... 8). Path and file name with suffix are required
/FO "X:\..\Filename.EXT"	Opens a new file for Plot Buffer 1

/FMAX s	Maximal repetition rate of projection (in Hz), standard is 30Hz, changing to higher rates can cause a change in laser class!
/FBOXUPDATE	Updates The File and Folder Box
/G n	Pen group n (1...4)
/GA n n = 1...16	Pen n off
/GE n n = 1...16	Pen n on
/GS n n = 1...16	Pen n on and all other off
/G0	All pens off
/G9	All pens on
/GLIST n1...n16	Select next pen from pen list
/H n	Projection height n (1...4) REMARK: Height is stored in the Projector
/HA h	Set absolute projection height 1/10mm
/HR dh	Set relative projection height 1/10mm
/H0	Set projection height to zero
/H?F	Set absolute projection height ask h in feet
/H?I	Set absolute projection height ask h in inches
/H?M	Set absolute projection height ask for h in meters
/HLIST h1...h16	Select next height from height list
/IDGLIST	Inverse of /DGLIST (without parameters)
/IDHLIST	Inverse of /DHLIST (without parameters)
/IX0	Horizontal mirroring at mx off
/IX1 mx	Horizontal mirroring at mx REMARK: Will Mirror on a virtual Line
/IY0	Vertical mirroring at my off
/IY1 my	Vertical mirroring at my
/K0	Partial view off REMARK: will not disable partial view from Laser Menu
/K1 s	Partial view on Reflection ON (2*size) 1/10 mm
/K1 dx dy	Partial view on Reflection ON (2* dx, dy) 1/10 mm

/K1 dx dy tt tt = 5 - 120 µs	Partial view on Reflection ON (2* dx, dy) 1/10 mm tt = TriggerMinTime *)
/K2 tt tt = 5 - 120 µs	Menu on Reflection enable vertical, tt = TriggerMinTime *)
/K3 tt tt = 5 - 120 µs	Menu on Reflection horizontal, tt = TriggerMinTime *)
/L0	Manual clipping off
/L1 x y dx dy	Manual clipping with Position x y and Size dx dy
/LAYER n n = 0...32	Set LPM 8 screen layer (n=0...32) Note: Layer 0 is transparent and will be seen at all times.
/LPMRESTART	Restart LPM instance
/M n	Movement to origin n (1...4) REMARK: Origins are stored in the Projector
/MA x y [z]	Set the absolute movement to x,y [optional height z] 1/10mm
/MR dx dy [dz] relative	Set the relative movement from dx,dy [optional height dz] 1/10mm
/M0	Set the absolute movement to 0, 0
/M?F ft.In16	Set the relative movement; Dialogue Box ask for delta xy in foot. REMARK: Input Format is : Feet Inch 1/16 with an fixed format
/M?I	Set the relative movement; Dialogue Box ask for delta xy in inches.
/M?M	Set the relative movement; Dialogue Box ask for delta xy in meters.
/MATCHX dx	Laser offset relative dx 1/10mm
/MATCHY dy	Laser offset relative dy 1/10mm
/MHT	Toggle mirror horizontally at centre-point
/MH0	Mirror horizontally at centre-point off
/MH1	Mirror horizontally at centre-point on
/MVT	Toggle mirror vertically at centre-point

/MV0	Mirror vertically at centre-point off
/MV1	Mirror vertically at centre-point on
/P0	Disables all plot buffers
/P9	Activates all plot buffers
/PA n	Disables plot buffer n, all other plot buffers keep those states
/PE n	Activates plot buffer n, all other plot buffers keep those states
/PNEXT	Activates the next (filled) plot buffer. If necessary, restarts from the beginning
/PPREV	Activates the previous (filled) plot buffer. If necessary, restarts from the end
/P# n	Set active buffer n = 1...99
/P& bbbbbbbb	Set active first 8 buffers, b = 0 | 1, 1= active
/PF n n n .. nmax	Set allocation of Picture Buffer to Field n=1..16; Activate MultiTool View
/PF0	Remove allocation Picture Buffer to Field; Deactivate MultiTool View
/REF	Start asynchronous referencing
Note: The rotation angle is processed in steps of 1/10 degrees, if the firmware version is 9.4 (or older). The rotation angle is processed in steps of 1/100 degrees, if the firmware version used is 9.5 or newer.	
/RA phi	Rotate absolute phi 1/100 deg around centre
/RR dphi	Rotate relative dphi 1/100 deg around centre
/R0	Rotate absolute 0 1/100 deg around centre
/R?D	Rotate relative around centre, ask for delta phi deg
/RZA phi	Rotate absolute phi 1/100 deg around (0,0)
/RZR dphi	Rotate relative dphi 1/100 deg around (0,0)
/RZ0	Rotate absolute 0 1/100 deg around (0,0)
/RZ?D	Rotate relative around (0,0), ask for delta phi deg
/SAVETRACK	Saves the active position in tracking mode

	and ends tracking mode
/SELECT	Starts object tracking in first plot buffer with picture
/SELECT FIRST	Starts object tracking in first active plot buffer with picture
/SELECT LAST	Starts object tracking in last active plot buffer with picture
/SELECT n	Starts object tracking in plot buffer n (1 <= n <= 8)
/SX dx	Offset X, used from extension modules dx 1/10 mm Note: Can be used for additional offset. The offset is does not appear on the screen, but only on the basis of projection!
/SY *dy*	Offset Y, used from extension modules dy 1/10 mm Note: Can be used for additional offset. The offset is does not appear on the screen, but only on the basis of projection!
/TACD p xo yo	Fine tuning for Auto Calibration for Point p = 1...4
/TOPLEVEL	Set LPM to top level window
/UV rox, roy, rcx, rxy, ph	Transmit Reference-Values (Test and Internal Use Only)
/UR	Resets referencing values that were entered with the command *CREF.
/WACD	Save Fine tuning Parameter to Memory
/WCFG	Save actual configuration
/WINDOW "Title"	Bring window with "windowtitle" to top
/WINEXIT n n = 0...4	Ends Windows as follows: n = 0: EWX_FORCE Shuts down Windows without saving data before. n = 1: EWX_LOGOFF: Exits all programs safely. After that the actual user is logged off. n = 2: EWX_POWEROFF:

	Shuts down Windows and turns off the PC. The system has to support the automatic power off. Under Windows 2000/XP the user has to have the "SE_SHUTDOWN_NAME" rights. n = 3: EWX_REBOOT: Restarts Windows. Under Windows 2000/XP the user has to have the "SE_SHUTDOWN_NAME" rights. n = 4: EWX_SHUTDOWN: Shuts down Windows. Exits all programs safely. Under Windows 2000/XP the user has to have the "SE_SHUTDOWN_NAME"-right.
/WINPOS n n = 0...9	Sets the windows status and position. n = 0: Hides this window and activates the next window. n = 1: Minimizes this window and activates the next window. n = 2: Activates and displays the window. When the window is minimized or maximized the original size and position is restored. n = 3: Activates the window and displays it in the current size and position. n = 4: Activates the window and maximizes it. n = 5: Activates the window and shows it as an Icon. n = 6: Shows the window as Icon. The recently active window stays the active one. n = 7: Displays the window with the current status. The recently active window stays the active

	one. n = 8: Displays the window in original size and position. The recently active window stays the active one. n = 9: Activates and displays the window. If the window is minimized or maximized then the original size and position is restored.

Key Combinations in LPM

In LPM 8, you can use a number of keyboard combinations, with which you can execute actions directly. You will see an overview of these combinations in the following lines:

Key combinations with [Ctrl]:

[Ctrl] + O	Opens the dialogue window "Open a plot file".
[Ctrl] + A	Opens the dialogue window "Add a plot file".
[Ctrl] + C	Closes the selected object and plot buffer.
[Ctrl] + L	Closes all loaded objects and plot buffers.

Key combinations with [ALT Gr]:

[ALT Gr] + Insert	Loads the file pointed in the file box into the next free plot buffer.

For shifting objects using the keyboard you can use following key combinations:

Please note that you have to select the object to be shifted by clicking on it!	
[Cursor]	1 mm
[Ctrl + Cursor]	10 mm
[Shift + Cursor]	100 mm
[Shift + Ctrl + Cursor]	1 mm

For clockwise or counter-clockwise rotations, you can use the following key combinations:	
[Page Down] or [Page Up]	1°
[Ctrl + Page Down] or [Ctrl + Page Up]	10°
[Shift + Page Down] or [Shift + Page Up]	90°
[Shift + Ctrl + Page Down] or [Shift + Ctrl + Page Up]	0,1°

Function keys:

F1	opens the LPM help file
F5	actualizes the entries in the file and folder box

F11	opens the dialogue window "Open File"
F12	opens the dialogue window "Add File"

Key combinations during the quick calibration with Z-EASY:

1, 2 and 3	changes the mouse speed
X	turns around the mouse movement direction in the x-axis
Y	turns around the mouse movement direction in the y-axis
C	swaps x- and y-axis

Laser menu

You are given the option of controlling your laser projector directly via a laser menu. This means that you can create a menu with the Text Editor or Notepad, which can be activated and selected via the reflection of the laser beam. You can reflect the laser beam back to the projector using the reflector coil enclosed in the delivery. The reflection is detected by the reflection sensor that activates the laser menu. This chapter is primarily aimed at experienced users and service technicians/engineers. However, we will explain the following features of the laser menu as clearly as possible.

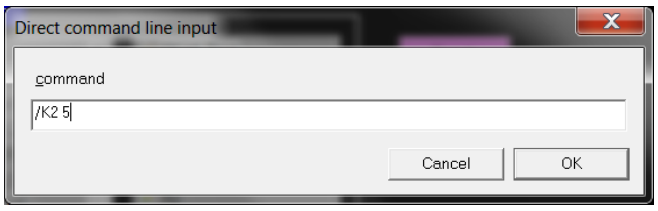
Turning on the laser menu

To activate the laser menu, open the configuration file LPM.INI. Find the section “IdlePolling” in the entry [Communication]. Set the entry “IdlePollingEnable” = 1. This parameter activates the processing of the report values between LPM and the laser projector. This boosts the network activity and should, therefore, only be switched on when the laser menu is being actively used. Save any changes in the LPM.INI file and close it.

Open LPM and wait for your laser projector to be registered. Load an object in an image memory.

Open the dialogue window to enter a direct command, e.g. right-click with the mouse on the blue LPM operating area, or by opening the menu entry OptionMenu Direct Command.

In the command line, enter the command "**/K2 t**" or "**/K3 t**", where “t” represents the time designated in [µs], which is used to activate the laser menu when reflecting a laser beam. 5 µs is given in the example below.



The reflection time “t” is based on the distance between projector and the area of reflection. Values lie, typically, between 5 and 25 µs. Please bear in mind that any reflecting objects may also activate the laser menu. In this instance, the figure for the valid reflection time needs to be increased. Where applicable, you will need to use special reflectors (i.e. bicycle reflectors).

Hold the reflector foil into the path of the laser beam. The laser menu should open shortly after. You can select each menu entry via the reflection.

Below is an example of a laser menu:



Figure 28: Example of a laser menu

Creating a laser menu

This chapter will teach you how to create your own laser menu. Think about which menu headings you want to show and set up a table for them.

Here is a possible example:

1. You want your object displayed in full screen.
2. You want to display only a specific section of your object.
3. You want to load a new file:
    a. You want to open the previous image from the file directory.
    b. You want to open the next image from the file directory.
4. Your object consists of three pen colours. You want to show all pen colours.
    a. You only want to show pen colour 1.
    b. You only want to show pen colour 2.
    c. you only want to show pen colour 3.
5. You want to project your object on different height levels.
    a. You want to display your object on level 0.
    b. You want to project your object at a height offset of 120mm.
    c. You want to project your object at a height offset of 240mm.
6. You want to rotate your object a specific amount.
    a. You want to rotate your object to the left.
    b. You want to rotate your object to the right.

The laser menu is set up in tabular form, using the following elements:

- Ndx
- Ket
- SR
- Value
- Text

Please note: Any commands or texts that follow double slashes (//) will be commented out and ignored by the projector. You have the option to add a comment for each function, making it easier to identify your commands. Please find the meaning of the aforementioned commands for the laser menu in the following table:

-1 // Ndx (0...49)	Index should always start with 0 (TopLevel), a maximum of 49 entries are possible.
-1 // Ket	Chain, refers to Index and, at the same time, points to the next element in the chain; -1 means end of chain; 0 is not allowed.
-1 // SR: 0:	The SR-entry dictates what function should be set up to activate the value-entry. An entry shown under “Value” refers to a sub menu index.
-1 // SR: 1:	Entry under "Value" is a report value.
-1 // SR: 2:	The vanishing point for the menu; the entry under “Value” states to which index point the laser menu should return to.
-1 // Value:	dictates whether the entered value is a report value, or whether it refers to the sub menu index.
-1 // Text	states the actions to be carried out. For most actions there are predefined signs available.
-1 // Comment	For the command you have created, write a comment that will simplify the identification. Please note that each comment written after a double slash (//), will be ignored (commented out).

Now that you have learned the meaning of the single elements, please find below the written laser menu, which corresponds with the requirements mentioned.

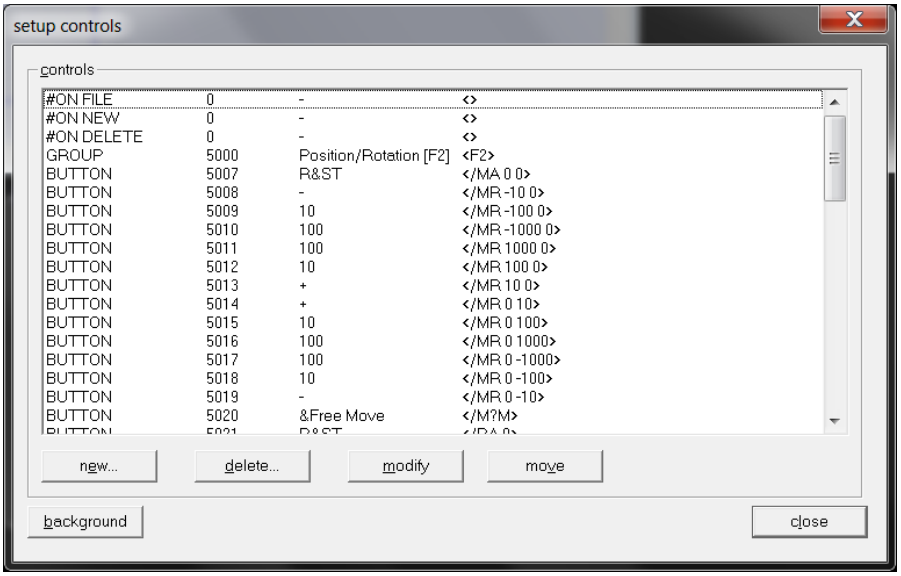
Open the Windows Text Editor or Notepad and copy the following list:

-1 //	Ndx	Ket	SR	Value	Text	Comment
	0	1	1	1	"b"	// object section
	1	2	1	2	"c"	// full-size object
	2	3	0	10	":"	// file selection-sub menu
	3	4	0	15	"Pen"	//pen colours-sub menu
	4	5	0	25	"Height"	// height-sub menu
	5	49	0	30	"a"	// symbol for rotation-sub menu
	10	11	1	10	"?"	// opens the next file in the current file directory
	11	49	1	11	">"	// opens the previous file in the current file directory
	15	16	1	15	"ALL"	// selects all pen colours
	16	17	1	16	"1"	// switches on only pen colour 1
	17	18	1	17	"2"	// switches on only pen colour 2
	18	49	1	18	"3"	// switches on only pen colour 3
	25	26	1	25	"0"	// height reset = 0 mm
	26	27	1	26	"120"	// height offset = 120 mm
	27	49	1	27	"240"	// height offset = 240 mm
	30	31	1	30	"RL"	// rotation to the left
	31	49	1	31	"RR"	// rotation to the right
	49	-1	2	0	"@"	// Escape

Save this file as .MNU format under any desired name; e.g. "Exerciselasermenu.mnu". Make a note of the storage path.

The next step is to match the entries you added under “Value” to LPM 8. You can only activate the respective functions once a control element is assigned to a report value.

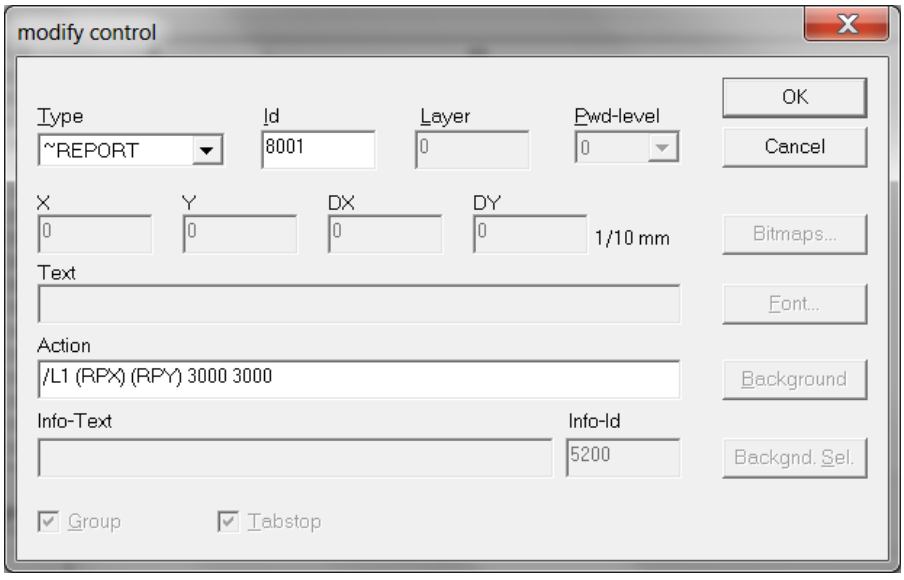
1. Open LPM, as well as the menu entry, “Control”, which you will find under File → Options → Control. Please note: you will need to enter a password to open this menu entry. The dialogue box (window) “set up control” is opened.



2. Click on the button “New...”. The dialogue box (window) “Modify control” is opened. Now you can systematically match up each report value to a function. A prior knowledge about the common **Projector commands** is required.

   Begin with the command for displaying only a section of an image. Please select the entry “~REPORT” under “Type”. Enter the following command into the action line: /L1 (RPX) (RPY) 3000 3000. The ID is derived as follows: The number created in the laser menu under “Value” is added to the standard value 8000.

   Therefore, as derived from the exercise laser menu: 8000 + 1 = 8001.



   Click “OK”. The entry should now also appear in the “Set up controls” window.

3. Add the next control element, which should show the object in full screen again. As before, click on the “New...” button and reselect “~REPORT”. Now insert the projector command /LO in the action line. The ID should come up as 8002. Click “OK”. The entry should be listed again.

4. Proceed, as described, on all “Values” to which an “SP”=1 is allocated, by matching the following projector command to it:

*NEXTPLOT	opens the next file in the current directory
*PREVPLOT	opens the previous file in the current directory
/G9	Switches on all pen colours.
/GS 1	Switches on pen colour 1 only
/GS 2	Switches on pen colour 2 only
/GS 3	Switches on pen colour 3 only
/HA 0	height offset 0
/HA 120	height offset 120 mm
/HA 240	height offset 240 mm
/RA 1000	rotation to the left
/RA -1000	rotation to the right

Please note:

If you are using several projectors, you need to address the individual commands directly to the specific projector that is set up to display the laser menu.

Example: If you want to display the laser menu via projector C, you need to set up the address command before the action command, *ADR C *NEXTPLOT.

5. Once you have matched all the control elements to the report values, with their respective action, the listings will appear in the dialogue window “Set up controls”. Close the window afterwards.

6. Finally, you can scale the letters on the laser menu presentation. For this, please download the file LPC.INI from your projector. For instructions on how to download the file, see “Identification on several projectors”.
7. Open the ELC.INI file with Text editor or Notepad. Search for the entry FontTimer. Behind it, state the time (in [µs]) after which to start the laser menu when reflecting the laser beam. Valid values lie between 50...100. Search for the entry FontScaling. Behind it, state a value (in mm) by how much you want to enlarge the icons and letters in the laser menu.
8. Save and close the file
9. In the window “File Up-and Download”, click on the “Upload” button. Enter the path of the saved LPC.INI file. After the upload is completed, click on the “Arrange re-boot” button.

Predefined characters

The projector comes with predefined characters. You can, however, change these to suit your requirements. All predefined characters and their meaning are listed in the following table:

A...Z	are shown as the corresponding capital letters A...Z
a	Rotation symbol
b	Switches on the object/image section.
c	Switches off the object/image section and displays the object in full screen.
d	Wolf-Logo
0...9	are shown as the corresponding numbers 0...9
Space bar	Leaves a space between the signs.
:	Sign for file menu
>	Sign for previous file (rectangular with arrow up)
?	Sign for next file (rectangular with arrow down)
;	Arrow up
<	Arrow down
=	Rectangular
@	Escape-Sign (Rectangular with an arrow down towering outside)

Troubleshooting

The projector won't start up, i.e. the status display stays dark:

Check the voltage on your power outlet. Are all fuses turned on? Is there enough current supply coming from the power cable?

Is the power cable connected properly to the projector and is the locking mechanism engaged?

If you are using an extension cable, is it switched on? Is the extension cable in proper working condition or does it need to be replaced?

Check the fuses. The fuses are situated near the power switch.

If necessary, replace faulty fuses with an equivalent model 2,5 A Slow blow 5 x 20mm.

The laser projector was working well during the power-on test, but no data transfer takes place afterwards:

Check the settings of the serial/Ethernet interface.

Check that the data that is being transferred to the laser projector is in correspondence with the designated field.

With the help of a separate computer, you can receive data that is being sent from the control PC. For this, use the Windows program **Hyper Terminal**.

Please note that you can use this program only in the serial communication mode.

Make sure that the values received correspond with the set-up of your projection surface.

The laser projector does not always receive data:

Check the data and power line for faults. The earth (grounding) connection for the laser projector should be identical to that of your machines.

Connection breaks down:

Using modern laptops or desk PC's without an integrated serial interface may cause problems when working with USB adapters. Should you require such an adapter, please check with Z-Laser.

Switch the Baud rate to 9600 Bit/s.

Is the data cable too long? Please use a shorter one.

Take care when routing the cable. The data cable must not be wired near other cable lines.

Check that there is sufficient space between them. The crossing of other cables and lines is not critical. However, avoid a parallel mounting of cables.

The yellow diode is switched on and data has been sent to the projector, yet there is no projection visible:

Check that the key switch is set in the "on" position.

Has the motion detector been activated?

You can turn the projection back on with the respective projector command.

Is the image data too small? Check the scaling and the data format.

Check the set-up of the laser projector. It is possible that the laser source is defective.

The red diode is blinking:

Critical temperature has been reached.

When the red diode reaches a projector temperature of 40°C, it will start to blink. Should nothing be done to correct this once the 45°C mark has been reached, the laser source and the galvo-motor will switch themselves off.

Corrective action for this is to fan cool air from the ground to the laser projector, or to connect the system to a water cooling device. In most cases, this problem can be sufficiently solved by supplying the system with fresh air. Please check with an air-conditioning or ventilation specialist.

The ideal ambient temperature lies between 15 – 30°C. The air flow should be at around 40-50m³/h.

When overheating, we recommend that you keep the projector free of voltage.

The desired projection accuracy is not achieved:

In any projection, the laser projector should have an accuracy over 0,6mm/M. If this is not the case, please check the points of your set-up.

If necessary, please re-check the measurements manually.

Make sure the diagonals on your projection surface are identical.

Check the heights of the 4 points: are they level with each other?

Is the image data too small? Check the scaling and the data format.

Check the set-up of the laser projector.

Are you working on a smooth projection surface?

Are there high vibrations in your working environment? Perhaps cranes are moving around in close proximity?

Was the projector mounted firmly and without vibration?

Is the projector perhaps installed too high or too low?

The projector switches itself off after some idle time:

The projector has a set time after which it is switched off. This can be set in the projector configuration file ELC.INI under IdleTimer.

The drift compensation points cannot be found:

Is the distance to the projector greater than 2000mm?

Is the reflector of the detection sensor on the projector dirty? If so, clean it with a soft, clean cloth, taking care not to leave any scratches on the surface.

Have you tilted or moved your working space in such a way that the projector can no longer reach the drift compensation points?

Are any objects standing in front of the drift compensation points, thereby blocking the detection?

After drift compensation, the laser projection switches itself off:

The search for the drift compensation has not been successful.

Clean the drift compensation points or replace them with new ones.

The green LED is blinking and the yellow one lights statically, yet no object is projected:

Switch off the laser projector, wait 60 seconds, then switch it back on.

The laser beam is faint and barely visible:

Intense sunlight makes the laser beam difficult to detect. To rectify this, shade the projection area to make the beam more visible.

Appendix

HPGL and Z Commands

The Hewlett Packard Graphic Language (short: HPGL) is a programming language developed by Hewlett-Packard for the executing of Plot-files. HPGL is a simple language. The commands consist of two capital letters which are followed by one or several arguments. In the following, you will see an overview of the most important HPGL-commands and their meaning, which are supported from the Firmware version 9.55. Some commands are only supported by direct, serial data transfer and are marked by a star (*). All commands are to be separated from each other by a line break or a semicolon “;”.

Small or capital letters are ignored. Spaces are ignored but are used for easier reading. Commands with X, Y, Z are only interpreted in the 3D Option. Values can be inserted as floating point number. However, as the comma sign (,) is used for the separation of single values you have to use instead the decimal point (.) for the insertion of a value with decimal numbers.

Example: 20.75.
The valid range of values lies between -32700 mm and +32700 mm. Decimal numbers should be rounded to 2 units.

HPGL Command Table

PA (Plot Absolute) Notation of the command : PA XVALUE , YVALUE ;	Plot Absolute draws a line from the last known coordinate to the coordinate given by PA x, y, (z) referring to the point of origin. The line is drawn visible after a PD (PenDown) command or invisible after a PU (PenUp) command.
PR (Plot Relative) Notation of the command : PR XVALUE , YVALUE ;	Plot Relative draws a line from the last known coordinate to the coordinate given by PR x, y, (z) referring to the last programmed coordinate. The line is drawn visible after a PD (PenDown) command or invisible after a PU (PenUp) command.
PD (Pen Down) Notation of the command : PD;	Switches to a visible object. The following coordinate is visibly drawn, depending on the preceding command either absolute or relative to the last programmed coordinate.
PU (Pen Up) Notation of the command : PU;	Switches to an invisible object. The following coordinate is not drawn visibly, depending on the preceding command either absolute or relative to the last programmed coordinate. Several coordinates after PU are ignored except from the last one.
SP (Select Pen) Notation of the command : SP n; Range of values : 1-16	The following coordinates are drawn in the pen colour programmed by SP.
CI (Circle) Notation of the command : CI r;	Draws a circle with the radius r around the last programmed coordinate.
AA (Arc Absolute) Notation of the command : AA x, y, j;	Draws an arc of a circle from the last known coordinate around the centre point x, y with the angle j. The angle value can be + positive or - negative. Negative is clockwise.
AR (Arc Relative) Notation of the command : AR x, y, j;	Draws an arc of a circle from the last known coordinate around the centre point x, y with the angle j. The coordinate is relative to the last programmed position. The angle is always absolute. The angle value can be + positive or - negative. Negative is clockwise.

IN (Initialize) Notation of the command : IN ;	Initializes the projector.
LB (Label) Notation of the command : PD; LB TEXT DT ; Range of value : Text is the string to be put out. DT is the programmed label terminator for text.	A text string is sent. The output is executed up to a programmed label terminator or until the next semicolon or line break (ETX). The maximal length of the text is 100 characters. Note: A semicolon cannot be indicated.
DT (Define label terminator) Notation of the command : DT c ; Range of value : c is a displayable ASCII character.	A text put out by LB can be programmed with a label terminator. For standards the label terminator end of text ETX (0x03) is used.
SI (Absolute character size) Notation of the command : SI width,height LB Text	Sets the character size for all following characters. Pre-defined character size is 20mm in X and Y-axis direction
SR (Relative character size) Notation of the command : SR width,height	Changing of the character size referring to the last programmed size or pre-defined size in percent.
DI (Absolute Direction) Notation of the command : DI g, a ; Range of value : g is opposite leg a is adjacent leg	The rotation is done by naming (a/g) and in Arctangent direction. Examples : Rotating about 90° DI 0,1; Rotating about 180° DI -1,0; Rotating about 30° DI 0,0.707;
DR (Relative Direction)	Like DI; but has percentage effects on the parameters.

Z-Command Table

There are some commands custom-built by Z-LASER, which are supported by any of our laser projectors in the serial communication mode.

ZH (Projection- height) Notation of the command : ZH h; h in [mm]	Coordinates programmed after ZH are displayed at the height offset given by ZH. A positive value indicates a height offset towards the laser projector.
ZF (Projection off) Notation of the command : ZF;	Switches off projection. The projection can be switched on with “ZN” again. A re-send of data is not necessary.
ZN (Projection on) Notation of the command : ZN;	Switches on projection which was switched off with ZF. Note: If the calibration has not finished successfully, the projection will not start automatically. A sending of ZN switches on the projection again but without calculating the drift compensation.
ZD (Drift compensation) Notation of the command : ZD;	Drift compensation eliminates thermal and mechanical drift. Note: If the calibration has not finished successfully, the projection will switch off. A sending of ZN switches on the projection again but without calculating the drift compensation.
ZDn dx dy (Scaling of the projection field) Notation of the command : ZDnVALUE dxVALUE dyVALUE	With the command ZD the laser projector can be influenced on its accuracy. The drift compensation basically eliminates thermal and electric drift, which stick to any electronic device. By searching the drift compensation points this drift is eliminated. The extension of the command ZD enables to move the actual drift compensation point. The movement has to be inserted. The moving has linear effects from the selected drift compensation point to all other drift compensation points! Note: The moving is temporary as long as the movement will be saved by the command ZDW. Note:

	To use this function in a useful way we recommend to place the drift compensation points on the later projection level very close to the clipping zone.
ZP (Pen selection) Notation of the command : ZP WERT ; Range of value : 1..16	Switches the projection so that only one pen colour is selected. The attribution of several colours is not supported. A data stream which has been divided by SP can be selected by this process. This optimizes the data transfer and you do not have to send the complete data to the projector again.
ZA (Rotation around point of origin with angle) Notation of the command : ZA XVALUE , YVALUE , RVALUE ;	A plot file is shifted about an offset and rotated. A previously programmed movement is replaced. Example: ZA 100,500,56; The object will be shifted about x=100 and y=500mm and rotated about 56°.
ZO (Shifting of the origin) Notation of the command : ZO XVALUE , YVALUE , ZVALUE ;	A plot file is shifted about an offset. The movement will be replaced by ZA. Example: ZA 100,500; The object will be shifted about x=100 and y=500mm.

ELC.INI

ClientAddress	A	Change the address of the projector by attaching a new letter between A...P. Please note that every letter can be selected only once within one group of projectors.
Baud	9600	Set here the data transfer rate [in bits/s]. Valid values are 9600, 19200, 38400, 57600 and 115200. This entry is only relevant for a serial communication mode and has to match with the entry in the LPM-configuration file LPM.INI.
Port	0x3F8	Gives the Port number of the interface, over which the input and output of data is to be done, as a hexadecimal code. 0x3F8 means COM Port 1, 0x2F8 means COM Port 2. This entry is only relevant for a serial communication mode.
IRQ	4	Interrupt Request Names a protocol, with which data can be transferred from a sender to a receiver. Every COM Port has an IRQ-number: COM Port 1 = 4 COM Port 2 = 3 This entry is only relevant for a serial communication mode. NEVER change this entry.
LanA	0	Internal network number. NEVER change this entry.
PlantId	0x00	Group number with which the laser projector can be made independently usable within a group when using network communication. Given as hexadecimal number.
FontTimer	50	Shows the output speed of the laser menu signs.
FontScaling	35	Shows the letter size in the laser menu. Displayed in [mm], with reference to the set system of coordinates.
RefIntern	0	Makes the **referencing** of the projectors possible. RefIntern = 0 the communication is bound to LPM. RefIntern = 1 the communication is not bound to LPM. This entry should not be changed.
DAPTolerance	60	Shows the maximal allowed variation when doing a multi-search of the drift compensation points in [Increments]. This entry should not be changed.
MultiCalActive	1	1 Multi-search when drift compensation is switched on.

		0 Multi-search when drift compensation is switched off.
ControllerBoardRev	C	Controller Board Revision Every Controller Board has its own command by which it is addressed. This entry mustn't be changed.
TimeOut	1000	Shows the time in [ms], which has to pass without data flow, so that the data sent before can be processed or that you can add further new data.
DisablePir	0	1 = Status display of the movement sensor switched off. 0 = Status display of the movement sensor switched on.
IdleTimer	600	Shows the time in [sec] which may pass in inactivity before the projector stops the image.

Accessories

For successfully operating the laser projector in connection with LPM, there is a large assortment of accessories available.

Extensions:

1. Bar code
2. Polling
3. ULB Polling
4. SAA Polling
5. Database
6. USB PIO
7. Teach
8. NC Trans
9. BWN Sort
10. Disto
11. DXF-Changer
12. BufferBox Poll
13. TigerStop

Import Modules:

1. HPGL
2. DXF
3. CNC (NUM, Siemens, Fanuc, Osai, Albatros)
4. Truss
5. BwES
6. BWO
7. Ply
8. Unitechnik

Teach

Introduction

LPM Teach allows the user to create a new image or alter a pre-existing image in LPM 8. The changes made in this extension DLL file are shown immediately by the projector. LPM Teach allows only the use of Plot Buffer 1, and all other buffers are unused. A mouse, with at least 3 keys and one roll wheel, or a keyboard is necessary to operate LPM Teach.

Integration into LPM

For integration into LPM 8, the following must be written into the LPM.INI:

[Modify]

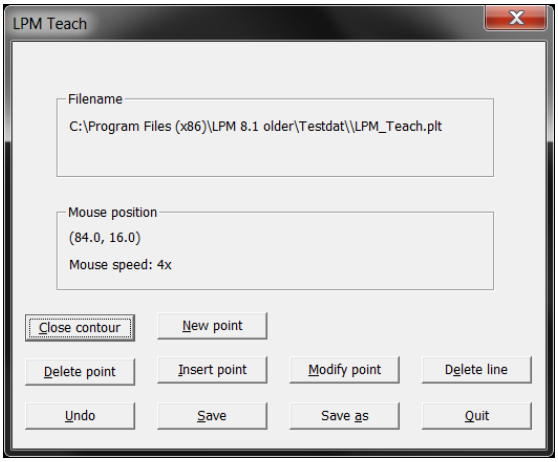
DLL= Programmpfad + LPM_Teach.dll
Length = 80.0

The entry “Length” gives the size of the displayed cross in [mm].

For LPM Teach to be initialised correctly in LPM 8 when an image is displayed, the command “/EXTERNLOAD” must be entered into the “#ON FILE” Control.

Starting Teach

To open LPM Teach, type the command “/EXTERNMODIFY 1”, or use a button for simplicity, after which the following menu is displayed:



At the same time, a cursor that looks like a triangular hour glass appears in the preview window and on the projection of LPM. It shows the current point in the drawing. The current coordinates can be found in the dialogue box.

The mouse speed is an internal conversion factor of the actual mouse movement during free drawing. The mouse wheel speed can be adjusted to 1/16x to 32x speed. A high speed is advantageous when large distances are to be covered on the projection surface, and a low speed, if very subtle changes are to be carried out.

If you want to move to the next point within the drawing, just click the left mouse button (or press a button) or press the Page Up key on the keyboard. To move to the last point, use the right mouse button or press the Page Down key. If you want to return to the starting point within the drawing, then hold the right mouse button down or the Pos1 key on the keyboard. The end is reached by a long press of the left mouse button or the End key.

Function of the buttons

The button **"Delete point"** ensures that the current point is deleted from the drawing. It is possible that the connected lines remain connected, but they no longer run on this point. If there is only one edge, i.e. 2 connecting points exist in the drawing, and then deleting is no longer possible.

Alternatively, pressing the "Del" key or clicking the left and the right mouse buttons at the same time can activate this feature.

The button **"Modify point"** activates the drawing mode, where you can change the coordinates of the current point. This mode can also be activated by pressing the centre mouse button, or by pressing the Return or the Space bar on the keyboard. The cursor becomes a cross in display mode. The point can now be moved by moving the mouse. The step size is dependent on the mouse speed and the strength of the movement. Cursor movement can also be made via the arrow keys on the keyboard. Each increment is moved 1 mm. However, it is possible to change the step size: Simultaneously pressing the "Ctrl" key and an arrow key raises the increment to 10 mm, pressing the "Shift" key and arrow keys raises the increment to 100 mm. If both “Shift” and "Ctrl" key, then the step size becomes 0.1 mm.

If you want to accept the current position in the drawing, it is sufficient to press the left mouse button or the Return key. To return to the stored position, so that the changes are not saved, press the right mouse button or the "Esc" key.

The button "**Add point**" ensures that another point is added.
If the cursor is currently at the end of a polygon, then the new point added to the end of this traverse, otherwise adds the point to the middle of the current and subsequent line.

Alternatively, press the "Insert" or the middle button of the mouse to activate this function. It is then immediately switched to drawing mode. When you are in this drawing mode, hold the middle mouse button or the "Insert" button again, and then the current point accepted and immediately adds a new point. This option is especially convenient, it increases the polygon.

The button "**New item**" is used to draw a new polygon. You have to define two points to project a new line.

The button "**Delete line**" is used to divide a polygon into two and remove the line

connecting the current point and the next point. If the cursor is at the end of a traverse, then it is of course not possible to use this button. The '#' key (in German keyboard layout) is provided as a keyboard short key for this function.

The button "**Undo**" undoes the last change to the drawing.

The "**Save**" button saves the drawing in the file name that is shown in the dialogue box above the button. There is only one HPGL-stream saved! So if you have, for example, previously opened an AutoCAD DXF file, this original file is now saved as a HPGL file and not as an AutoCAD file. This leads to errors later when re-opening the file! Therefore you should use the "**Save As**" button.

The button "**Save As**" is used to save the current drawing in a new file. It is only saved as a HPGL Stream! It opens another dialogue box where the user can select the file name. If the file already exists, you will see a security warning asking if the file should be overwritten.

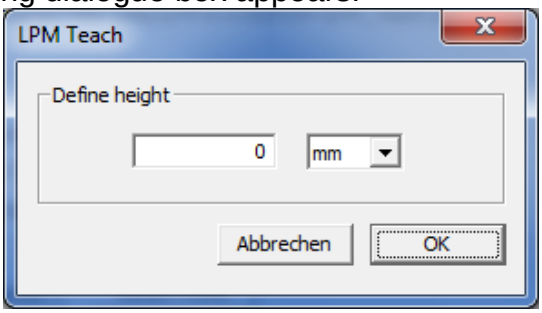
The "Exit" button closes the Teach window. You will also see the projected cursor disappear.

Define the drawing height

In order to define drawing height, you must enter the following in LPM.INI:

[Modify]
UseLPMTeachHeight=1

When you save the following dialogue box appears:



Enter the projection height in [inches or millimetres]. The amount entered will overwrite the previous value of 0 and is included in the HPGL file as a Z-height.

Maintenance of the Laser Projector

1. Cleaning the housing:

The housing should be cleaned regularly to remove dust or other residue. This can be done by using a clean, dry cloth. Extra care should be taken when cleaning the remunerated laser aperture plate and reflective elements. The laser aperture plate should be cleaned with a clean microfibre cloth, without applying too much pressure to the housing content.

2. Cleaning the reflectors:

The reflectors can be cleaned with any cleaning agent. Please take care not to allow cement dust particles on the reflector. This very fine dust destroys the surface of the glass reflector and they become “blind”. Some customers use compressed air, which, in principle, is OK. However, when there is a high content of cement dust, this can ruin the surface in time. Other customers use acetone or Isopropanol on a cotton bud (Q-tip). The procedure for this is to lightly spray the acetone into the reflector, gently washing out the cement dust. The cotton bud is then used to dab it dry afterwards.

3. **Cleaning the filter mats**:

The filter mats, positioned at the laser projector's air in- and outlet, are to be cleaned at least once a month. When heavily soiled, clean more regularly. At the back of the laser projector is a chart designated to manually document the completed cleaning work conducted.



Figure 29: Projector with filter mats



Figure 30: Projector without filter mats

Laser projectors of the LP-HFD2 series, that don't have special water-cooling equipment, have two cover panels at the back of the unit, for the air in- and outlet. These will both be equipped with filter mats on delivery. During standard operation, the filter mat at the air outlet can be removed to improve the cooling capacity.

4. Electrical contacts:

The electrical contacts situated on the projector can be cleaned with customary, oxide-free cleaning agents.

5. Universal bracket:

Check that the bracket is firmly seated. Other than that, no other servicing is required.

6. Water cooling:

Laser projectors that are connected to a re-cooling system with a closed water circuit need to be checked regularly for tightness. A manometer, connected to the re-cooling system, display the current water pressure. The display should be read during normal operation of the machine and should show a pressure that reads between one and two bar. Higher pressure could result in an automatic release of cooling liquid. Pressure that is too low, or air ingress in the cooling agent can cause the cooling capacity to malfunction. Further tips can be found in the cooling system handbook.

7. Overheating:

When the inner parts of the equipment reach a temperature of 40°C, a red LED at the front of the device will start to blink. When the equipment reaches a temperature of 45°C, the projection will turn off as a protection precaution!

8. Control software LPM:

After commissioning, the control software is basically maintenance free and should need no updates. The operating system should, however, be brought up to date with specified Microsoft updates, where necessary.

9. Data protection:

After a successful installation, a copy of the last set up condition should be saved under the index ..\LPM via SAVE. Please note: the automatic save of the quick calibration function does NOT save any changes in the operating parameters.

Comment:

Should the laser projector be moved during maintenance work, you may need to re-install it. This may require the assistance of a certified Z Laser technician.

Technical Data

LP-HFD2

Housing

Dimensions (l; w; h) [mm]:	500 x 200 x 141 mm (181 mm incl. fan) 19.685 x 7.874 x 5.551 in (7.126 incl. fan)
Protection class electrical :	3 (protective low voltage)
Weight :	7.3 kg (plus ca. 1.4 kg for separate power supply)

Voltage Supply

Operating voltage :	24 VDC ±5%
Power consumption (typical) :	50 W (max. 100 W)
Fuses :	2 x TT 2.50 A, 5x20 mm

Data Transfer

Communication type :	Ethernet / Serial 1. Ethernet TP, 2. 3.
Serial :	RS-232 IV24 (max. cable length 15 m)
Ethernet :	Ethernet TP 100 Base TX Cat5/Cat6
Protocol :	TCP/IP NetBIOS over IPX/SPX
Profi Net :	external optional, other fieldbus systems on request

Wavelength
Output power 7 mW (1) 14 mW 7 mW (1) 28 mW
Laser class (on EN 60825) 2M 3R 2M 3R

Laser

Laser colour :	Red 638 nm	Green 520 nm
Laser power :	7 mW (1) or 14 mW	7 mW (1) or 28 mW
Laser class :	2M or 3B	

Mounting Location

Storage temperature:	-5° C up to +60 °C
Operating condition	+5 °C up to +50 °C (with passive cooling) +5 °C up to +55 °C (with cooling air hose) +5 °C up to +60 °C (with adaptive water cooling)
Humidity :	<80% relative, non-condensing

Projection Area

Optical angle :	Max. 80°
Optical resolution :	819 Inc / Degree
Accuracy :	Accuracy (2) (depends on projection

	distance) ± 0.25 mm/m for standard and ±0.1 mm/m for high precision for 33°C operating temperature
Axis resolution: :	16 bit

Repetition Frequency

Maximum:	50Hz
Software :	LPM 9 (Laser Projection Manager)
Graphic format without LPM 8:	HpGl / HpGl-Subset

Installation Protocol

Installation protocol

	Place of Installation :	***Dealers information :***
Company Name :		
Company Address :		
Contact Person :		
Contact Fon / Fax / mail :		
Application :		
Description :		
Special features :		
Date of installation :		

Projectors details :

Serial No.:		
Type :		Purchased by dealer :
Date of purchase :		

Installation environment :

Single Projector :	❍ Yes ❍ No	Connected Projectors Serial No.:
LPM Version :	❍ 6.3____ ❍ 7.____ ❍ 8.____	
Registration No.:		
Serial :	❍ RS 232 ❍ RS 485 ❍ LWL	Cable length :
Ethernet Network :	❍ Crossover (P2P)	Cable length :
	❍ Switch/ Hub	
Network protocol :	❍ **IPX/SPX**	❍ **TCP/IP**
	Plant ID :	Dynamic IP :
	Lan Address :	Static IP :
	Ethernet Frame :	Subnet Mask :
		Gateway :
		Port :
Client release :		

explanatory notes

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Installation protocol

Installed Software :

Operation System : Windows : ❍ 98 ❍ NT ❍ 2000 ❍ XP ❍ Vista Home ❍ Vista Ultimate

Service Pack : _____ Server : _____

	Module name :	**Module Version :**
LPM :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
LPM Modules :		
Hardware accessories :		
Hardware accessories :		
Hardware accessories :		
Hardware accessories :		

Registration Key : _____

Control Software : _____

Name / Manufacturer : _____

Version : _____

contact Person : _____

phone : _____

Data stored : ❍ LPM ❍ Client ❍ sent to service@z-laser.de

location : _____

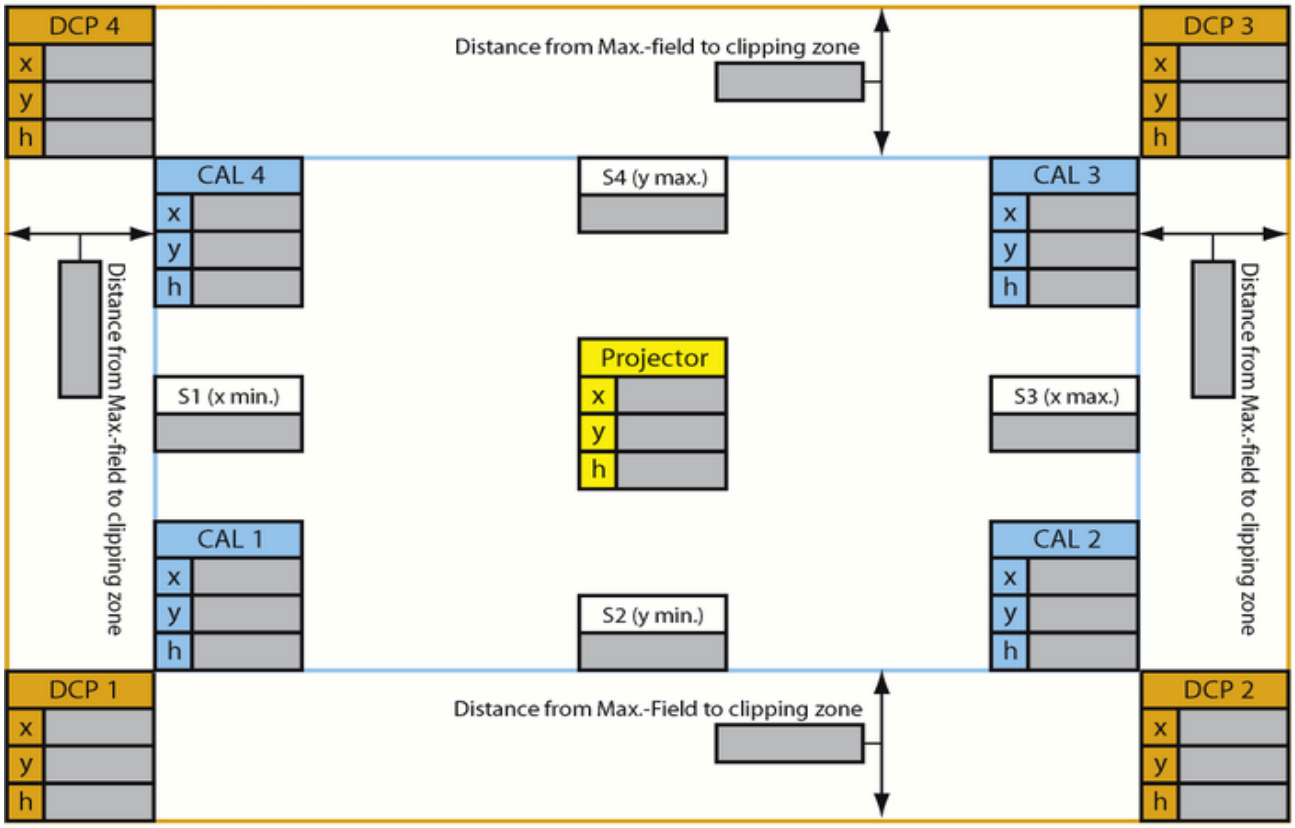
customer contact : _____

Z-Laser Optoelektronik (c) 2009

Installation protocol

Technician		Date	
Company		City	
Country			
Application			
Machine			
Type		Serial number	
Drift compensation	☐ active ☐ not active	Drift compensation points	☐ 2 Points ☐ 4 Points
Number of projectors		Firmware	

Geometrical data

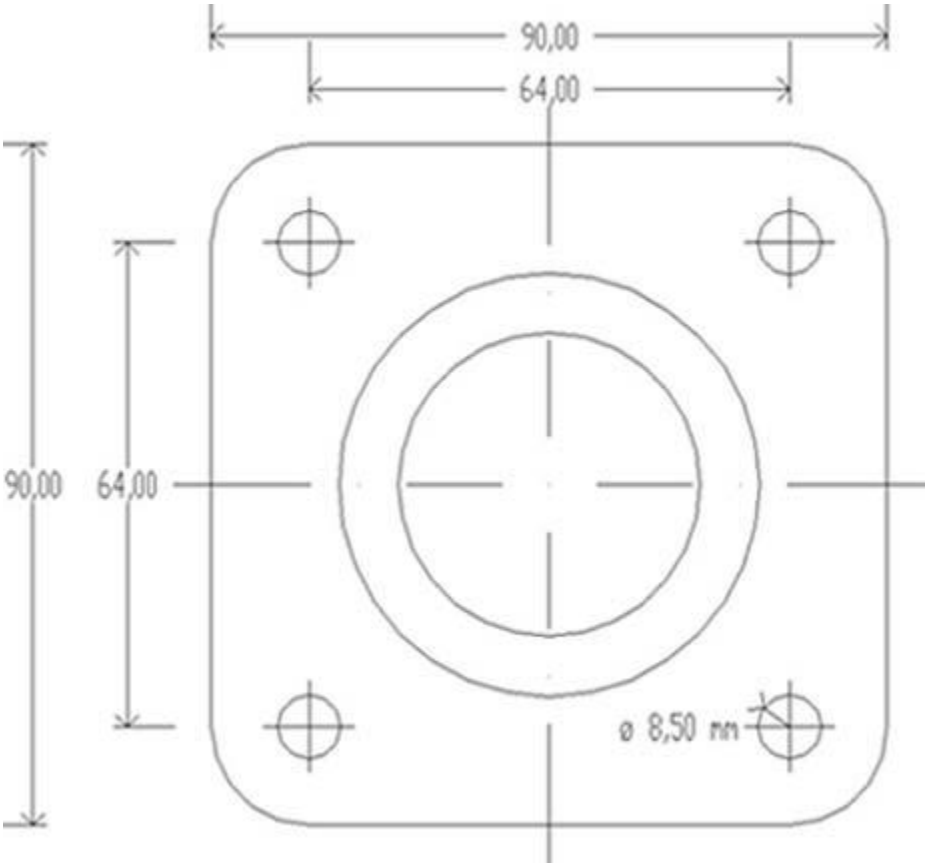


CAL = Calibration marker — = Clipping zone

DCP = Drift compensation points — = Max.-field

Drilling Pattern

The fixing of the laser projectors should be made with the accompanying cardanic mounting kit. The following drawing shows the geometry of the mounting flange.



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